

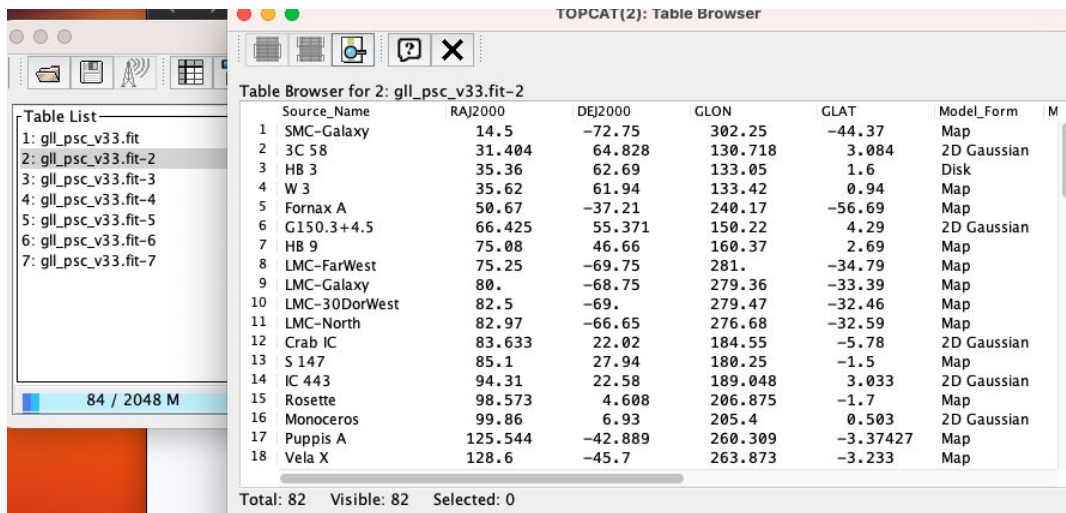
Applying ML/Deep Learning Techniques to Gamma Ray Astronomy

Krishiv Bhatia

Update: 12/08/2023

Update on Progress

- Refreshed Jerry's YouTube presentation
- Downloaded the 4FGL-DR4 FITS file from https://fermi.gsfc.nasa.gov/ssc/data/access/lat/14yr_catalog/
- Installed the TOPCAT tool on my laptop
- Viewed the 4FGL-DR4 FITS file from TOPCAT.



The screenshot shows the TOPCAT(2): Table Browser window. On the left, a 'Table List' sidebar shows several FITS files, with 'gll_psc_v33.fit-2' selected. The main panel displays a table titled 'Table Browser for 2: gll_psc_v33.fit-2'. The table has columns for Source_Name, RAJ2000, DEJ2000, GLON, GLAT, Model_Form, and M. It lists 18 sources, including SMC-Galaxy, 3C 58, HB 3, W 3, Fornax A, G150.3+4.5, HB 9, LMC-FarWest, LMC-Galaxy, LMC-30DorWest, LMC-North, Crab IC, S 147, IC 443, Rosette, Monoceros, Puppis A, and Vela X. At the bottom, it shows 'Total: 82', 'Visible: 82', and 'Selected: 0'.

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Model_Form	M
1	SMC-Galaxy	14.5	-72.75	302.25	-44.37	Map	
2	3C 58	31.404	64.828	130.718	3.084	2D Gaussian	
3	HB 3	35.36	62.69	133.05	1.6	Disk	
4	W 3	35.62	61.94	133.42	0.94	Map	
5	Fornax A	50.67	-37.21	240.17	-56.69	Map	
6	G150.3+4.5	66.425	55.371	150.22	4.29	2D Gaussian	
7	HB 9	75.08	46.66	160.37	2.69	Map	
8	LMC-FarWest	75.25	-69.75	281.	-34.79	Map	
9	LMC-Galaxy	80.	-68.75	279.36	-33.39	Map	
10	LMC-30DorWest	82.5	-69.	279.47	-32.46	Map	
11	LMC-North	82.97	-66.65	276.68	-32.59	Map	
12	Crab IC	83.633	22.02	184.55	-5.78	2D Gaussian	
13	S 147	85.1	27.94	180.25	-1.5	Map	
14	IC 443	94.31	22.58	189.048	3.033	2D Gaussian	
15	Rosette	98.573	4.608	206.875	-1.7	Map	
16	Monoceros	99.86	6.93	205.4	0.503	2D Gaussian	
17	Puppis A	125.544	-42.889	260.309	-3.37427	Map	
18	Vela X	128.6	-45.7	263.873	-3.233	Map	

Update on Progress

- Downloaded ATNF Pulsar Catalogue tar gz file. Followed README to produce the executable psrcat file.
- Followed README
- Psrcat -h works but I get error in the db_file command
- What is this for? How to fix error?

```
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % ls
README          evalwrap.c      libpsrcat.a     psrcat.db
defineParams.c  evalwrap.o      makeit          psrcat.h
defineParams.o  externalCall.c  myrefs.bbl      psrcat_ref
displayOutput.c externalCall.o   parseParameters.c readCatalogue.c
displayOutput.o galcoord_ecliptic.c parseParameters.o readCatalogue.o
evaldefs.h      galcoord_ecliptic.o plotParams.c    wc_strncmp.c
evalkern.c      galcoord_ecliptic.o plotParams.o    wc_strncmp.h
evalkern.h      galcoord_ecliptic.o psrcat          wc_strncmp.o
evalkern.o      glitch.db       psrcat.c
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % psrcat -h
-all          Merge with $PSRCAT_RUNDIR/obs*.db
-allref       Lists the entire bibliographic reference file
-bib           Provide bibliographic reference
-bibliography  Lists references within the catalogue
-c            List of parameters to obtain from the database
-check        Checks the catalogue for incorrect entries
-db_file      Database path and name
-descend      Sort in descending order
-e            Produce short ephemeris files suitable for TEMPO
-e2           Produce long ephemeris files
-e3           Produce ephemeris file with selected parameters
-exactMatch   Require exact match for string comparisons
-h            This help information
-l            Logical string for filtering unwanted pulsars
              e.g. p0 > 5
-listref      Provide bibliographic references at end of table
-merge        Merges multiple catalogue files. For example
              -merge "mycat*.db" will merge the public catalogue
              with all the catalogues with filenames mycat*.db in the
              local directory.
-nohead       Do not print a header at the top of the table
-nonumber     Do not print line numbers in output
-nocand       Do not include pulsar candidates
-nointerim    Do not include interim solutions
-null        Null string for displaying for unknown parameter
```

```
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % psrcat -db_file ./psrcat.db -c "name p0 dm" -l "p0 > 3"
zsh: illegal hardware instruction psrcat -db_file ./psrcat.db -c "name p0 dm" -l "p0 > 3"
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar %
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % psrcat -db_file ./psrcat.db
zsh: illegal hardware instruction psrcat -db_file ./psrcat.db
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar %
```


Next Steps

- Could I have access to the Python scripts that you used with the 3FGL catalog
- Then, if I understand correctly, I need to reproduce your results using PyTorch?
- And then convert to using Neural Networks for Deep Learning?

Update: 01/08/2024

Progress

- 3FGLReproductionBasicLR Python code pointed to 3FGL Catalog file gll_psc_v16.fit.
- Updated it to point to 4FGL catalog file gll_psc_v33.fit.
- Tested with updated feature name
- Modified it for 1 hidden layer NN and DNN (2 hidden layers NN)

Results

Model	AGN Sensitivity	PSR Sensitivity	Accuracy of Best P Fit
Basic LR (0 hidden layer NN)	0.9724137931034482	1.0	0.9733333333333334
Neural Network (1 hidden layer)	0.9620689655172414	1.0	0.9633333333333334
Deep Neural Network (2 hidden layers)	0.9724137931034482	1.0	0.9733333333333334

Features Used

- Used the following features from columns in 4FGL catalog file gll_psc_v33.fit
 - ["PL_Index", "LP_Index", "PLEC_IndexS", "Variability_Index", "Unc_PL_Flux_Density", "Unc_LP_Flux_Density", "Unc_Energy_Flux100", "Unc_PLEC_Flux_Density", "Unc_Flux1000", "LP_beta", "Frac_Variability", "LP_SigCurv", "Signif_Avg"]
 - Found the corresponding 3FGL columns in the 4FGL list
 - Did some trial and error for other columns and checked model performance
- Column list is in pages 32-33 of Abdollahi_2020_ApJS_247_33 paper

Columns

- Printed out from code

1 DataRelease	21 PL_Flux_Density	41 Unc_PLEC_Exp_Index	61 ASSOC_4FGL
2 RAJ2000	22 Unc_PL_Flux_Density	42 PLEC_SigCurv	62 TEVCAT_FLAG
3 DEJ2000	23 PL_Index	43 PLEC_EPeak	63 ASSOC_TEV
4 GLON	24 Unc_PL_Index	44 Unc_PLEC_EPeak	64 CLASS1
5 GLAT	25 LP_Flux_Density	45 Npred	65 ASSOC1
6 Conf_68_SemiMajor	26 Unc_LP_Flux_Density	46 Flux_Band	66 ASSOC_PROB_BAY
7 Conf_68_SemiMinor	27 LP_Index	47 Unc_Flux_Band	67 ASSOC_PROB_LR
8 Conf_68_PosAng	28 Unc_LP_Index	48 nuFnu_Band	68 RA_Counterpart
9 Conf_95_SemiMajor	29 LP_beta	49 Sqrt_TS_Band	69 DEC_Counterpart
10 Conf_95_SemiMinor	30 Unc_LP_beta	50 Variability_Index	70 Unc_Counterpart
11 Conf_95_PosAng	31 LP_SigCurv	51 Frac_Variability	71 Flags
12 ROI_num	32 LP_EPeak	52 Unc_Frac_Variability	72 ASSOC_FGL
13	33 Unc_LP_EPeak	53 Signif_Peak	73 ASSOC_FHL
Extended_Source_Name	34 PLEC_Flux_Density	54 Flux_Peak	74 ASSOC_GAM1
14 Signif_Avg	35 Unc_PLEC_Flux_Density	55 Unc_Flux_Peak	75 ASSOC_GAM2
15 Pivot_Energy	36 PLEC_IndexS	56 Time_Peak	76 ASSOC_GAM3
16 Flux1000	37 Unc_PLEC_IndexS	57 Peak_Interval	77 CLASS2
17 Unc_Flux1000	38 PLEC_ExpfactorS	58 Flux_History	78 ASSOC2
18 Energy_Flux100	39 Unc_PLEC_ExpfactorS	59 Unc_Flux_History	
19 Unc_Energy_Flux100	40 PLEC_Exp_Index	60 Sqrt_TS_History	
20 SpectrumType			

Feature Transformation

- FGL4Dataset feature transformations

```
def __init__(self, wanted_type_col, fluxindices, wantedtypes, wantedfeaturenames, pointsourcecata):  
    # Create x/input data  
    xdata = []  
    ydata = []  
    # Select sources that are a part of the wanted types  
    for source in pointsourcecatalogue.data:  
        for feature_names in wantedfeaturenames:  
            print("{0} = {1}".format(*args: feature_names, source[feature_names]))  
            if source[wanted_type_col] in wantedtypes:  
                unit = []  
                # Append wanted params. Applied a log transformation to paras with highly skewed distributions.  
                unit.append(float(source[wantedfeaturenames[0]]))  
                unit.append(float(source[wantedfeaturenames[1]]))  
                unit.append(float(source[wantedfeaturenames[2]]))  
                unit.append(float(source[wantedfeaturenames[3]]))  
                unit.append(math.log(source[wantedfeaturenames[4]]))  
                if (source[wantedfeaturenames[4]] > 0) else float(source[wantedfeaturenames[4]])  
                unit.append(math.log(source[wantedfeaturenames[5]]))  
                if (source[wantedfeaturenames[5]] > 0) else float(source[wantedfeaturenames[5]])  
                unit.append(math.log(source[wantedfeaturenames[6]]))  
                if (source[wantedfeaturenames[6]] > 0) else float(source[wantedfeaturenames[6]])  
                unit.append(math.log(source[wantedfeaturenames[7]]))  
                if (source[wantedfeaturenames[7]] > 0) else float(source[wantedfeaturenames[7]])  
                unit.append(math.log(source[wantedfeaturenames[8]]))  
                if (source[wantedfeaturenames[8]] > 0) else float(source[wantedfeaturenames[8]])  
                unit.append(float(source[wantedfeaturenames[9]]))  
                unit.append(float(source[wantedfeaturenames[10]]))  
                unit.append(float(source[wantedfeaturenames[11]]))  
                unit.append(float(source[wantedfeaturenames[12]]))
```

Apply log transformations
to params with highly
skewed distributions

PL_Index = 2.5443360805511475

LP_Index = 2.5455660820007324

PLEC_IndexS = 2.539679765701294

Variability_Index = 25.699934005737305

Unc_PL_Flux_Density = 3.7675866251677195e-14

Unc_LP_Flux_Density = 4.284851688957439e-14

Unc_Energy_Flux100 = 4.238916617889388e-13

Unc_PLEC_Flux_Density = 3.952874743566419e-14

Unc_Flux1000 = 2.748007377206818e-11

LP_beta = -0.009841139428317547

Frac_Variability = 0.5357248187065125

LP_SigCurv = 0.0

Signif_Avg = 8.640633583068848

Neural Networks Configurations Used

- Used

- 0 hidden layer Neural Networks - Linear Regression
- 1 hidden layer NN
- Deep Neural Networks (2 hidden layers)

0 hidden layer NN - Linear Regression

```
Linear(in_features=13, out_features=1, bias=True)
```

NN (1 hidden layer)

```
Sequential(  
  (0): Linear(in_features=13, out_features=60, bias=True)  
  (1): ReLU()  
  (2): Linear(in_features=60, out_features=1, bias=True)  
)
```

DNN (2 hidden layers)

```
Sequential(  
  (0): Linear(in_features=13, out_features=250, bias=True)  
  (1): ReLU()  
  (2): Linear(in_features=250, out_features=40, bias=True)  
  (3): ReLU()  
  (4): Linear(in_features=40, out_features=1, bias=True)  
)
```


Other Changes & Github

Classes

- Separated the large code into classes for code modularity, improvement
- Created separate classes for:
 - **Driver:** 3FGLReproductionBasicLR_v1.py, 4FGLReproductionBasic{LR, NN, DNN}.py,
 - **Feature transformation:** FGL3Dataset.py / FGL4Dataset.py
 - **Models:** LogisticRegression.py, NeuralNetwork.py, DeepNeuralNetwork.py
 - **Utility functions:** utils.py (e.g. run_iteration)

- 3FGLReproductionBasicLR_v1.py
 - FGL3Dataset.py
 - LogisticRegression.py
 - utils.py

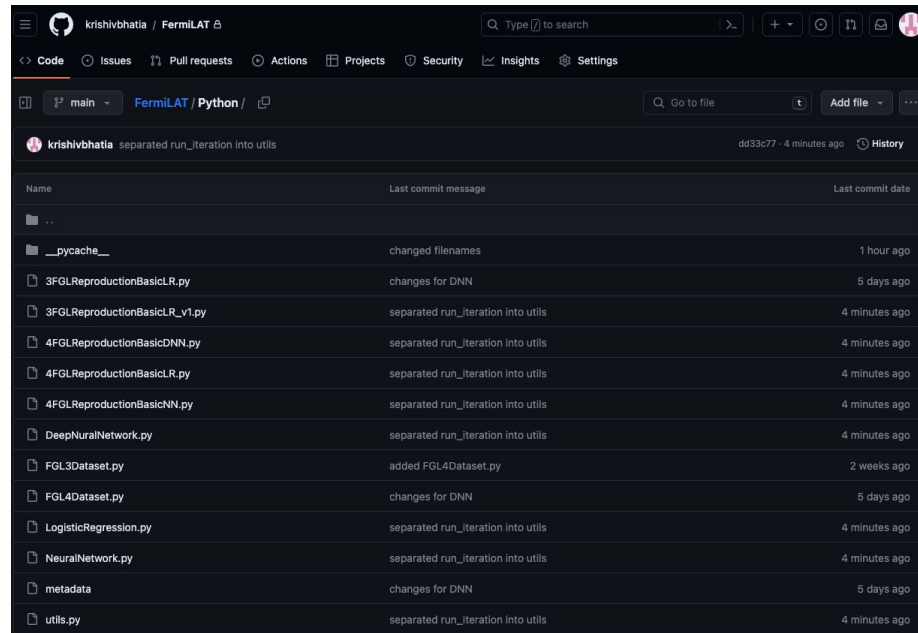
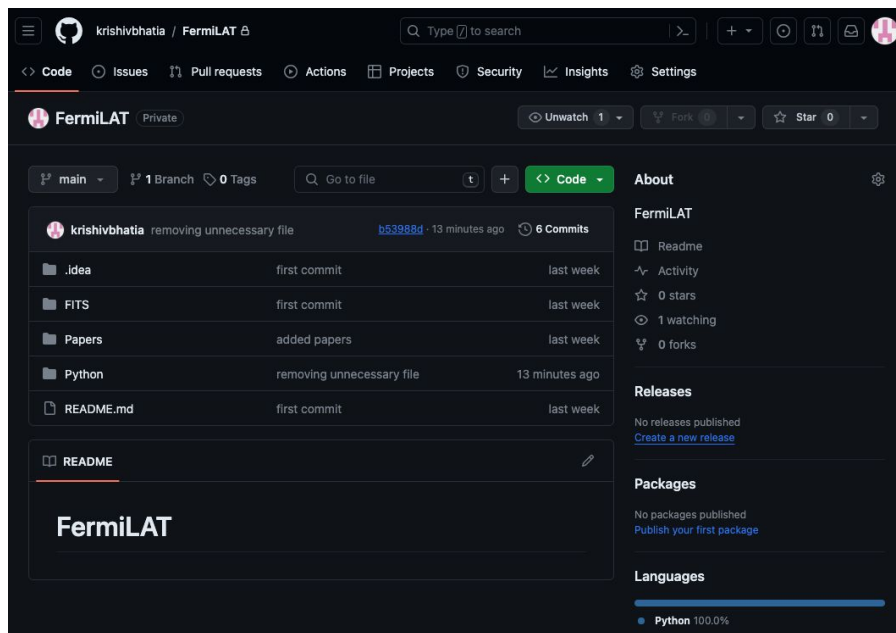
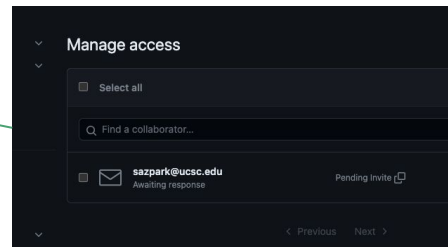
- 4FGLReproductionBasicLR.py
 - FGL4Dataset.py
 - LogisticRegression.py
 - utils.py

- 4FGLReproductionBasicLR.py
 - FGL4Dataset.py
 - LogisticRegression.py
 - utils.py
- 4FGLReproductionBasicDNN.py
 - FGL4Dataset.py
 - DeepNeuralNetwork.py
 - utils.py

Github Repo

- Checked into github
 - <https://github.com/krishivbhatia/FermiLAT/>

Sent you collaborator invite



Results: Linear Regression

Files for Linear Regression

- Python Scripts
 - 4FGLReproductionBasicLR.py
 - FGL4Dataset.py
 - LogisticRegression.py
 - utils.py
- Execution
 - python3 4FGLReproductionBasicLR.py

1-layer NN (No hidden layer)
Basic LR



```
main ▾ FermiLAT / Python / LogisticRegression.py
vrbhatia changes for DNN
1 loc) · 1.79 KB
Blame Raw [ ] [ ] [ ]
import math
import torch
import torch.nn as nn

class LogisticRegression(nn.Module):
    def __init__(self, input_features):
        print("(len(input_features[0][0]) = ", len(input_features[0][0]))
        print("input_features[0][0] = ", input_features[0][0])
        super(LogisticRegression, self).__init__()
        self.model = nn.Linear(len(input_features[0][0]), 1)
        print()
        print(self.model)
        print()

    def forward(self, x):
        y_predicted = torch.sigmoid(self.model(x))
        return y_predicted
```

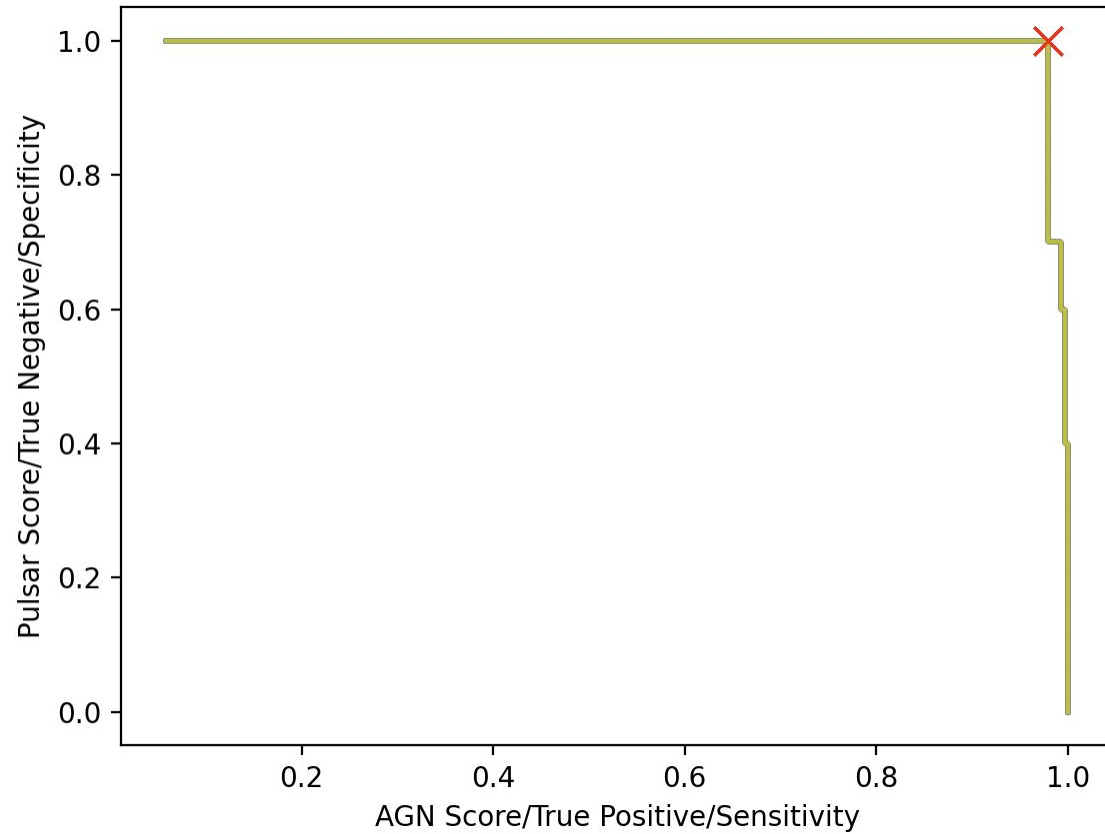
Results

- Linear(in_features=13, out_features=1, bias=True)
- **Results (200 iterations, lr = 0.00001)**
 - optimal_p = 0.9352
 - Best True Positive (AGN Successfully Identified): 282 (290)
 - Best AGN/True Positive/Sensitivity : 0.9724137931034482
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Results

- Linear(in_features=13, out_features=1, bias=True)
- **Results (300 iterations, lr = 0.00001)**
 - optimal_p = 0.9352
 - Best True Positive (AGN Successfully Identified): 282 (290)
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 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Plot



Results: NN - 1 hidden layer

Files for NN (1 hidden layer)

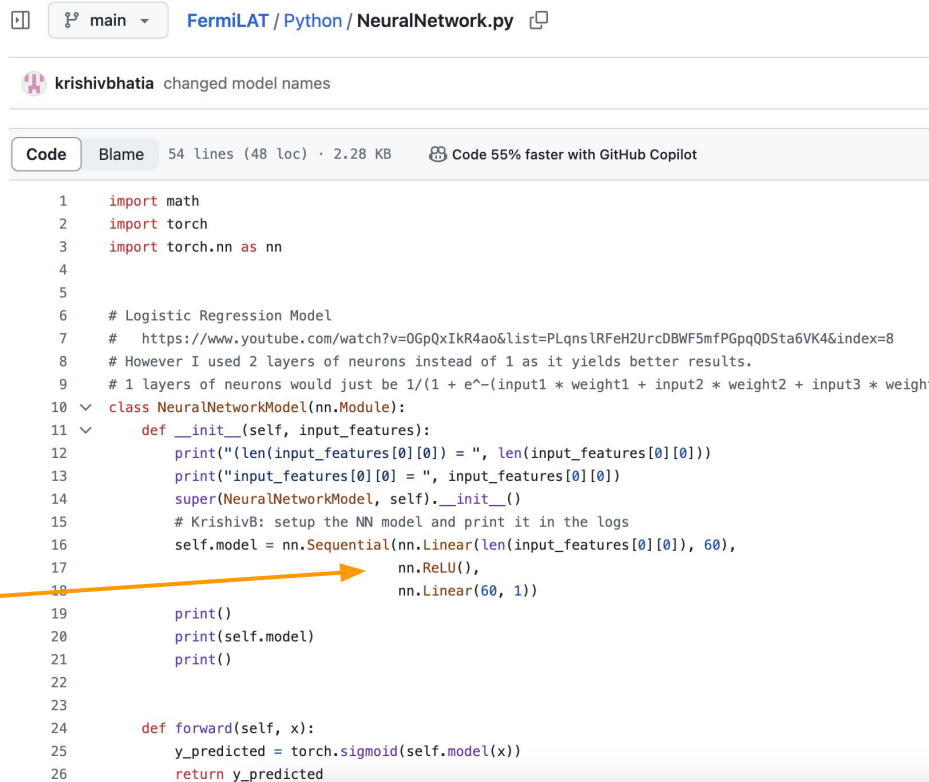
- Python Scripts

- 4FGLReproductionBasicNN.py
- FGL4Dataset.py
- NeuralNetwork.py
- utils.py

- Execution

- python3 4FGLReproductionBasicNN.py

1 hidden layers NN
ReLU activation function



```
1 import math
2 import torch
3 import torch.nn as nn
4
5
6 # Logistic Regression Model
7 # https://www.youtube.com/watch?v=0GpQxIkR4ao&list=PLqns1RFeH2UrcDBWF5mfPGpqQDSta6VK4&index=8
8 # However I used 2 layers of neurons instead of 1 as it yields better results.
9 # 1 layers of neurons would just be 1/(1 + e^-(input1 * weight1 + input2 * weight2 + input3 * weight3))
10 class NeuralNetworkModel(nn.Module):
11     def __init__(self, input_features):
12         print("(len(input_features[0][0]) = ", len(input_features[0][0]))
13         print("input_features[0][0] = ", input_features[0][0])
14         super(NeuralNetworkModel, self).__init__()
15         # KrishivB: setup the NN model and print it in the logs
16         self.model = nn.Sequential(nn.Linear(len(input_features[0][0]), 60),
17                                   nn.ReLU(),
18                                   nn.Linear(60, 1))
19
20         print()
21         print(self.model)
22         print()
23
24     def forward(self, x):
25         y_predicted = torch.sigmoid(self.model(x))
26         return y_predicted
```

Results

- **Model: Sequential(**

(0): Linear(in_features=13, out_features=60, bias=True)

(1): ReLU() ← Activation Function

(2): Linear(in_features=60, out_features=1, bias=True))

1 hidden layer



- **Results (300 iterations, lr = 0.0001)**

- optimal_p = 0.9525
- Best True Positive (AGN Successfully Identified): 279 (290)
- Best AGN/True Positive/Sensitivity : 0.9620689655172414
- Best True Negative (PSR Successfully Identified): 10 (10)
- Best Pulsar/True Negative/Specificity : 1.0
- Accuracy of Best P Fit : 0.9633333333333334

Results

- **Model: Sequential**
 - (0): Linear(in_features=13, out_features=60, bias=True)
 - (1): ReLU()
 - (2): Linear(in_features=60, out_features=1, bias=True))
- **Results (60 iterations, lr = 0.0001)**
 - optimal_p = 0.9525
 - Best True Positive (AGN Successfully Identified): 279 (290)
 - Best AGN/True Positive/Sensitivity : 0.9620689655172414
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9633333333333334

Results

- **Model: Sequential**
 - (0): Linear(in_features=13, out_features=40, bias=True)
 - (1): Tanh() ← tested with tanh() activation function also
 - (2): Linear(in_features=40, out_features=1, bias=True))
- **Results (300 iterations, lr = 0.0001)**
 - optimal_p = 0.9544
 - Best True Positive (AGN Successfully Identified): 278 (290)
 - Best AGN/True Positive/Sensitivity : 0.9586206896551724
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.96

Results: DNN - 2 hidden layers

Files for DNN: 2 hidden layers

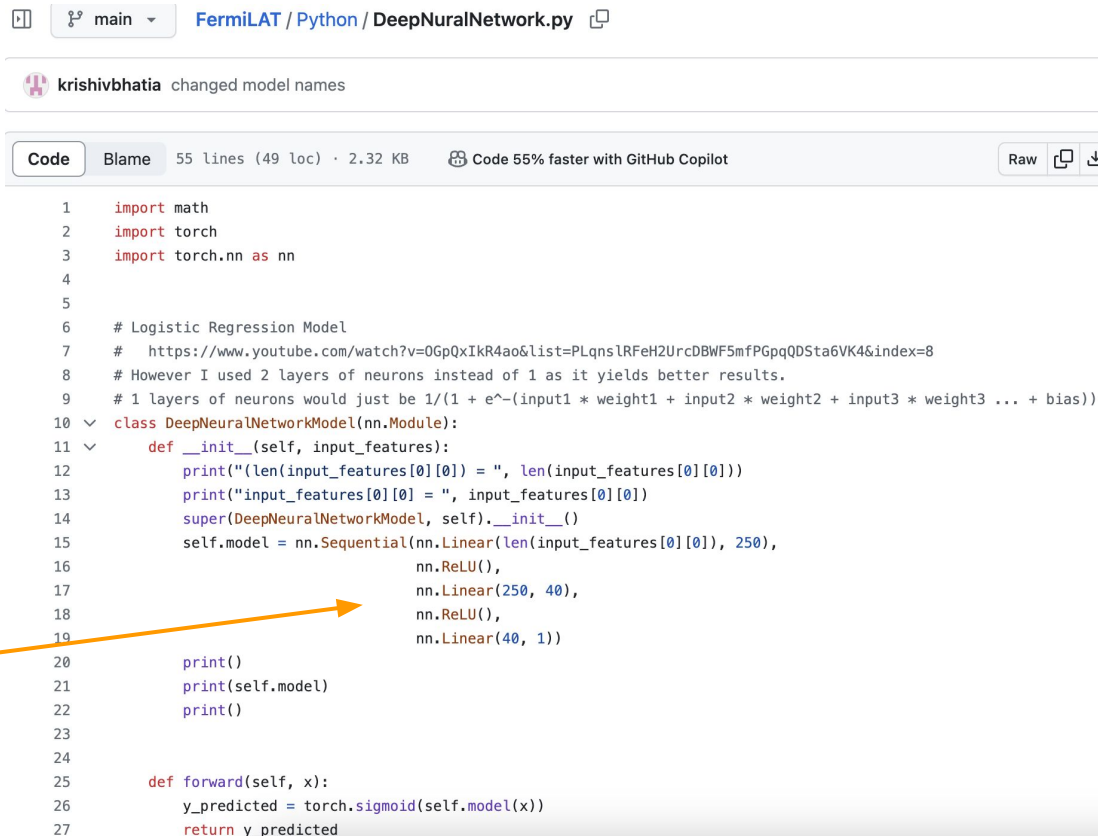
- Python Scripts

- 4FGLReproductionBasicDNN.py
- FGL4Dataset.py
- DeepNeuralNetwork.py
- utils.py

- Execution

- python3 4FGLReproductionBasicDNN.py

2 hidden layers NN
ReLU activation function



```
1 import math
2 import torch
3 import torch.nn as nn
4
5
6 # Logistic Regression Model
7 # https://www.youtube.com/watch?v=OGpQxIkR4ao&list=PLqnsIRFeH2UrcDBWF5mfPGpqQDSta6VK4&index=8
8 # However I used 2 layers of neurons instead of 1 as it yields better results.
9 # 1 layers of neurons would just be 1/(1 + e^-(input1 * weight1 + input2 * weight2 + input3 * weight3 ... + bias))
10 class DeepNeuralNetworkModel(nn.Module):
11     def __init__(self, input_features):
12         print("len(input_features[0][0]) = ", len(input_features[0][0]))
13         print("input_features[0][0] = ", input_features[0][0])
14         super(DeepNeuralNetworkModel, self).__init__()
15         self.model = nn.Sequential(nn.Linear(len(input_features[0][0]), 250),
16                                   nn.ReLU(),
17                                   nn.Linear(250, 40),
18                                   nn.ReLU(),
19                                   nn.Linear(40, 1))
20
21         print()
22         print(self.model)
23         print()
24
25     def forward(self, x):
26         y_predicted = torch.sigmoid(self.model(x))
27         return y_predicted
```

Results

- **Model: Sequential(**

- (0): Linear(in_features=13, out_features=250, bias=True)

- (1): ReLU() ← activation function

- (2): Linear(in_features=250, out_features=40, bias=True)

- (3): ReLU() ← activation function

- (4): Linear(in_features=40, out_features=1, bias=True))

2 hidden layers



- **Results (200 iterations, lr = 0.0001)**

- optimal_p = 0.9475
 - Best True Positive (AGN Successfully Identified): 282 (290)
 - Best AGN/True Positive/Sensitivity : 0.9724137931034482
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Results

- **Model: Sequential()**
 - (0): Linear(in_features=13, out_features=250, bias=True)
 - (1): ReLU()
 - (2): Linear(in_features=250, out_features=40, bias=True)
 - (3): ReLU()
 - (4): Linear(in_features=40, out_features=1, bias=True))
- **Results (300 iterations, lr = 0.0001)**
 - optimal_p = 0.9475
 - Best True Positive (AGN Successfully Identified): 282 (290)
 - Best AGN/True Positive/Sensitivity : 0.9724137931034482
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Questions

- Hardness ratios of LAT energy bands
 - 3FGL Catalog had many bands:
 - "Flux100_300", "Flux300_1000", "Flux1000_3000", "Flux3000_10000", "Flux10000_100000"
 - 4FGL Catalog has only one energy band column present:
 - Energy_Flux100
 - From Abdollahi 2020, definition: *Energy flux from 100MeV to 100GeV obtained by spectral fitting*
 - Where do we get other bands to calculate hardness ratios?

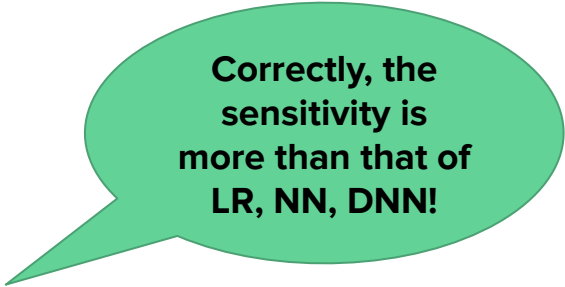
Next Steps

- Random Forest & Decision Tree implementations
 - Found Random Forest & Decision Tree implementations using PyTorch
 - <https://discuss.pytorch.org/t/how-can-i-use-knn-random-forest-models-in-pytorch/42464/2>
 - https://github.com/ValentinFigue/Sklearn_PyTorch
 - Test class shows how to use it:
 - https://github.com/ValentinFigue/Sklearn_PyTorch/blob/master/tests/random_forest_tests.py
 - https://github.com/ValentinFigue/Sklearn_PyTorch/blob/master/tests/binary_tree_tests.py
 - Use it to implement Random Forest & Decision Trees in PyTorch for 4FGL Catalog?

Update: 01/15/2024

Progress

- Completed Random Forest & Decision Tree implementations using PyTorch for 4FGL Catalog file
 - Random Forest
 - Highest sensitivity = 98.05749805749805
 - Decision Trees
 - Highest sensitivity = 98.13519813519814



Correctly, the sensitivity is more than that of LR, NN, DNN!

Background

- Referenced Random Forest & Decision Tree implementations using PyTorch
 - https://github.com/ValentinFigue/Sklearn_PyTorch
- Random Forest test class shows how to use it:
 - https://github.com/ValentinFigue/Sklearn_PyTorch/blob/master/tests/random_forest_tests.py
 - `random_forest = Sklearn_PyTorch.TorchRandomForestRegressor(nb_trees=10, nb_samples=2, max_depth=3)`
 - `random_forest.fit(train_features, train_labels)`
- Decision Tree test class shows how to use it:
 - https://github.com/ValentinFigue/Sklearn_PyTorch/blob/master/tests/binary_tree_tests.py
 - `binary_tree = Sklearn_PyTorch.TorchDecisionTreeClassifier(max_depth=1)`
 - `binary_tree.fit(train_features, train_labels)`

Code Updates

- Checked in code in github for:

Random Forest

- 4FGLReproduction_RF.py
 - FGL4Dataset.py
 - random_forest.py
 - utils.py

Decision Trees

- 4FGLReproduction_decision_tree.py
 - FGL4Dataset.py
 - decision_tree.py
 - decision_node.py
 - utils.py

Github Repo - Python package

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 **krishivbhatia** modifications for directories 3adc19f · 4 minutes ago [History](#)

Name	Last commit message	Last commit date
 ..		
 DecisionTrees	modifications for Random Forest and Decsion Trees	4 minutes ago
 LogisticRegression	modifications for Random Forest and Decsion Trees	4 minutes ago
 NeuralNetworks	modifications for Random Forest and Decsion Trees	4 minutes ago
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 Utils	modifications for Random Forest and Decsion Trees	4 minutes ago
 <code>__init__.py</code>	modifications for Random Forest and Decsion Trees	4 minutes ago

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 **krishivbhatia** modifications for Random Forest and Decsion Trees 5f26b63 · 5 minutes ago [History](#)

Name	Last commit message	Last commit date
 ..		
 4FGLReproduction_RF.py	modifications for Random Forest and Decsion Trees	5 minutes ago
 __init__.py	modifications for Random Forest and Decsion Trees	5 minutes ago
 random_forest.py	modifications for Random Forest and Decsion Trees	5 minutes ago

Github Repo - Decision Trees package

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Name	Last commit message	Last commit date
 ..		
 4FGLReproduction_DecisionTree.py	modifications for Random Forest and Decsion Trees	5 minutes ago
 __init__.py	modifications for Random Forest and Decsion Trees	5 minutes ago
 decision_node.py	modifications for Random Forest and Decsion Trees	5 minutes ago
 decision_tree.py	modifications for Random Forest and Decsion Trees	5 minutes ago

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 **krishivbhatia** changes in Random Forest and Decision Trees code for FermiLAT 6ebb528 · 9 minutes ago [History](#)

Name	Last commit message	Last commit date
 ..		
 FGL3Dataset.py	modifications for Random Forest and Decsion Trees	4 hours ago
 FGL4Dataset.py	modifications for Random Forest and Decsion Trees	4 hours ago
 __init__.py	modifications for Random Forest and Decsion Trees	4 hours ago
 metadata	modifications for Random Forest and Decsion Trees	4 hours ago
 utils.py	changes in Random Forest and Decision Trees code for FermiLAT	9 minutes ago

Code Details

- Method of creating classifier and invoking fit method:

4FGLReproduction_RF.py

```
# Krishiv Bhatia: Invoke TorchRandomForestClassifier
random_forest = TorchRandomForestClassifier( nb_trees: 200, nb_samples: 2500, max_depth: 15)
print("random forest = ", random_forest.nb_trees, random_forest.nb_samples, random_forest.max_depth)

# Krishiv Bhatia: split train dataset into features and labels
train_features, train_labels = dataset_to_features_labels(train_dataset)
random_forest.fit(torch.FloatTensor(train_features), torch.LongTensor(train_labels))
```

4FGLReproduction_decision_tree.py

```
# Krishiv Bhatia: Invoke TorchDecisionTreeClassifier
decision_tree = TorchDecisionTreeClassifier(20)
print("TorchDecisionTreeClassifier = ", decision_tree.max_depth)

# Krishiv Bhatia: split train dataset into features and labels
train_features, train_labels = dataset_to_features_labels(train_dataset)
decision_tree.fit(torch.FloatTensor(train_features), torch.LongTensor(train_labels))
```

Transforms dataset into
separate feature & labels
arrays (next slide)

Random Forest

Decision Trees

Transformations

- Transformed the tensor feature and labels array into regular arrays

```
main ▾ FermiLAT / Python / Utils / utils.py

Blame 150 lines (126 loc) · 4.75 KB Code 55% faster with GitHub Copilot

def mean(values):
    """
    Mean function.
    """
    m = 0.0
    for value in values:
        m = m + value/len(values)
    return m

def dataset_to_features_labels(dataset):
    # Krishiv Bhatia: train or test dataset is of the form:
    #   tensor array(train/test features), tensor array (output label)
    # or like this:
    #   tensor array(feature 0, feature 1, ... feature 12), tensor array(output label)
    # You can print it like this:
    #   for i in train_dataset:
    #       print(i)
    # An example:
    #   (tensor([ 2.1602,  1.8308,  1.6083, 13.2215, -32.9276, -32.4171, -29.4589,
    #            -32.3813, -24.3150,  0.4812,  0.0000,  2.7429,  5.4350]), tensor([1.]))
    #
    # Convert tensor feature array in dataset to numpy feature array
    train_features = ([i[i[0]].detach().numpy() for i in dataset])
    # print("train_features = ", train_features)
    # Convert tensor label array in dataset to regular array
    train_labels = ([torch.LongTensor.item(i[1]) for i in dataset])
    # print("train_labels = ", train_labels)
    return train_features, train_labels
```

References

- <https://stackoverflow.com/questions/49768306/pytorch-tensor-to-numpy-array>
- <https://stackoverflow.com/questions/54268029/how-to-convert-a-pytorch-tensor-into-a-numpy-array>
- <https://stackoverflow.com/questions/67711752/how-to-extract-tensors-to-numpy-arrays-or-lists-from-a-larger-pytorch-tensor>
- <https://stackoverflow.com/questions/47588682/how-to-cast-a-1-d-intensor-to-int-in-pytorch>
- <https://stackoverflow.com/questions/57727372/how-do-i-get-the-value-of-a-tensor-in-pytorch>

i[0] gets tensor feature array
i[1] gets tensor label array

Random Forest Results

- `random_forest = TorchRandomForestClassifier(50, 2000, 10)`
 - `correct = 1262/1287`
 - `sensitivity = 98.05749805749805`
- `random_forest = TorchRandomForestClassifier(200, 2000, 15)`
 - `correct = 1258/1287`
 - `sensitivity = 97.74669774669775`
- `random_forest = TorchRandomForestClassifier(200, 1500, 15)`
 - `correct = 1262/1287`
 - `sensitivity = 98.05749805749805`

Random Forest Results

- `random_forest = TorchRandomForestClassifier(15, 500, 10)`
 - `correct = 1261/1287`
 - `sensitivity = 97.97979797979798`
- `random_forest = TorchRandomForestClassifier(100, 500, 10)`
 - `correct = 1255/1287`
 - `sensitivity = 97.51359751359752`
- `random_forest = TorchRandomForestClassifier(50, 1000, 15)`
 - `correct = 1258/1287`
 - `sensitivity = 97.74669774669775`
- `random_forest = TorchRandomForestClassifier(50, 1500, 10)`
 - `correct = 1259`
 - `sensitivity = 97.82439782439782`

Decision Tree Results

- `decision_tree = TorchDecisionTreeClassifier(10)`
 - `correct = 1263`
 - `sensitivity = 98.13519813519814`
- `decision_tree = TorchDecisionTreeClassifier(20)`
 - `correct =`
 - `sensitivity =`
- `decision_tree = TorchDecisionTreeClassifier(30)`
 - `correct = 1263`
 - `sensitivity = 98.13519813519814`
- `decision_tree = TorchDecisionTreeClassifier(80)`
 - `correct = 1263`
 - `sensitivity = 98.13519813519814`

Update: 01/22/2024

Progress

- Completed the problem of Energy Band Level Hardness ratios
- Column: Flux_Band
- Printed it out. Numpy array with 8 numbers
 - `source[flux_col] = [1.318705452994784e-12, 3.918336766162156e-10, 3.155801175935835e-10, 1.0454052207231612e-10, 2.0413597895396762e-11, 6.700519479541089e-12, 1.909341825270805e-12, 2.2682772056883754e-16]`

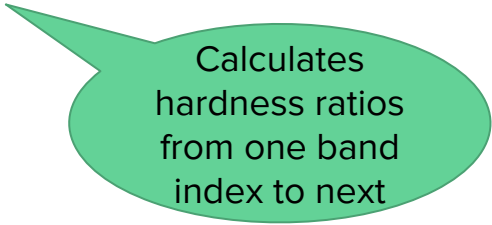
Transformation

- In FGL4Dataset.py
- Converted numpy array to list

- `flux_list = res = source[flux_col].tolist()`

- Calculated hardness ratios

```
# Calculate hardness ratios (also considered in the paper)
for index in range(1, len(flux_list)):
    unit.append((flux_list[index] - flux_list[index-1] + 0.0) /
                (flux_list[index] + flux_list[index-1] + 0.0))
xdata.append(unit)
```



Calculates
hardness ratios
from one band
index to next

Logistic Regression

- Model:
 - **Linear(in_features=20, out_features=1, bias=True)**
- Iterations = 30/100
 - Best P Threshold Value : 0.972
 - Score : 1.0586206896551724
 - Best Score : 1.9655172413793105
 - Best True Positive (AGN Successfully Identified): 280
 - AGN count : 290
 - Best AGN/True Positive/Sensitivity : 0.9655172413793104
 - Best True Negative (PSR Successfully Identified): 10
 - PSR count : 10
 - Best Pulsar/True Negative/Specificity : 1.0
 - **Accuracy of Best P Fit : 0.9666666666666667**
 - Best correct : 290
 - No of samples : 300

Neural Networks

- Model:
 - Sequential(
 - (0): Linear(in_features=20, out_features=60, bias=True)
 - (1): ReLU()
 - (2): Linear(in_features=60, out_features=1, bias=True))
- Iterations = 30
 - Best P Threshold Value : 0.976
 - Score : 1.0551724137931036
 - Best Score : 1.9586206896551723
 - Best True Positive (AGN Successfully Identified): 278
 - AGN count : 290
 - Best AGN/True Positive/Sensitivity : 0.9586206896551724
 - Best True Negative (PSR Successfully Identified): 10
 - PSR count : 10
 - Best Pulsar/True Negative/Specificity : 1.0
 - **Accuracy of Best P Fit : 0.96**
 - Best correct : 288
 - No of samples : 300

Deep Neural Networks

- Model:
 - **Sequential**
 - (0): **Linear(in_features=20, out_features=250, bias=True)**
 - (1): **ReLU()**
 - (2): **Linear(in_features=250, out_features=50, bias=True)**
 - (3): **ReLU()**
 - (4): **Linear(in_features=50, out_features=1, bias=True)**
 -)
- Results
 - Best P Threshold Value : 0.971
 - Score : 1.0586206896551724
 - Best Score : 1.9724137931034482
 - Best True Positive (AGN Successfully Identified): 282
 - AGN count : 290
 - Best AGN/True Positive/Sensitivity : 0.9724137931034482
 - Best True Negative (PSR Successfully Identified): 10
 - PSR count : 10
 - Best Pulsar/True Negative/Specificity : 1.0
 - **Accuracy of Best P Fit : 0.9733333333333334**
 - Best correct : 292
 - No of samples : 300

Deep Neural Networks

- DNN
- Model:
 - Sequential(
 - (0): Linear(in_features=20, out_features=250, bias=True)
 - (1): ReLU()
 - (2): Linear(in_features=250, out_features=40, bias=True)
 - (3): ReLU()
 - (4): Linear(in_features=40, out_features=1, bias=True))
 - Best P Threshold Value : 0.967
 - Score : 1.0517241379310345
 - Best Score : 1.9689655172413794
 - Best True Positive (AGN Successfully Identified): 281
 - AGN count : 290
 - Best AGN/True Positive/Sensitivity : 0.9689655172413794
 - Best True Negative (PSR Successfully Identified): 10
 - PSR count : 10
 - Best Pulsar/True Negative/Specificity : 1.0
 - **Accuracy of Best P Fit : 0.97**
 - Best correct : 291
 - No of samples : 300

Random Forests

- Model:
 - **# Trees: 200**
 - **# Samples: 2500**
 - **Depth: 15**
- Results
 - Correct = 1258
 - Total = 1286
 - Sensitivity = 97.82439782439782

Random Forests

- Model:
 - **# Trees: 200**
 - **# Samples: 1500**
 - **Depth: 15**
- Results
 - Correct = 1258
 - Total = 1286
 - Sensitivity = 97.74669774669775

Random Forests

- Model:
 - **# Trees: 200**
 - **# Samples: 500**
 - **Depth: 15**
- Results
 - Correct = 1254
 - Total = 1286
 - Sensitivity = 97.43589743589743

Decision Trees

- Decision Trees
- Model:
 - **Depth: 20**
 - Correct = 1259
 - Total = 1286
 - Sensitivity = 97.82439782439782

Next Steps

- Could you kindly clarify what needs to be done with the following tasks
 - associated/unassociated sources
 - AGN types

Update: 02/09/2024

Progress

- Simplified Logistic Regression & Neural Networks Python code to train and then test on the entire 30% test dataset
- This resolves the earlier problem where the test was showing only PSR count = 10 & AGN count = 290
- New scripts:
 - LR: 4FGLReproductionBasicLRnew.py
 - NN: 4FGLNNnew.py
 - DNN: 4FGLDNNnew.py
 - Only difference in the above scripts is the model trained and tested

Training

```
torch.set_default_tensor_type(torch.DoubleTensor)
torch.manual_seed(82)
LogRegModel = LogisticRegression(train_dataset)
# Stochastic Gradient Descent. I could use Adams but Adams is worse than regular SGD unless you
optimizer = torch.optim.SGD(LogRegModel.parameters(), lr=0.0001)
print("optimizer = ", optimizer.__class__)
# Binary Cross Entropy Loss which is the loss method that would most likely be used in this scenario
criterion = nn.BCELoss()
tot_iter = 100
print("total iterations = ", tot_iter)
for iteration in range(0, tot_iter):
    dev_inputs = dev_labels = []
    print("Iteration # ", iteration)
    total_samples = len(train_dataset)
    # KrishivB: Put run_iteration code in utils.py, imported it, and call here
    for epoch in range(30):
        (dev_inputs, dev_labels) = run_iteration(LogRegModel, iteration, train_dataset, total_samples, criterion,
                                                optimizer, dev_inputs, dev_labels, append_dev_input_label: True)
```

Define model once

Iterations for loop

Train model in iterations

Testing

Enumerate over
test dataset

Predict

Get actual
value

Check if predicted
matches with actual

```
correct = correct_agm = correct_psr = total_agm = total_psr = 0
for i, (inputs, labels) in enumerate(test_dataset):
    y_predicted = LogRegModel(inputs) # Insert/fit Model for prediction here
    predicted_item = y_predicted.data.item()
    print(" i, y_predicted, y_predicted.data, predicted_item = ", i, y_predicted, y_predicted.data, predicted_item)
    predicted_val = 0 if (math.isnan(predicted_item)) else round(predicted_item)
    label_val = labels.item()
    total_agm += 1 if label_val == 1 else 0
    total_psr += 1 if label_val == 0 else 0
    match = (predicted_val == label_val)
    if match:
        correct += 1
        correct_agm += 1 if label_val == 1 else 0
        correct_psr += 1 if label_val == 0 else 0
    print("i, predicted_item, predicted_val, label_val, correct, match = ",
          i, predicted_item, predicted_val, label_val, correct, match)
```


Results

Logistic Regression

- # iterations = 50
- Results
 - Correct = 1261
 - correct_agn, correct_psr = 1241 20
 - total, total_agn, total_psr = 1287 1244 43
 - **total sensitivity = 0.9797979797979798**
 - agn sensitivity = 0.997588424437299
 - psr sensitivity = 0.46511627906976744

- # iterations = 750
- Results
 - Correct = 1261
 - correct_agn, correct_psr = 1239 22
 - total, total_agn, total_psr = 1287 1244 43
 - **total sensitivity = 0.9797979797979798**
 - agn sensitivity = 0.9959807073954984
 - psr sensitivity = 0.5116279069767442

- # iterations = 100, 150, 200
- Results
 - Correct = 1262
 - correct_agn, correct_psr = 1240 22
 - total, total_agn, total_psr = 1287 1244 43
 - **total sensitivity = 0.9805749805749806**
 - agn sensitivity = 0.9967845659163987
 - psr sensitivity = 0.5116279069767442

- # iterations = 1000, 1250
- Results
 - Correct = 1261
 - correct_agn, correct_psr = 1238 23
 - total, total_agn, total_psr = 1287 1244 43
 - **total sensitivity = 0.9797979797979798**
 - agn sensitivity = 0.9951768488745981
 - psr sensitivity = 0.5348837209302325

Neural Network

- # iterations = 10
- Results
 - Correct = 1261
 - correct_agm, correct_psr = 1241 20
 - total, total_agm, total_psr = 1287 1244 43
 - total sensitivity = 0.9797979797979798
 - agm sensitivity = 0.997588424437299
 - psr sensitivity = 0.46511627906976744

- # iterations = 50
- Results
 - Correct = 1265
 - correct_agm, correct_psr = 1240 25
 - total, total_agm, total_psr = 1287 1244 43
 - total sensitivity = 0.9829059829059829
 - agm sensitivity = 0.9967845659163987
 - psr sensitivity = 0.5813953488372093

- # iterations = 100
- Results
 - Correct = 1264
 - correct_agm, correct_psr = 1239 25
 - total, total_agm, total_psr = 1287 1244 43
 - total sensitivity = 0.9821289821289821
 - agm sensitivity = 0.9959807073954984
 - psr sensitivity = 0.5813953488372093

- # iterations = 150
- Results
 - Correct = 1265
 - correct_agm, correct_psr = 1237 28
 - total, total_agm, total_psr = 1287 1244 43
 - total sensitivity = 0.9829059829059829
 - agm sensitivity = 0.9943729903536977
 - psr sensitivity = 0.6511627906976745

Deep Neural Network

- # iterations = 10
- Results
 - Correct = 1260
 - correct_agn, correct_psr = 1241 19
 - total, total_agn, total_psr = 1287 1244 43
 - total sensitivity = 0.9790209790209791
 - agn sensitivity = 0.997588424437299
 - psr sensitivity = 0.4418604651162791

- # iterations = 25
- Results
 - Correct = 1266
 - correct_agn, correct_psr = 1240 26
 - total, total_agn, total_psr = 1287 1244 43
 - total sensitivity = 0.9836829836829837
 - agn sensitivity = 0.9967845659163987
 - psr sensitivity = 0.6046511627906976

- # iterations = 100
- Results
 - Correct = 1267
 - correct_agn, correct_psr = 1239 28
 - total, total_agn, total_psr = 1287 1244 43
 - total sensitivity = 0.9844599844599845
 - agn sensitivity = 0.9959807073954984
 - psr sensitivity = 0.6511627906976745

- # iterations = 150
- Results
 - Correct = 1267
 - correct_agn, correct_psr = 1236 29
 - total, total_agn, total_psr = 1287 1244 43
 - total sensitivity = 0.9829059829059829
 - agn sensitivity = 0.9935691318327974
 - psr sensitivity = 0.6744186046511628

Observations

- I found overall AGN sensitivity to be > 0.995 throughout but PSR sensitivity to be around 0.5.
- With the NN & DNN model, PSR sensitivity started going up with more training (as # iterations increased).
- In the case of NN with 150 iterations (each iteration having 30 epochs), PSR sensitivity increased to 0.65.
- In the case of DNN, with 25 iterations, PSR sensitivity increased to 0.60 and with 100 iterations, increased to 0.65, and increased to 0.67 with 150 iterations.
- So we may need to find what is the problem with PSR?
 - Is it related to the choice of model, features, or training?

Update: 02/19/2024

Progress

- Fix the problem of low Pulsar count
- Added “msp” in wantedtypes list



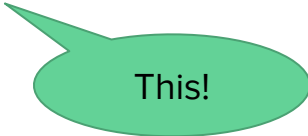
This!

```
|# KrishivB: Added AGN in caps  
|wantedtypes = ["PSR", "psr", "YNG", "yng", "MSP", "msp", "FSRQ", "fsrq", "BLL", "bll", "BCU", "bcu", "RDG", "rdg",  
|               "NLSY1", "nlsy1", "AGN", "agn", "ssrq", "sey"]
```

Progress

- Added “MSP”, “msp” to the **source[wanted_type_col]** check in FGL4Dataset.py
- It was missing. Only “PSR”, “psr” were present.

```
# Calculate hardness ratios (also considered in the paper)
for index in range(1, len(flux_list)):
    unit.append((flux_list[index] - flux_list[index-1] + 0.0) /
                (flux_list[index] + flux_list[index-1] + 0.0))
xdata.append(unit)
if source[wanted_type_col] in ("PSR", "psr", "MSP", "msp"):
    ydata.append([0]) # 0 is for pulsar
else:
    ydata.append([1]) # 1 is for AGN
```



This!

Logistic Regression

- # iterations = 25
- Results
 - Correct = 1260
 - correct_agn, correct_psr = 1172 88
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9699769053117783**
 - agn sensitivity = 0.9890295358649789
 - psr sensitivity = 0.7719298245614035

- # iterations = 200, 500
- Results
 - Correct = 1264
 - correct_agn, correct_psr = 1177 87
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9730561970746728**
 - agn sensitivity = 0.9932489451476794
 - psr sensitivity = 0.7631578947368421

- # iterations = 50, 100, 150
- Results
 - Correct = 1264
 - correct_agn, correct_psr = 1177 87
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9730561970746728**
 - agn sensitivity = 0.9932489451476794
 - psr sensitivity = 0.7631578947368421

- # iterations = 300
- Results
 - Correct = 1265
 - correct_agn, correct_psr = 1178 87
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9738260200153964**
 - agn sensitivity = 0.9940928270042194
 - psr sensitivity = 0.7631578947368421

Neural Network

- # iterations = 25
- Results
 - Correct = 1261
 - correct_agn, correct_psr = 1173 88
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.970746728252502**
 - agn sensitivity = 0.9898734177215189
 - psr sensitivity = 0.7719298245614035

- # iterations = 200
- Results
 - Correct = 1261
 - correct_agn, correct_psr = 1178 83
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.970746728252502**
 - agn sensitivity = 0.9940928270042194
 - psr sensitivity = 0.7280701754385965

- # iterations = 50, 100
- Results
 - Correct = 1257
 - correct_agn, correct_psr = 1176 81
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9676674364896074**
 - agn sensitivity = 0.9924050632911392
 - psr sensitivity = 0.7105263157894737

- # iterations = 300
- Results
 - Correct = 1262
 - correct_agn, correct_psr = 1180 82
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9715165511932256**
 - agn sensitivity = 0.9957805907172996
 - psr sensitivity = 0.7192982456140351

Deep Neural Network

- # iterations = 25
- Results
 - Correct = 1260
 - correct_agn, correct_psr = 1172 88
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9699769053117783**
 - agn sensitivity = 0.9890295358649789
 - psr sensitivity = 0.7719298245614035

- # iterations = 200
- Results
 - Correct = 1260
 - correct_agn, correct_psr = 1175 85
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9699769053117783**
 - agn sensitivity = 0.9915611814345991
 - psr sensitivity = 0.7456140350877193

- # iterations = 100
- Results
 - Correct = 1258
 - correct_agn, correct_psr = 1176 82
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.968437259430331**
 - agn sensitivity = 0.9924050632911392
 - psr sensitivity = 0.7192982456140351

- # iterations = 300
- Results
 - Correct = 1260
 - correct_agn, correct_psr = 1173 87
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9699769053117783**
 - agn sensitivity = 0.9898734177215189
 - psr sensitivity = 0.7631578947368421

Random Forest

- **Model**

- # Trees: 200
- # Samples: 500
- Depth: 15

- **Results**

- Correct = 1245
- sensitivity = 95.84295612009238

- **Model**

- # Trees: 300
- # Samples: 1500
- Depth: 15

- **Results**

- Correct = 1251
- sensitivity = 96.30484988452656

- **Model**

- # Trees: 200
- # Samples: 1000
- Depth: 15

- **Results**

- Correct = 1246
- sensitivity = 95.91993841416475

Update: 02/26/2024

Progress

- Classified unassociated sources as PSR or AGN
 - Initially, trained and tested models with regular dataset
 - Then, read dataset with field Class1 == null
 - Used trained model to predict whether the unassociated sources are PSRs or AGNs
- Also, plotted measure of outlyingness of the unassociated sources from the class.

Outlyingness Measure

- Predictions ranged from [0.0, 1.0]
- AGNs
 - Predictions were in range [0.5, 1.0]
 - The more near it was to 1.0, the closer it was to the class
 - The farther away it was from 1.0 (or closer to 0.5), the more outlying it was from the class
- PSRs
 - Predictions were in range [0.0, 0.5]
 - The more near it was 0.0, the closer it was to the class
 - The farther away it was from 0.0 (or closer to 0.5), the more outlying it was from the class

Code change

- New files

- Logistic Regression
 - 4FGL_LR_cls1_null.py
- Neural Networks
 - 4FGL_NN_cls1_null.py
 - 4FGL_DNN_cls1_null.py
- Random Forests
 - 4FGL_RF_cls1_null.py
- Utils
 - FGL4Dataset_cls1_null.py



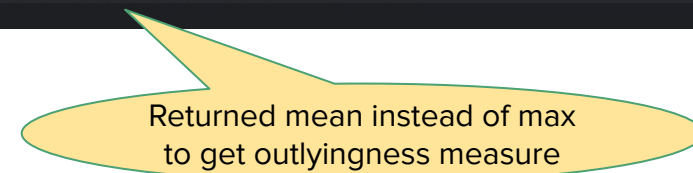
Read dataset with field Class1 == null

- Updated files

- Random Forests
 - random_forest.py



```
predictions = []
for tree, index_features in zip(self.trees, self.trees_features):
    sampled_vector = torch.index_select(vector, dim=0, index=index_features)
    predictions.append(tree.predict(sampled_vector))
print("predictions=", predictions)
print("sum(predictions)={}, len(self.trees)={}, sum(predictions)/len(self.trees)={}",
      sum(predictions), len(self.trees), sum(predictions)/len(self.trees))
return sum(predictions)/len(self.trees)
# return max(set(predictions), key=predictions.count)
```



Returned mean instead of max
to get outlyingness measure

Code change

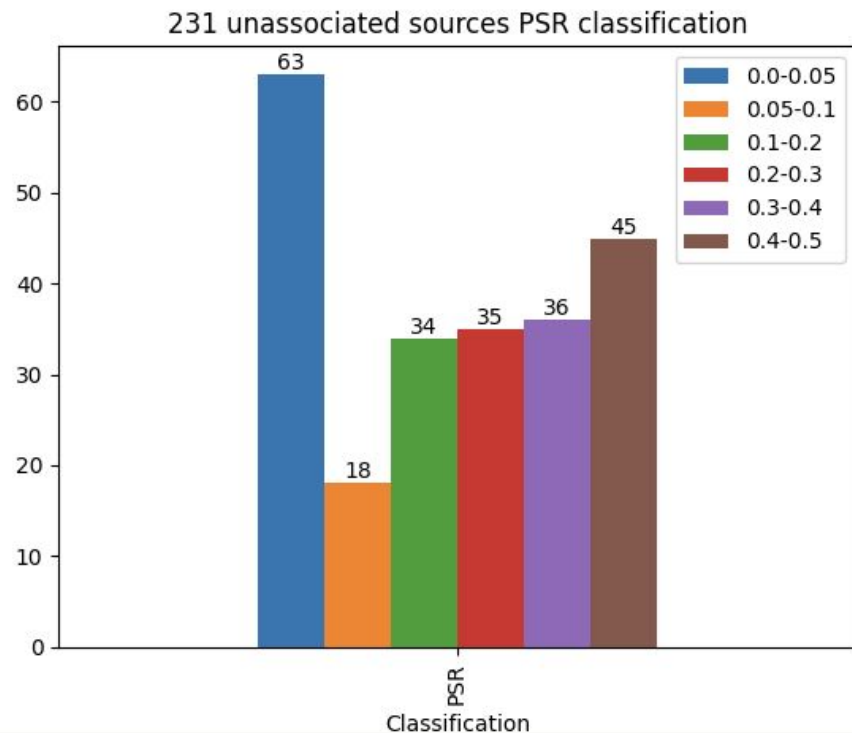
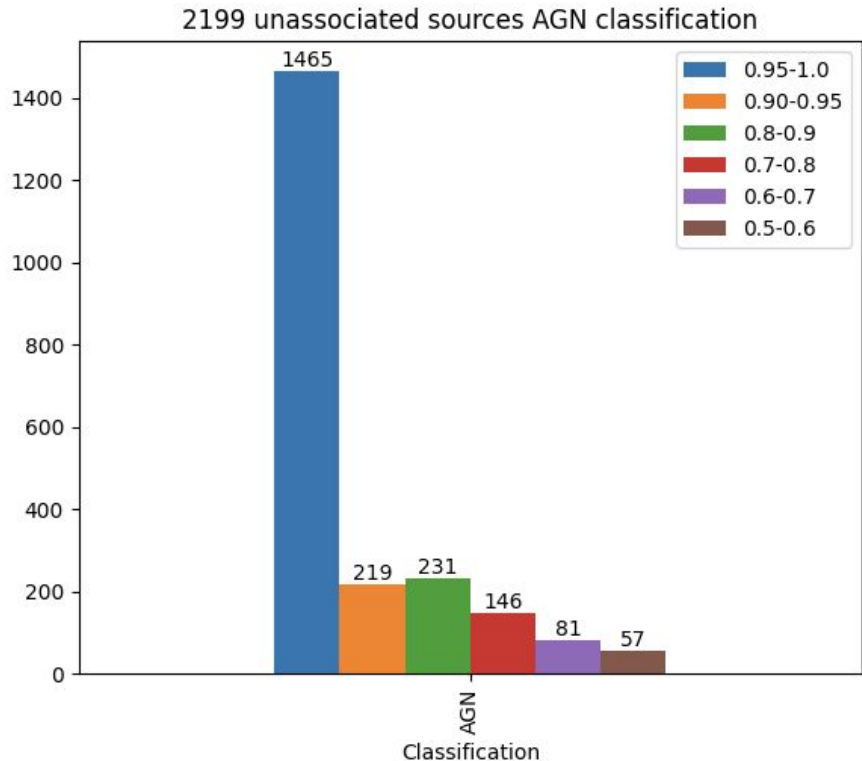
- FGL4Dataset_cls1_null.py

```
class FGL4Dataset_Cls1_Null(Dataset):
    # Krishiv Bhatia
    def __init__(self, wanted_type_col, flux_col, wanted_types, wanted_feature_names, poi
        # Create x/input data
        xdata = []
        ydata = []
        # Select sources that are a part of the wanted types
        for source in point_source_catalogue.data:
            # for feature_names in wanted_feature_names:
            #     print("{0} = {1}".format(feature_names, source[feature_names]))
            # print("(type(source[flux_col]) = ", type(source[flux_col]))
            flux_list = res = source[flux_col].tolist()
            # print("source[flux_col] = {0}".format(flux_list))
            print("source[wanted_type_col] = ", source[wanted_type_col])
            if source[wanted_type_col] == '':
                unit = []
                # Append wanted params. Applied a log transformation to paras with highly
                unit.append(float(source[wanted_feature_names[0]]))
                unit.append(float(source[wanted_feature_names[1]]))
                unit.append(float(source[wanted_feature_names[2]]))
                unit.append(float(source[wanted_feature_names[3]]))
```

Read dataset with field Class1 == null

Logistic Regression

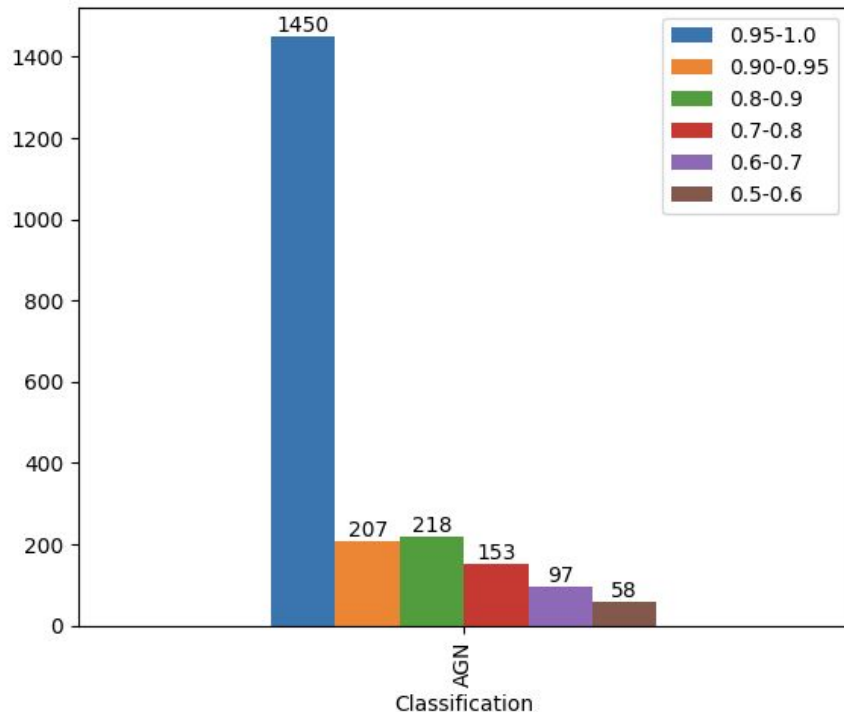
- # iterations = 50 Total: 2430 unassociated sources



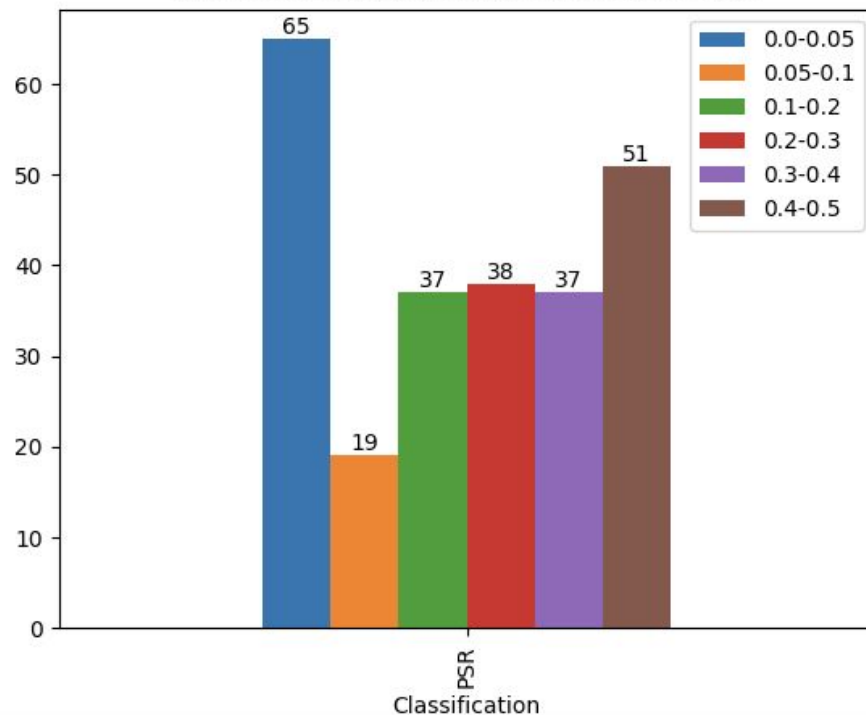
Logistic Regression

- # iterations = 100 Total: 2430 unassociated sources

2183 unassociated sources AGN classification



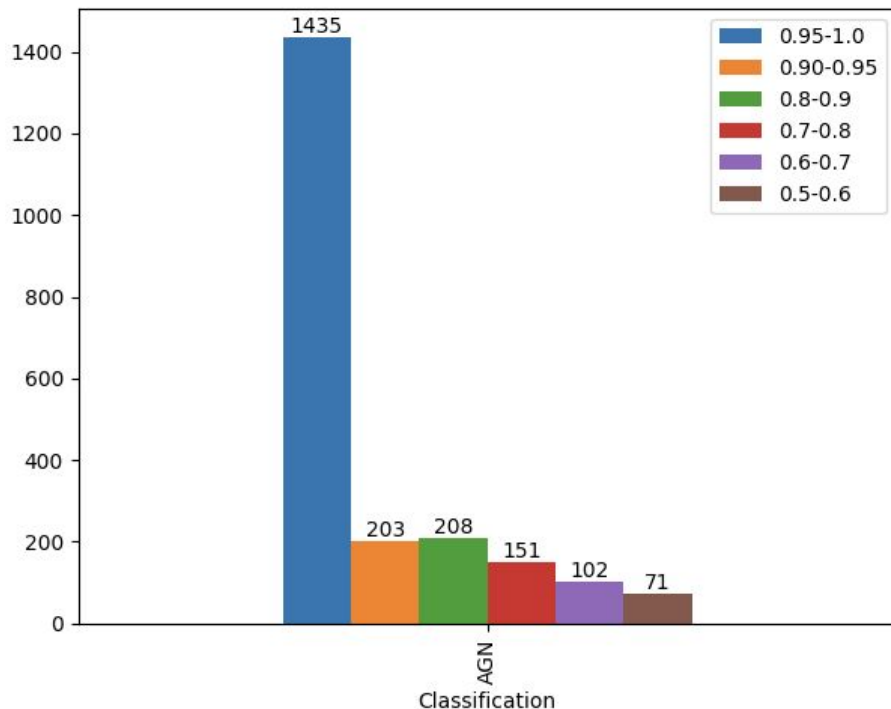
247 unassociated sources PSR classification



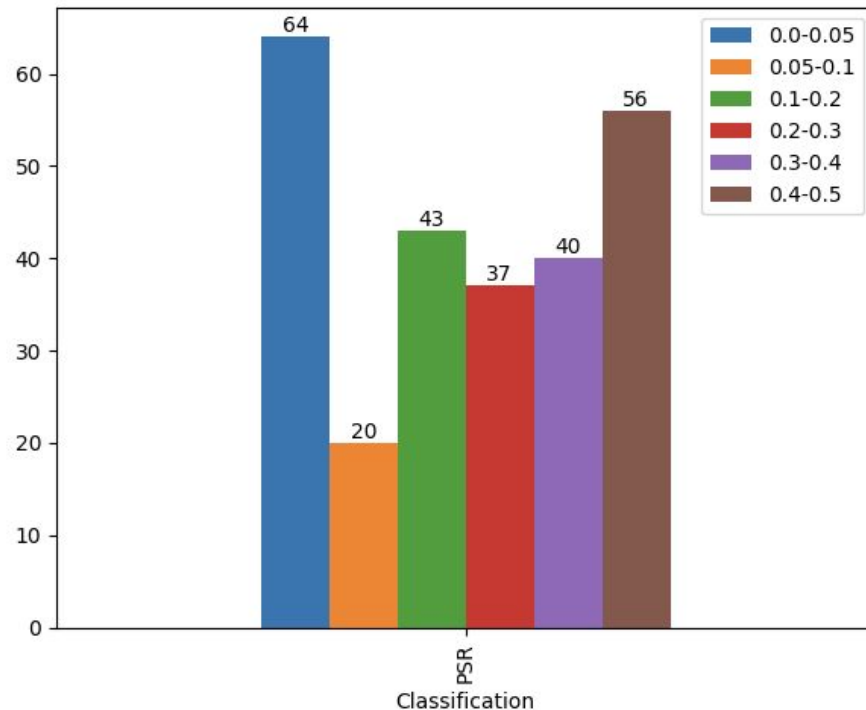
Logistic Regression

- # iterations = 200 Total: 2430 unassociated sources

2170 unassociated sources AGN classification



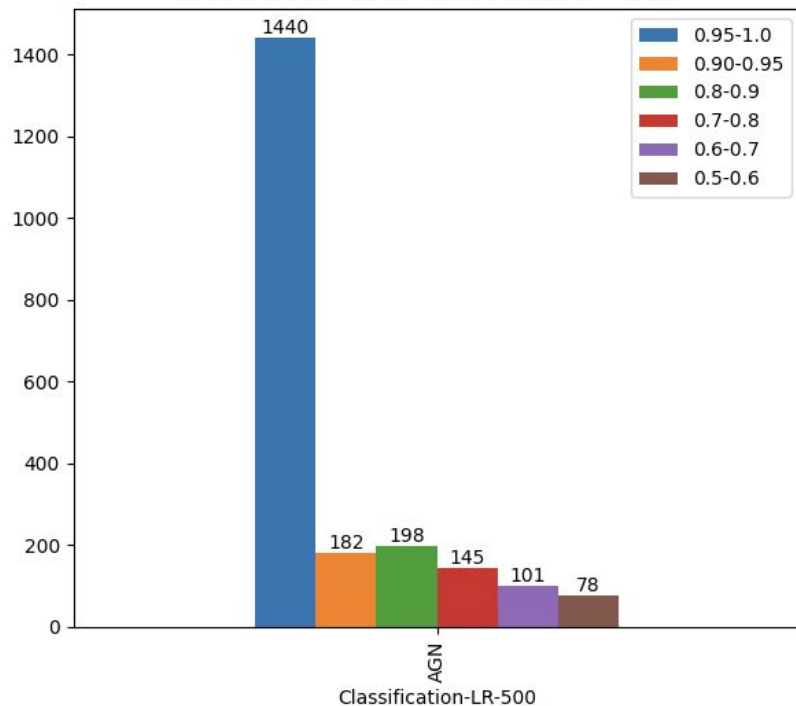
260 unassociated sources PSR classification



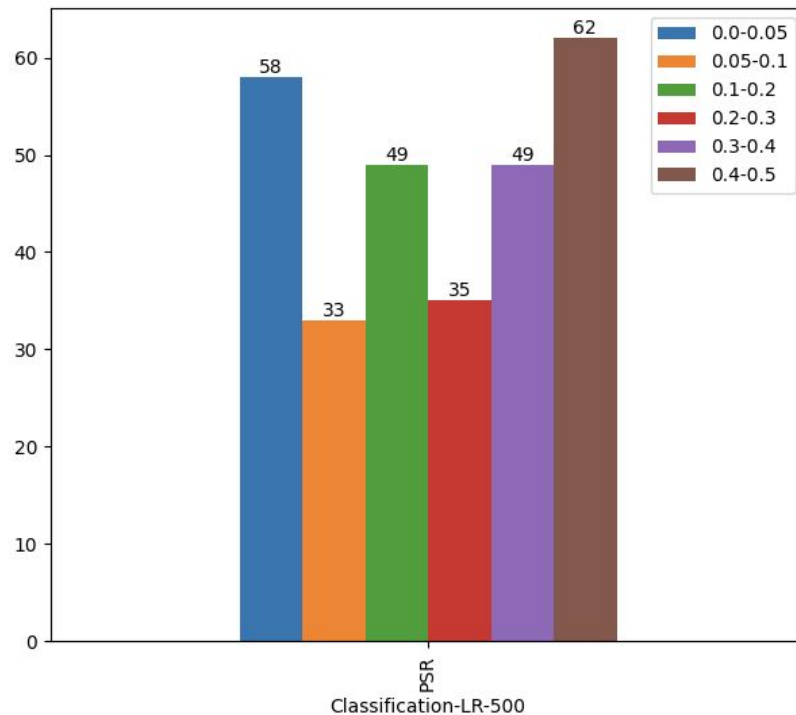
Logistic Regression

- # iterations = 500 Total: 2430 unassociated sources

2144 unassociated sources AGN classification

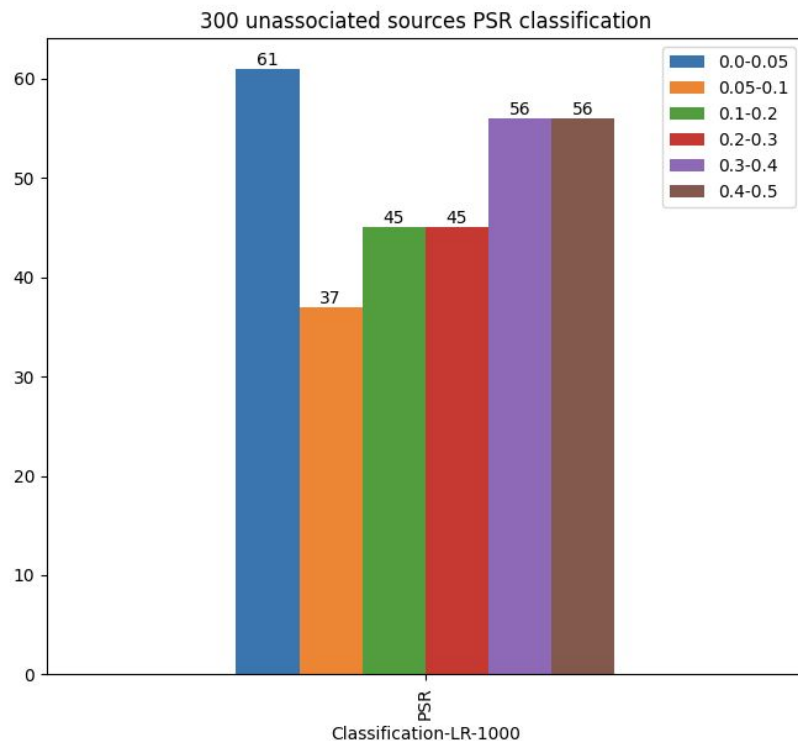
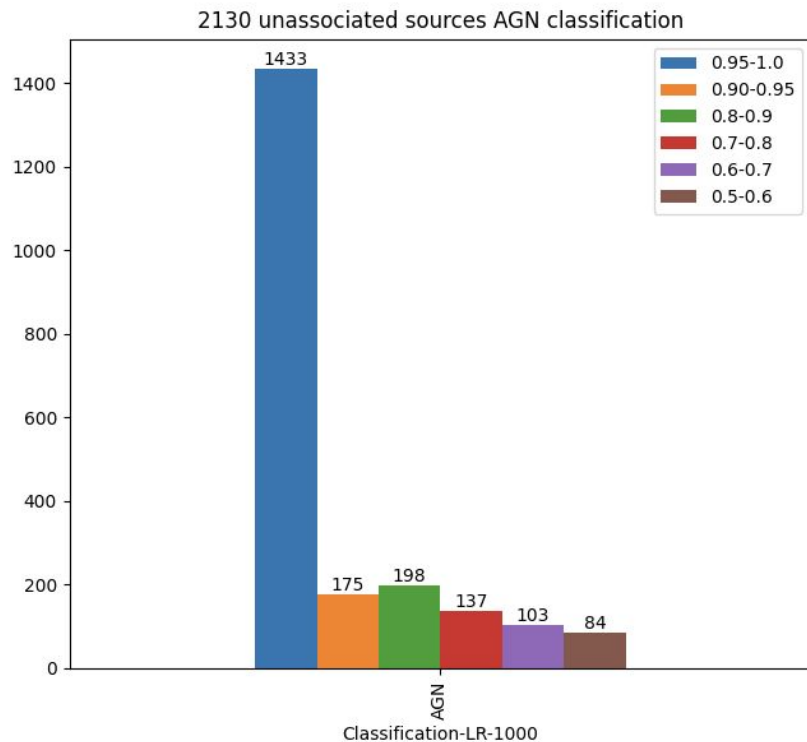


286 unassociated sources PSR classification



Logistic Regression

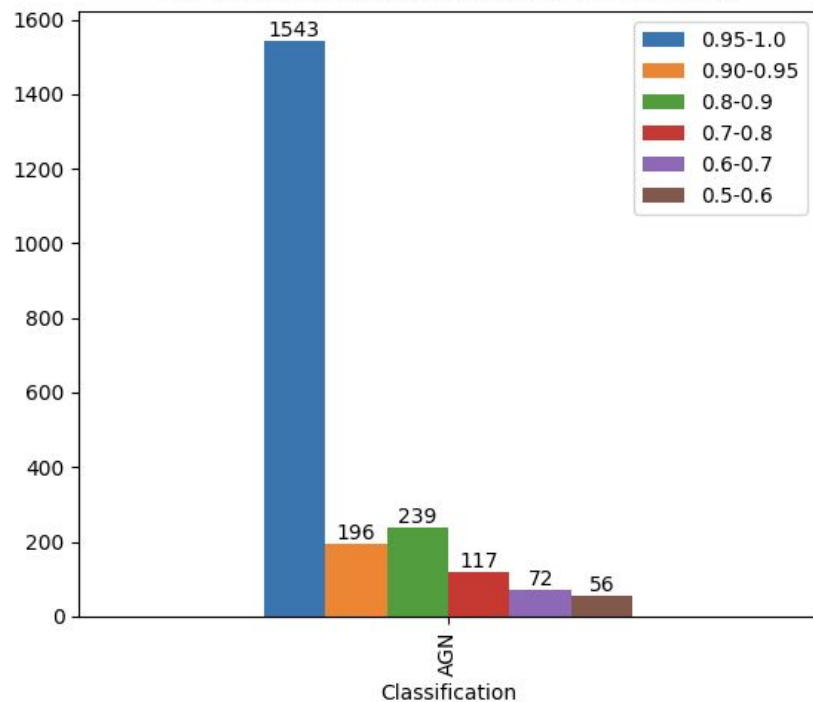
- # iterations = 1000 Total: 2430 unassociated sources



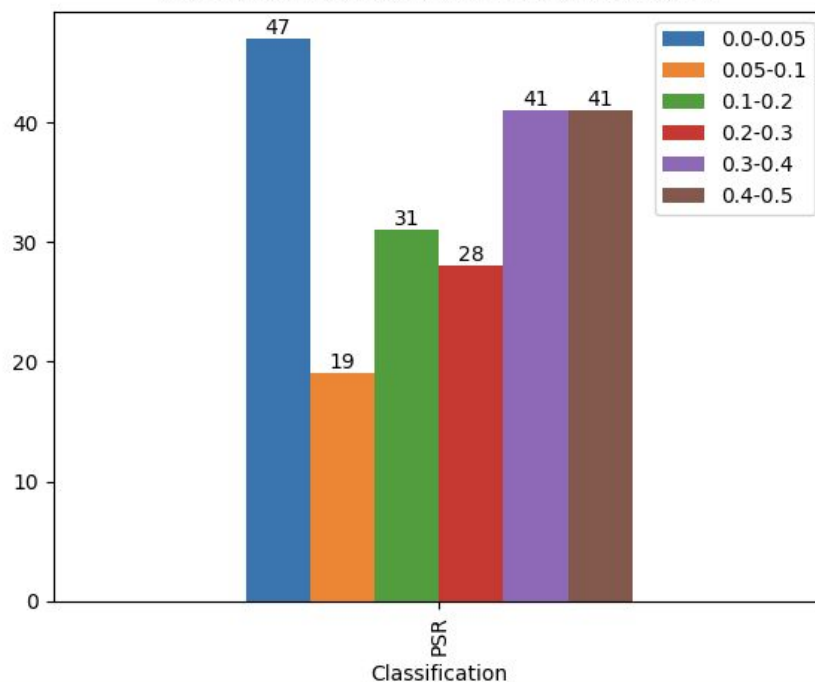
Neural Networks

- # iterations = 50 Total: 2430 unassociated sources

2223 unassociated sources AGN classification



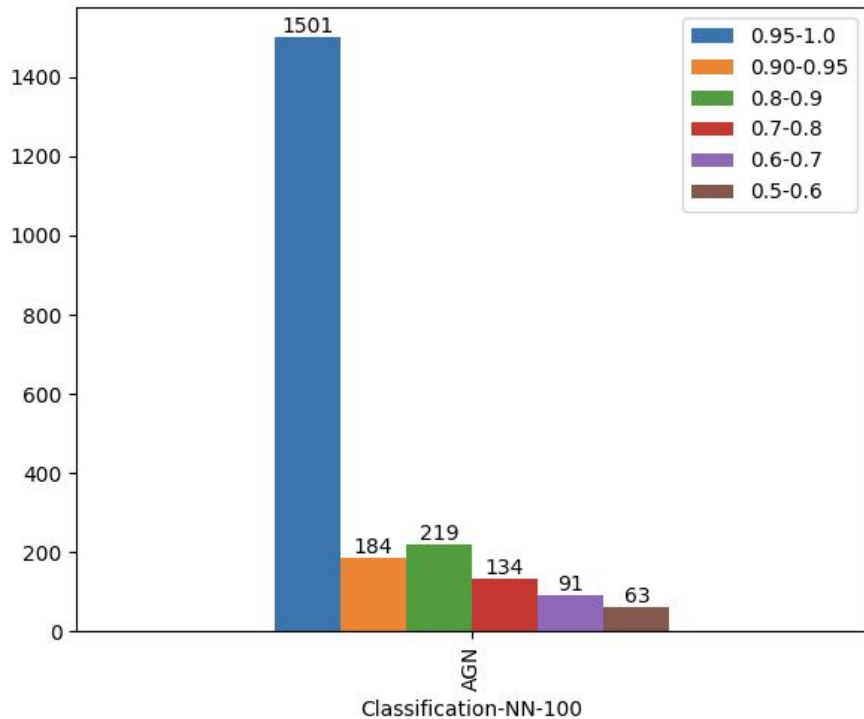
207 unassociated sources PSR classification



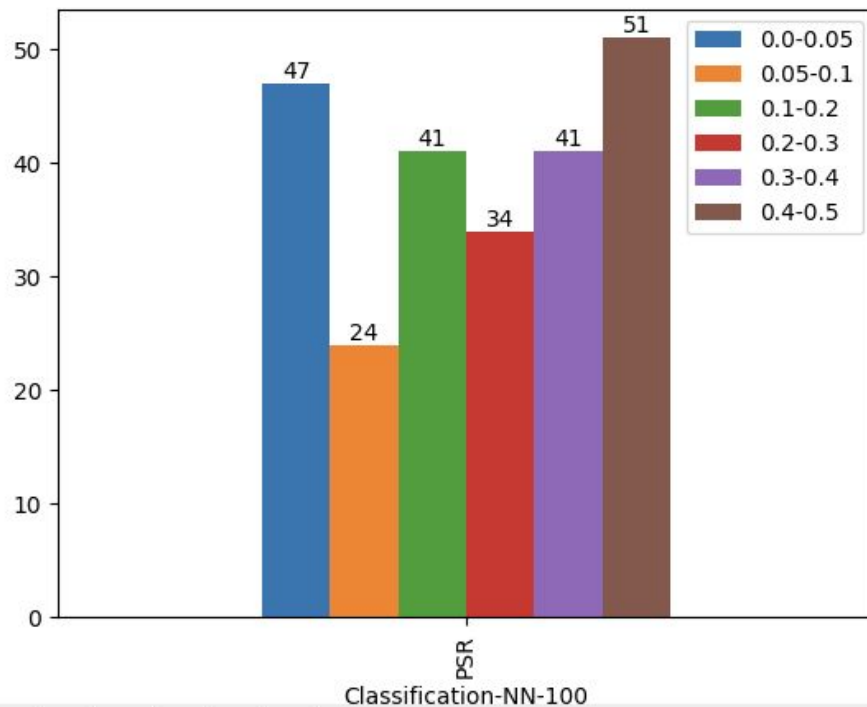
Neural Networks

- # iterations = 100 Total: 2430 unassociated sources

2192 unassociated sources AGN classification



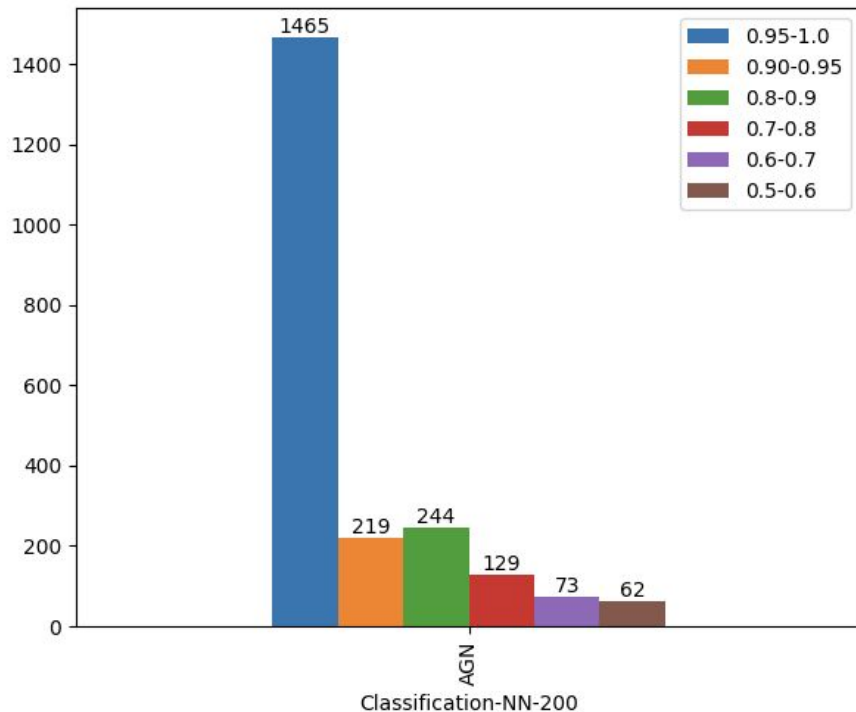
238 unassociated sources PSR classification



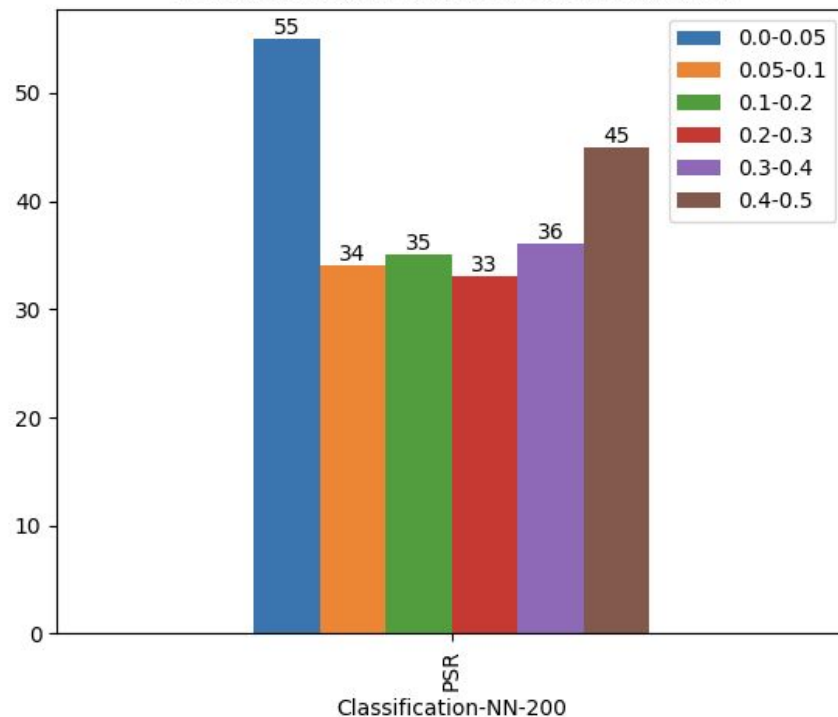
Neural Networks

- # iterations = 200 Total: 2430 unassociated sources

2192 unassociated sources AGN classification



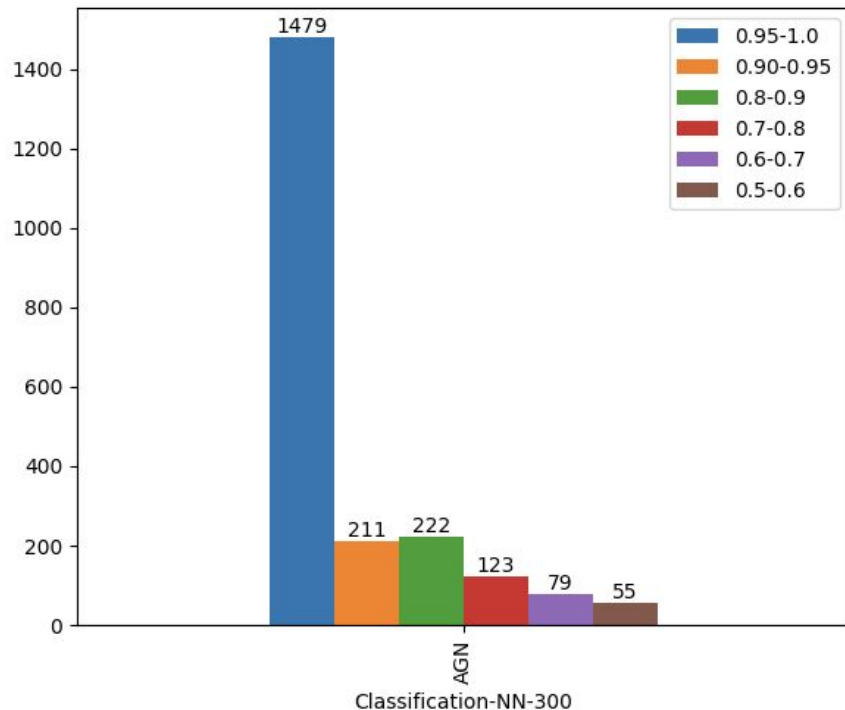
238 unassociated sources PSR classification



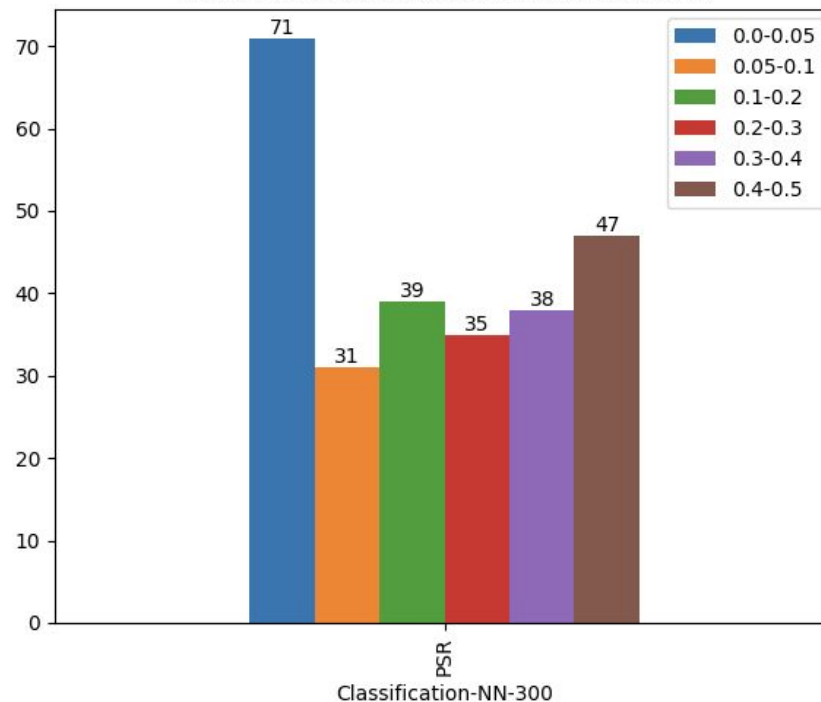
Neural Networks

- # iterations = 300 Total: 2430 unassociated sources

2169 unassociated sources AGN classification



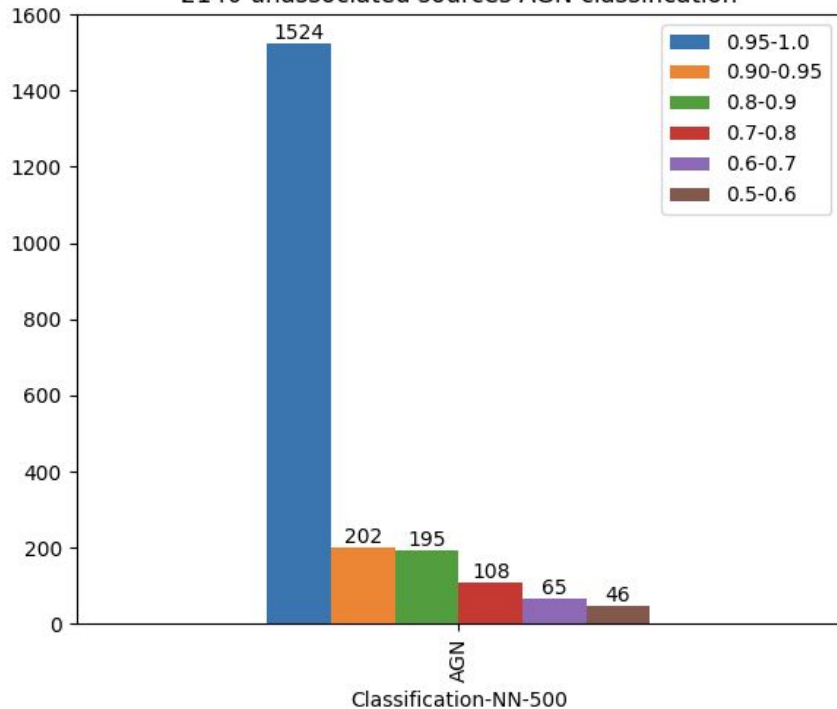
261 unassociated sources PSR classification



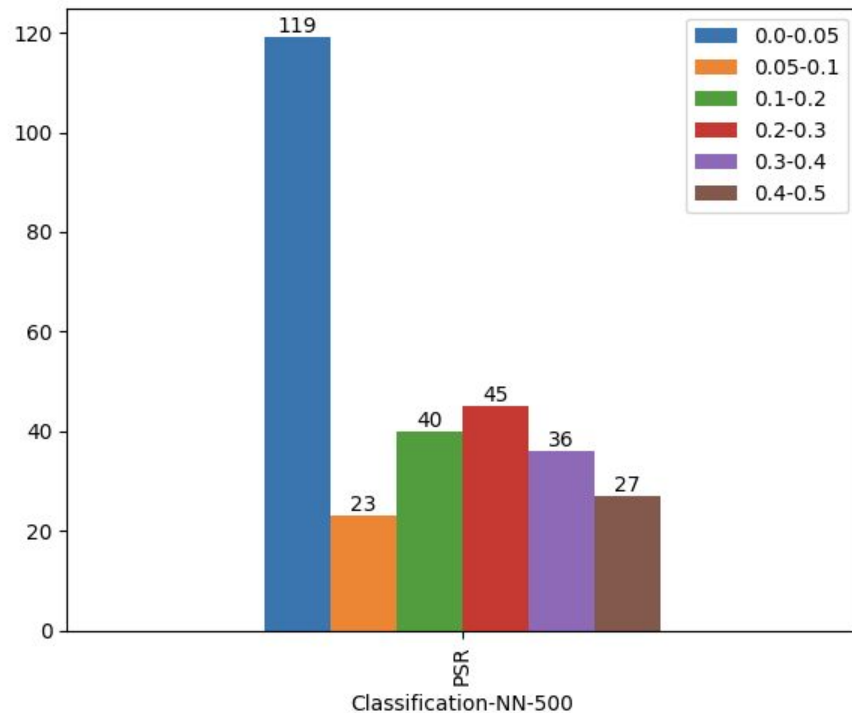
Neural Networks

- # iterations = 500 Total: 2430 unassociated sources

2140 unassociated sources AGN classification



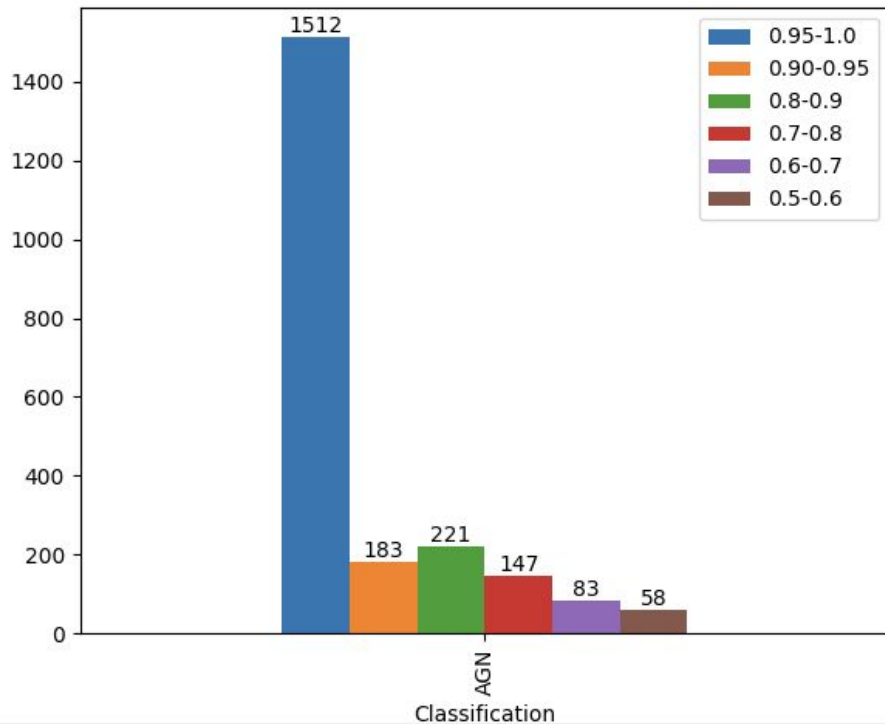
290 unassociated sources PSR classification



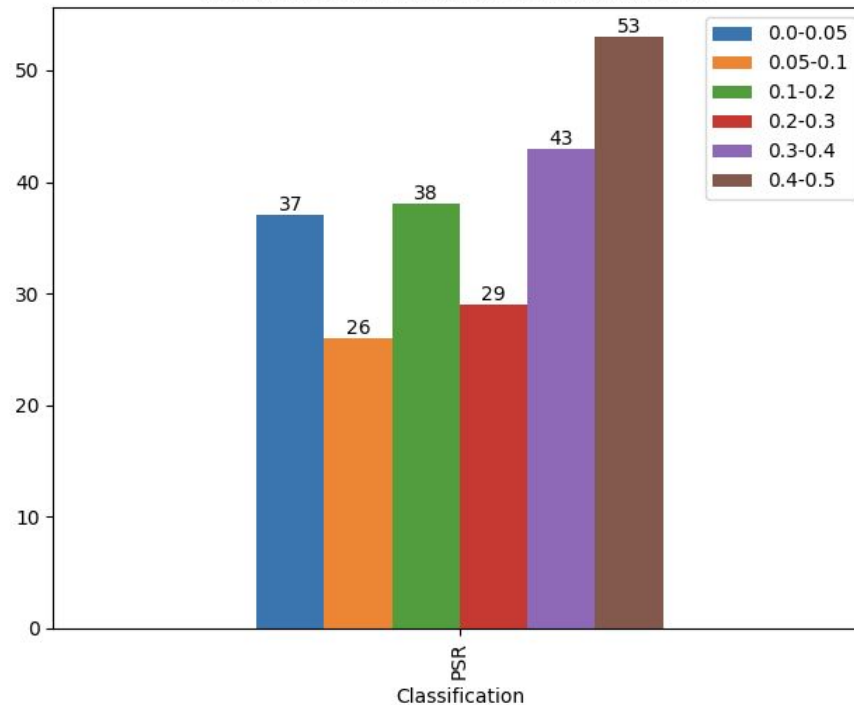
Deep Neural Networks

- # iterations = 50 Total: 2430 unassociated sources

2204 unassociated sources AGN classification



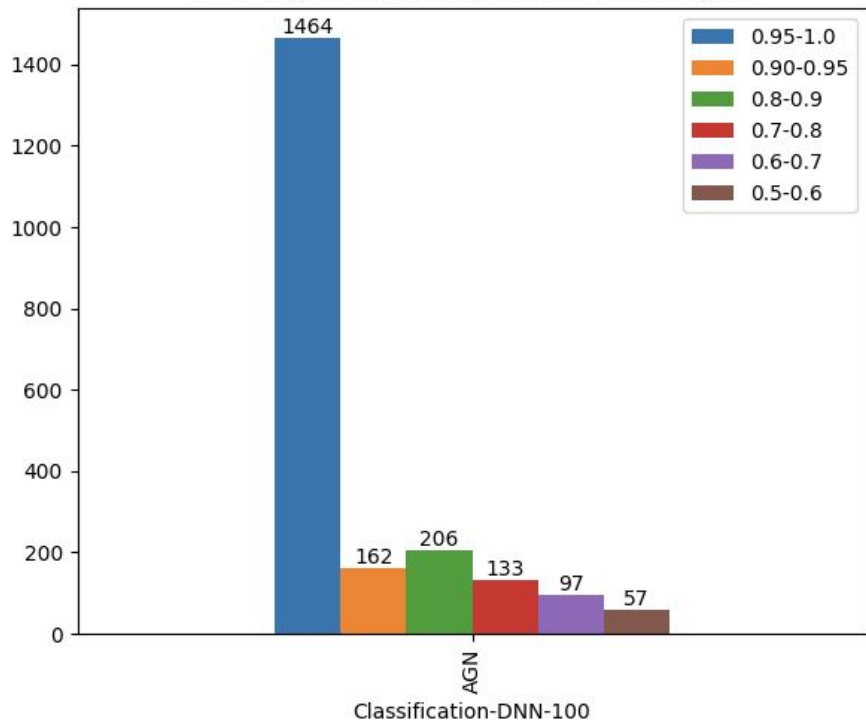
226 unassociated sources PSR classification



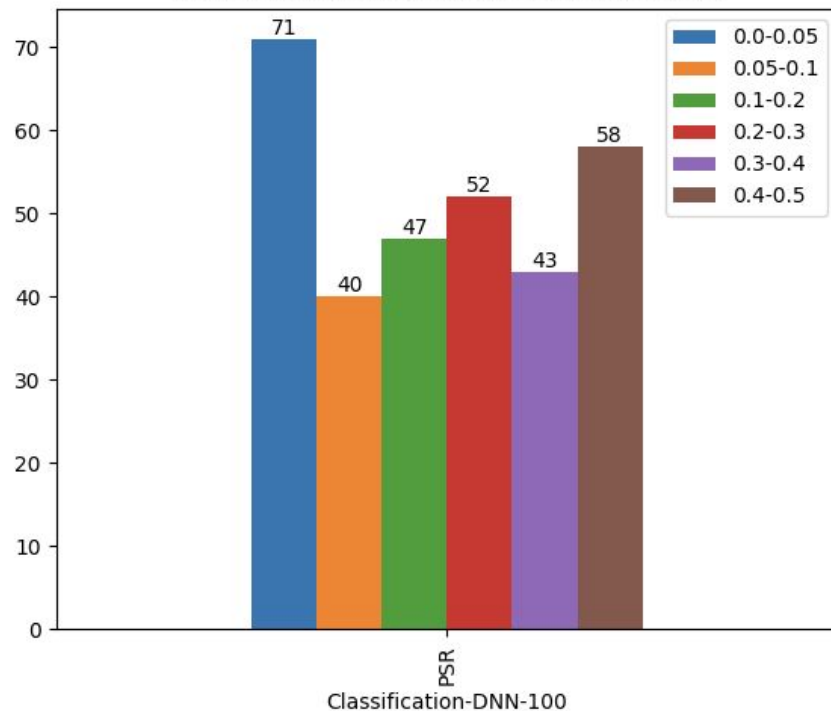
Deep Neural Networks

- # iterations = 100 Total: 2430 unassociated sources

2119 unassociated sources AGN classification



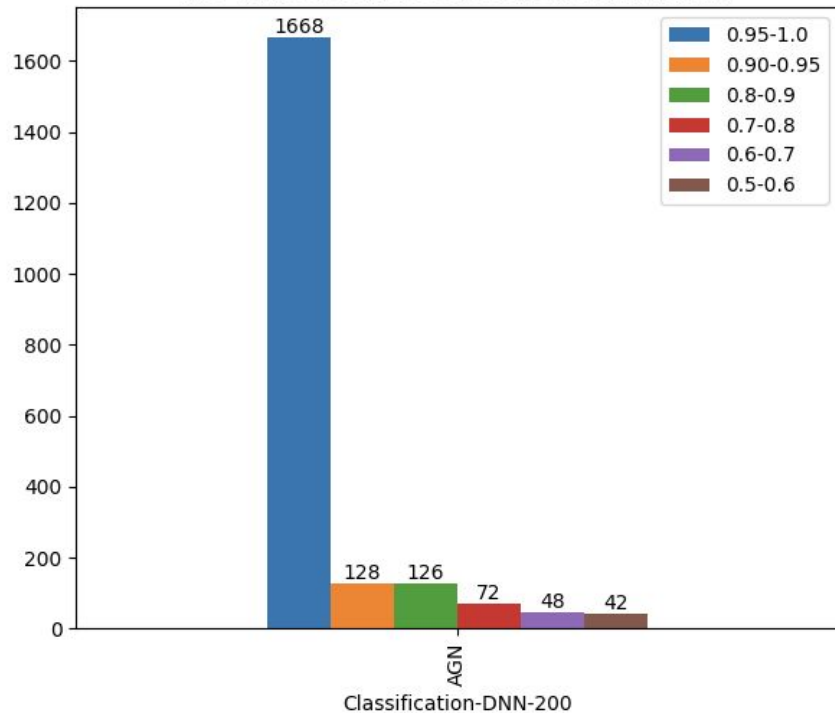
311 unassociated sources PSR classification



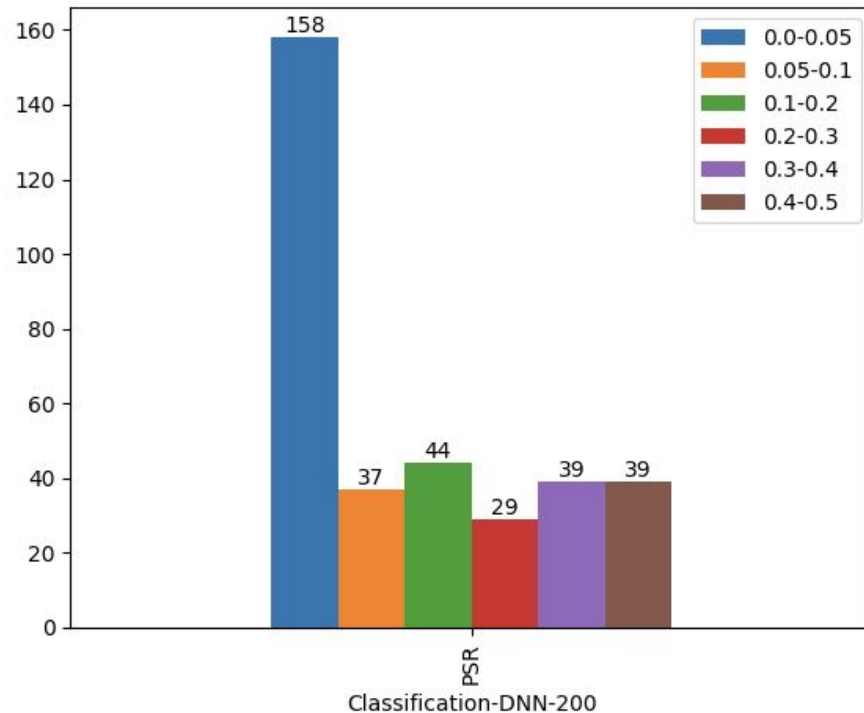
Deep Neural Networks

- # iterations = 200 Total: 2430 unassociated sources

2084 unassociated sources AGN classification

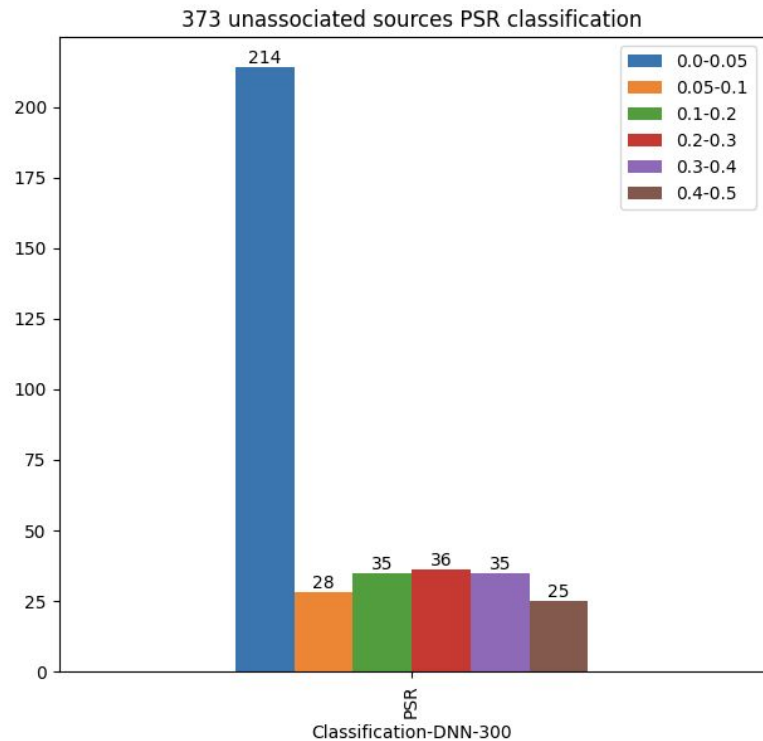
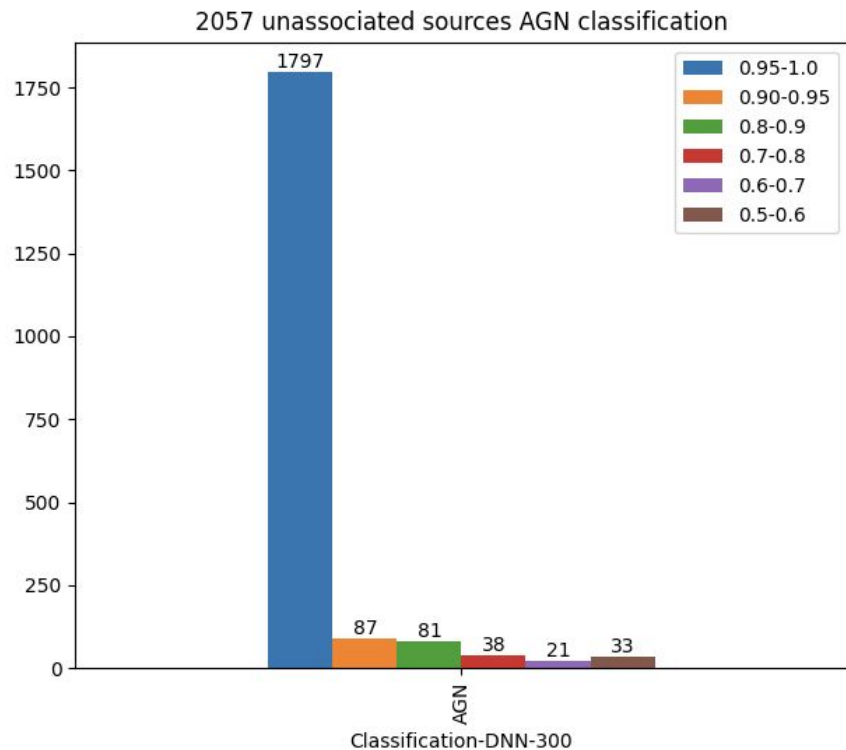


346 unassociated sources PSR classification



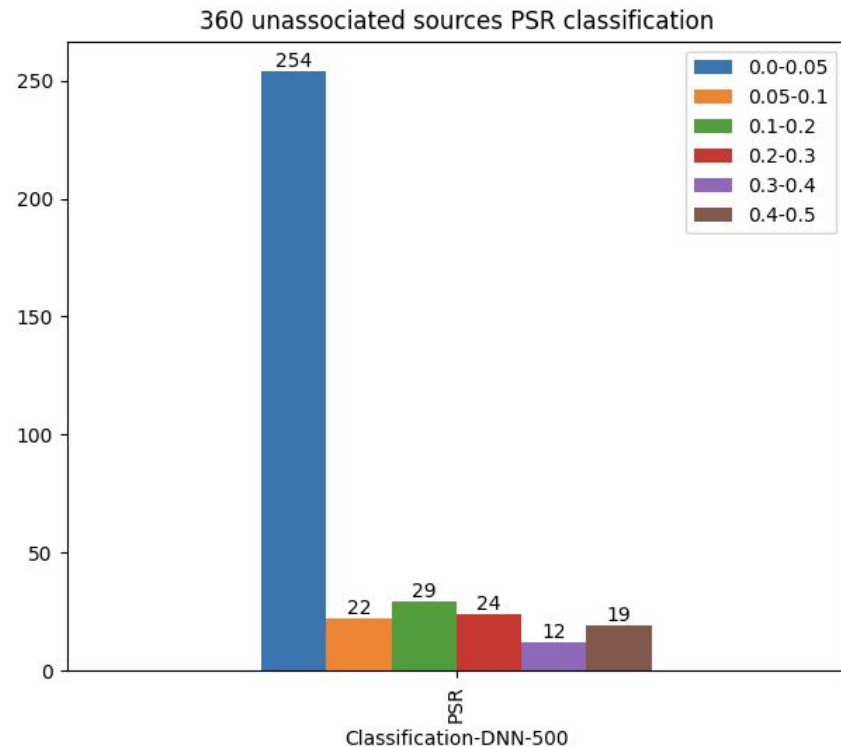
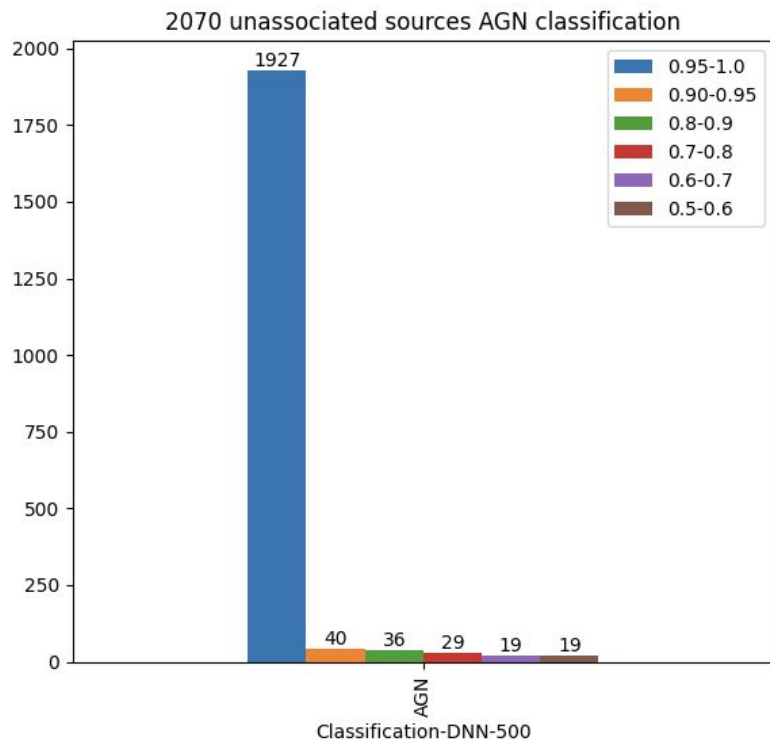
Deep Neural Networks

- # iterations = 300 Total: 2430 unassociated sources



Deep Neural Networks

- # iterations = 500 Total: 2430 unassociated sources

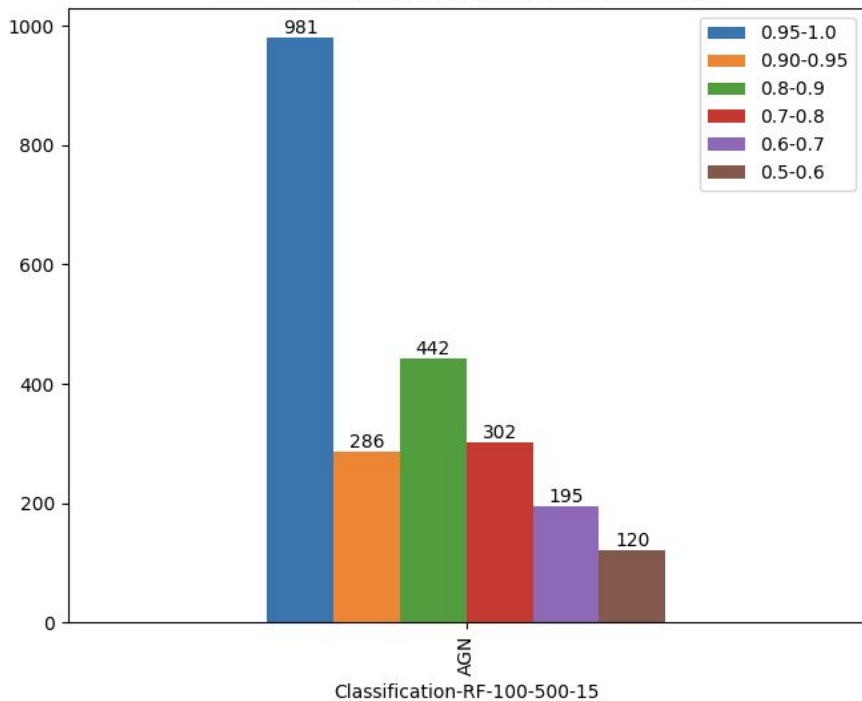


Random Forests

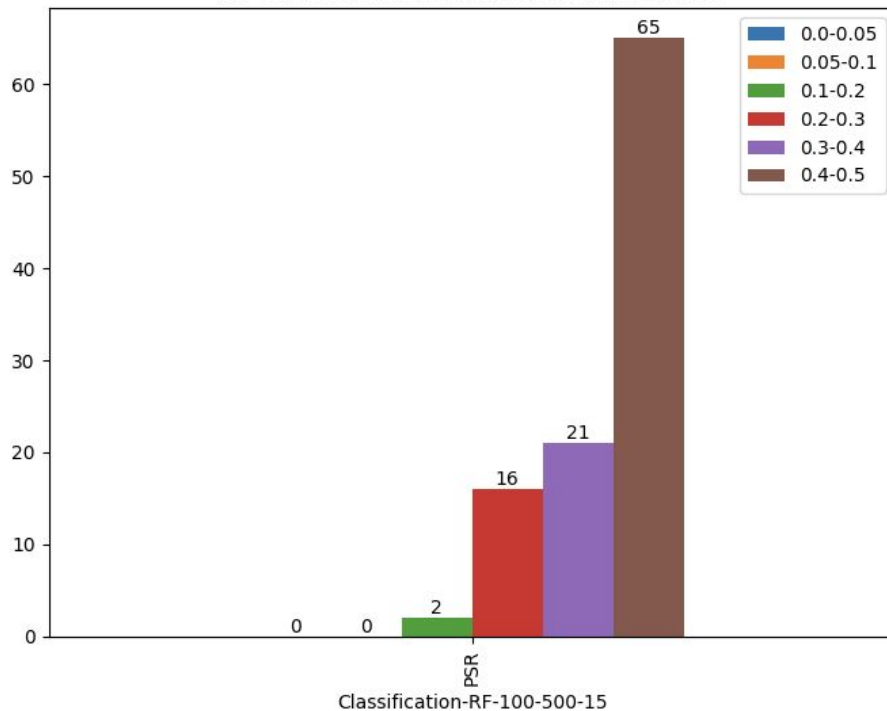
- #trees: 100, # samples: 500, depth: 15

Total: 2430 unassociated sources

2326 unassociated sources AGN classification



104 unassociated sources PSR classification



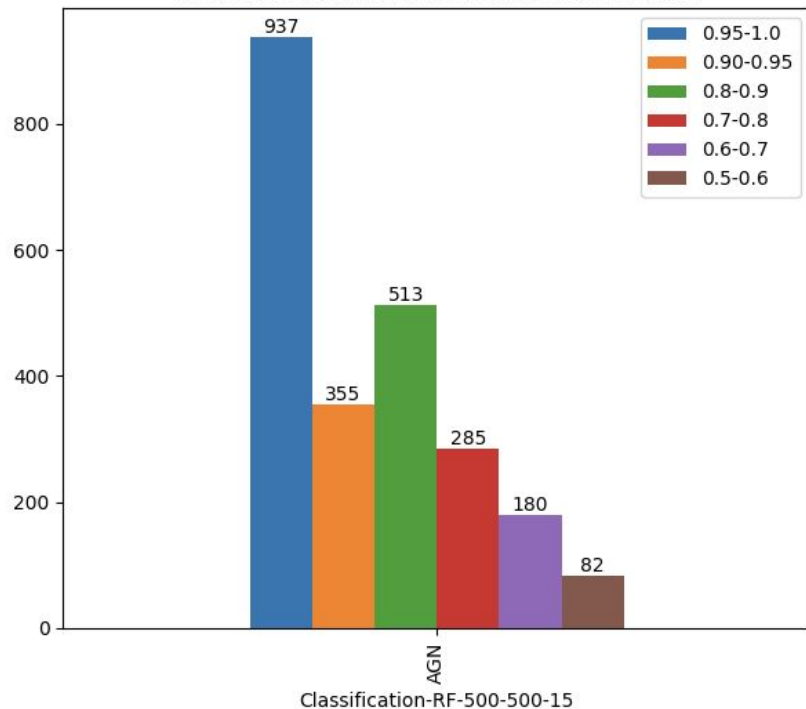
[illegible]

Random Forests

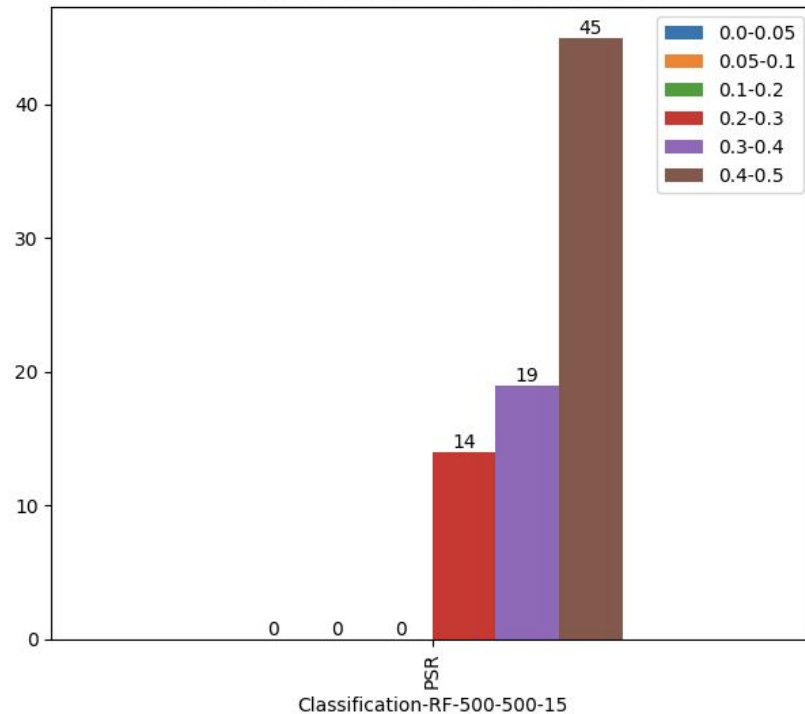
- #trees: 500, # samples: 500, depth: 15

Total: 2430 unassociated sources

2352 unassociated sources AGN classification



78 unassociated sources PSR classification

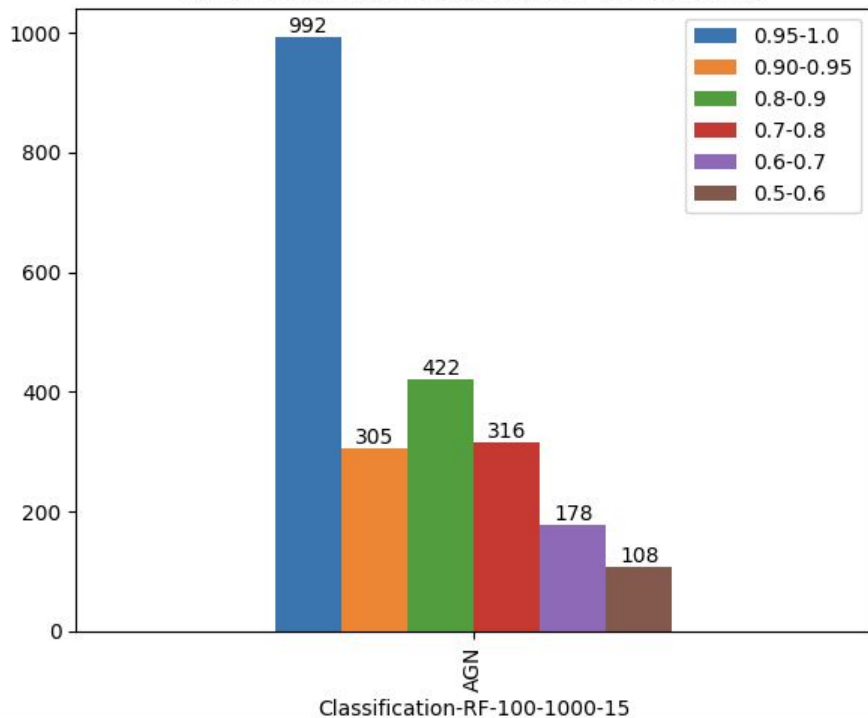


Random Forests

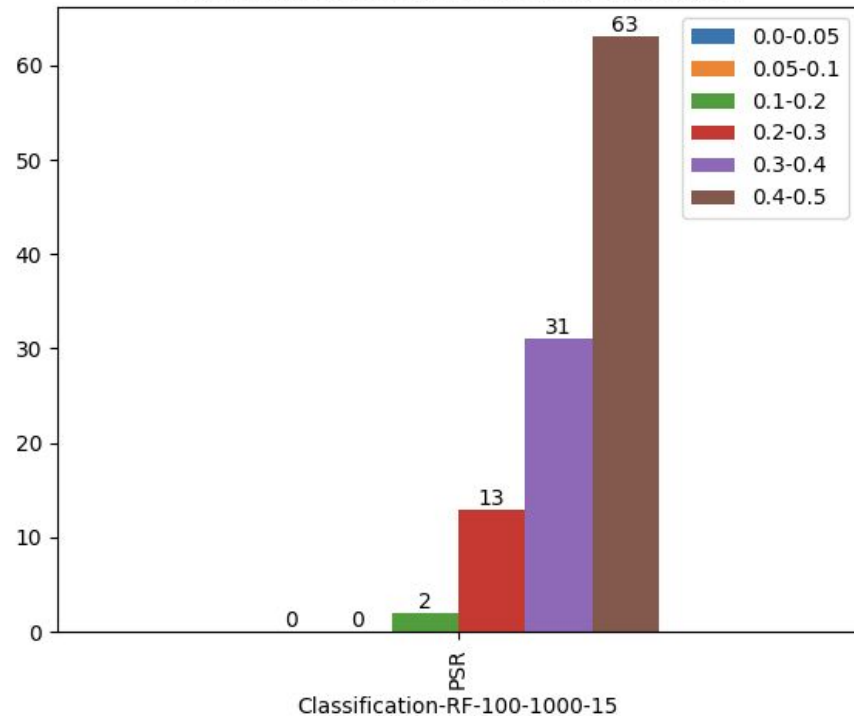
- #trees: 100, # samples: 1000, depth: 15

Total: 2430 unassociated sources

2321 unassociated sources AGN classification



109 unassociated sources PSR classification

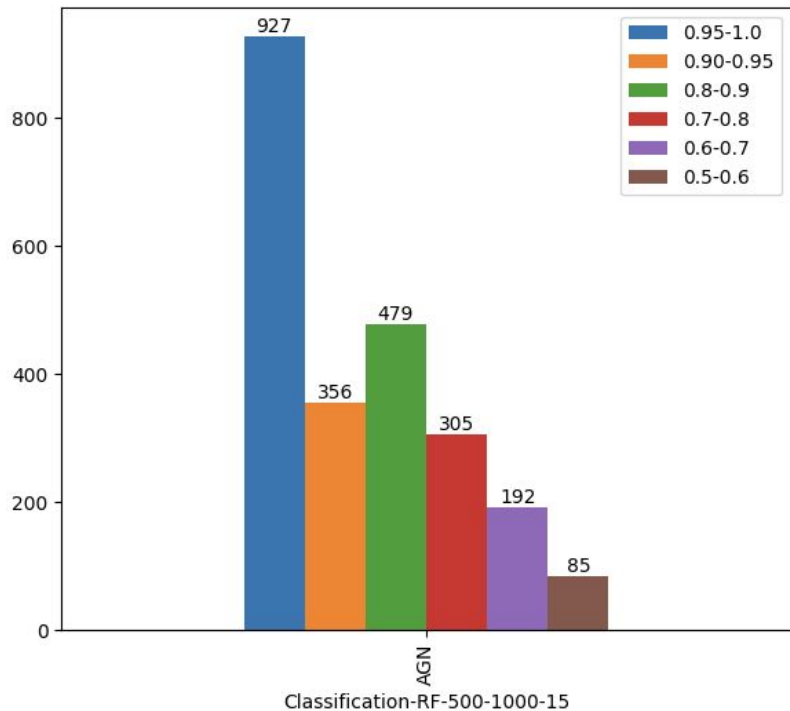


Random Forests

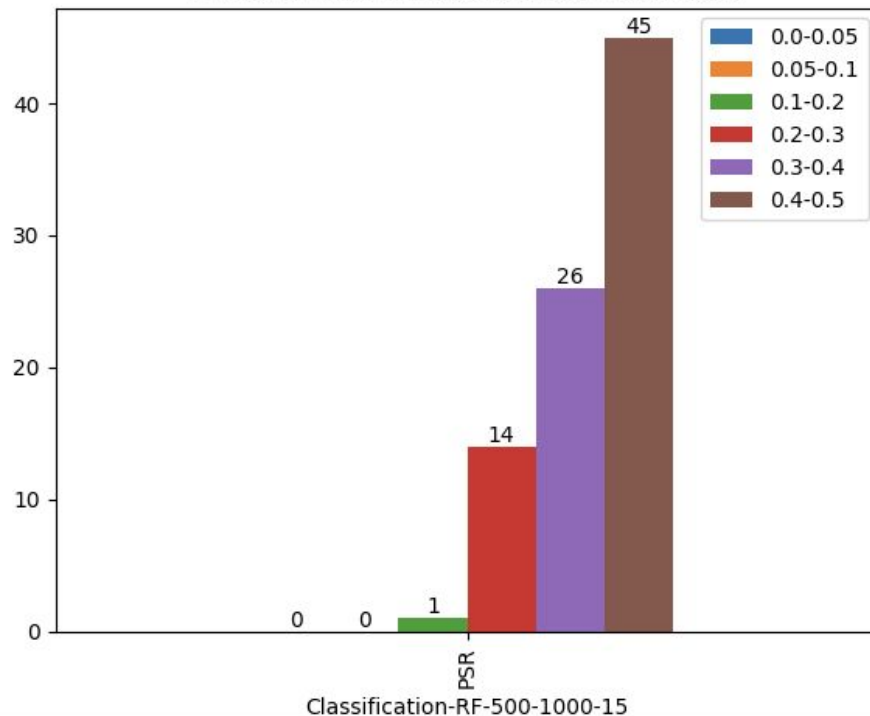
- #trees: 500, # samples: 1000, depth: 15

Total: 2430 unassociated sources

2344 unassociated sources AGN classification



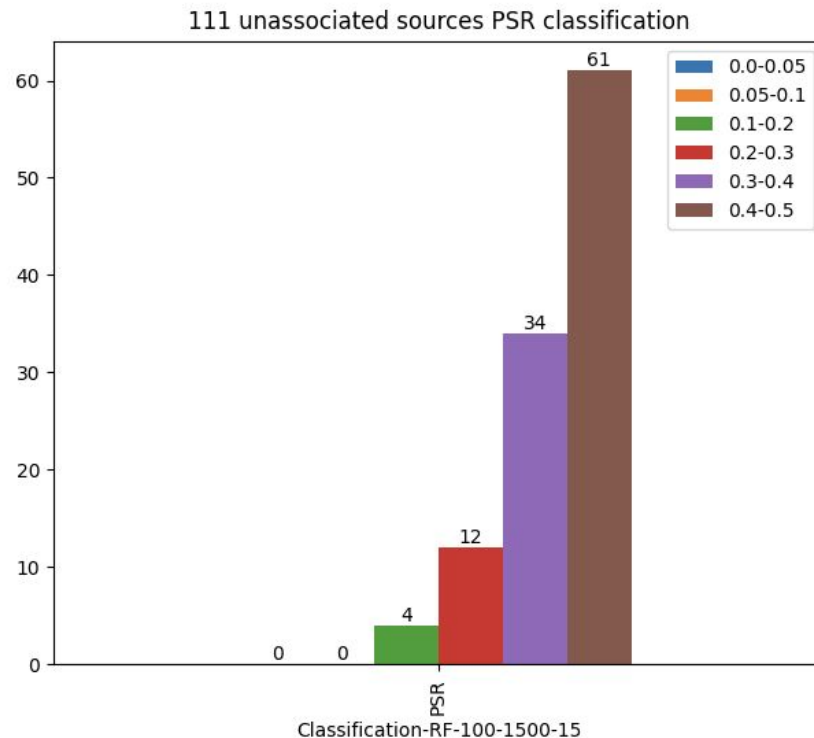
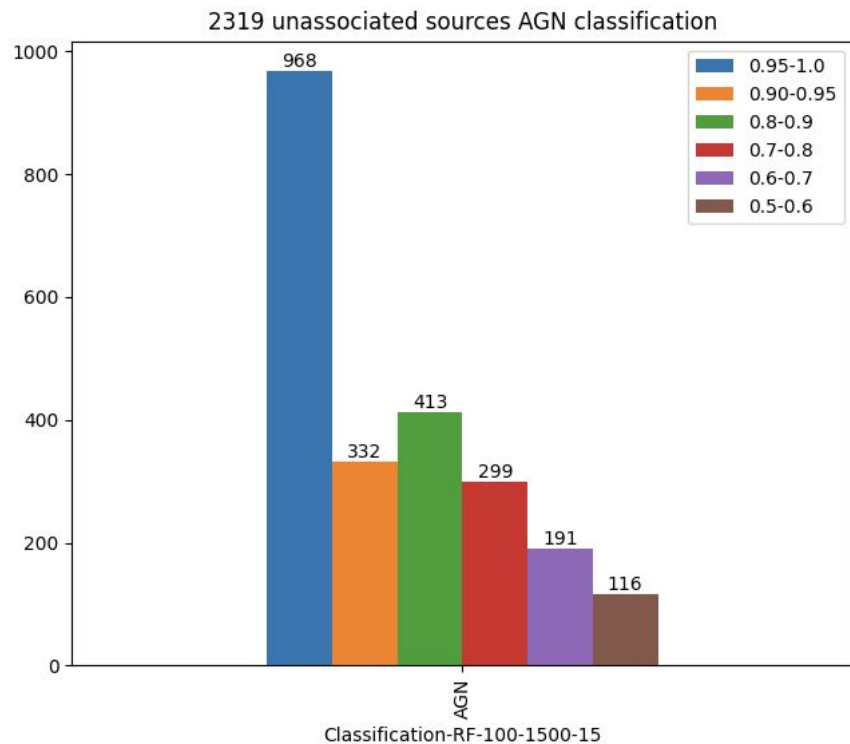
86 unassociated sources PSR classification



Random Forests

- #trees: 100, # samples: 1500, depth: 15

Total: 2430 unassociated sources

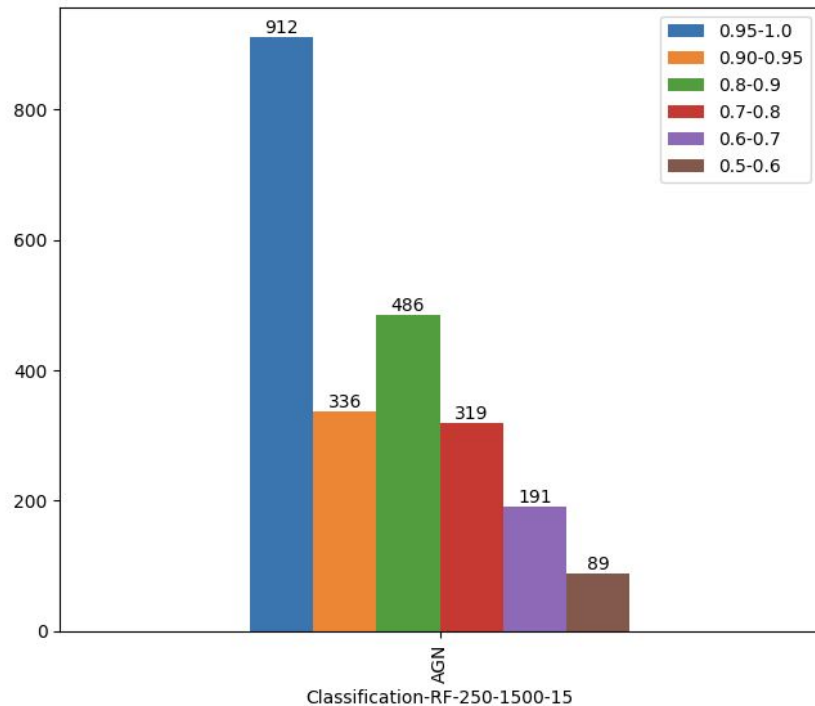


Random Forests

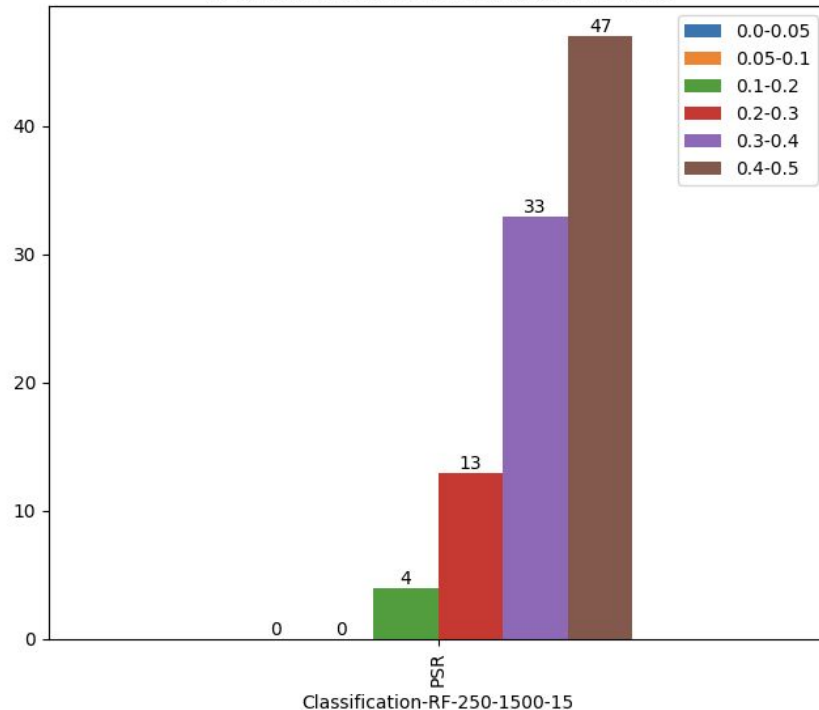
- #trees: 250, # samples: 1500, depth: 15

Total: 2430 unassociated sources

2333 unassociated sources AGN classification



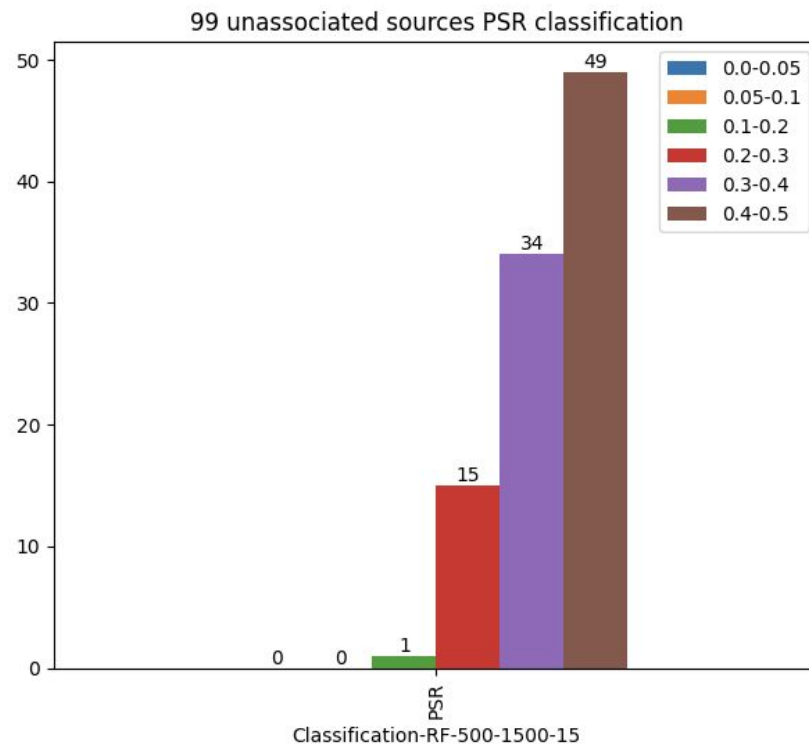
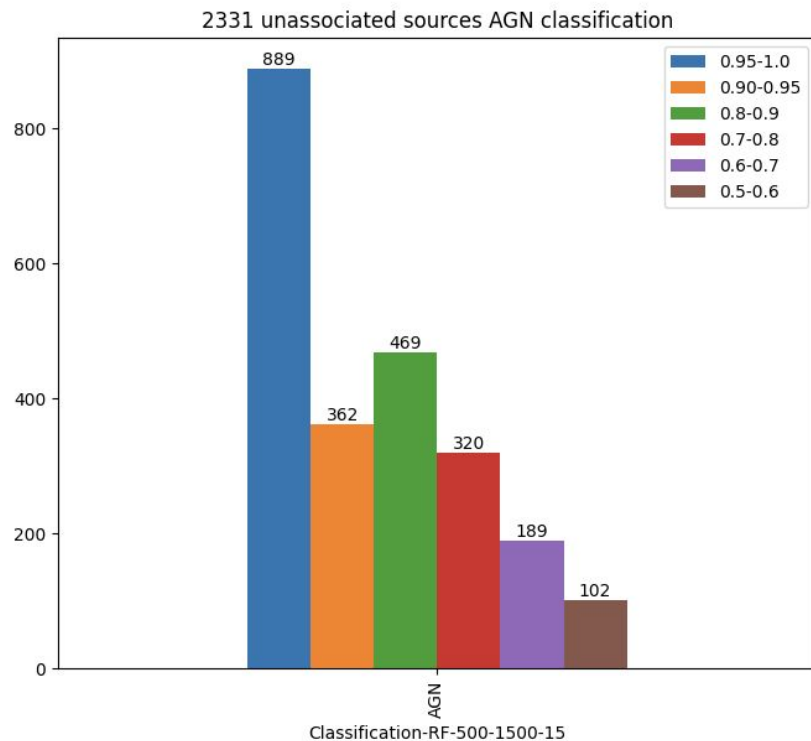
97 unassociated sources PSR classification



Random Forests

- #trees: 500, # samples: 1500, depth: 15

Total: 2430 unassociated sources

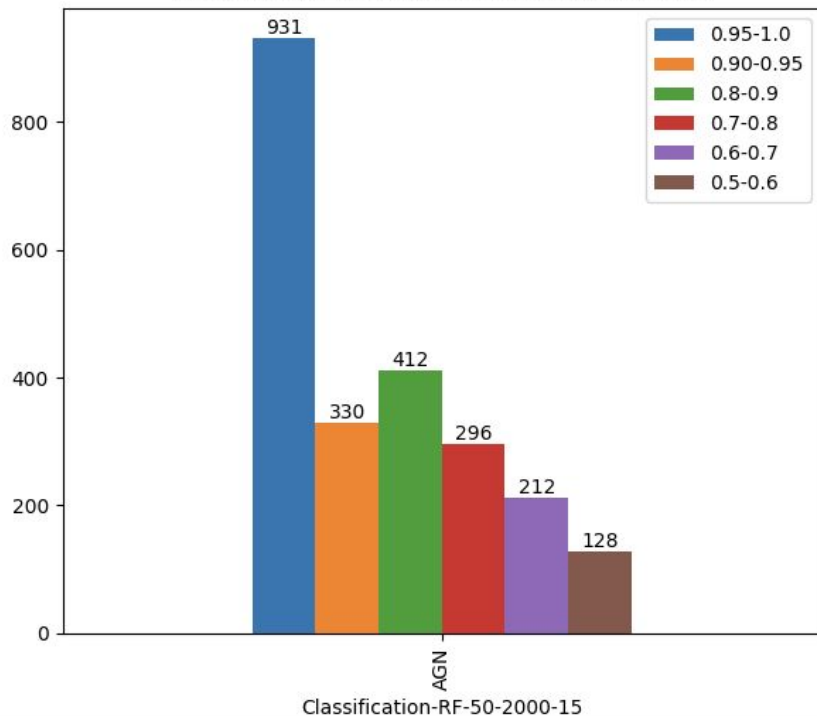


Random Forests

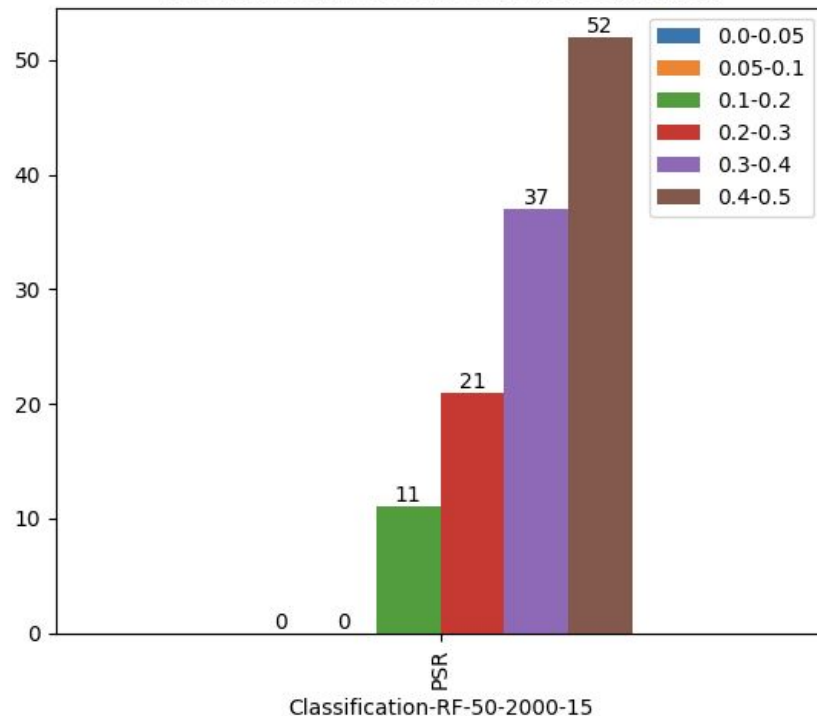
- #trees: 50, # samples: 2000, depth: 15

Total: 2430 unassociated sources

2309 unassociated sources AGN classification



121 unassociated sources PSR classification

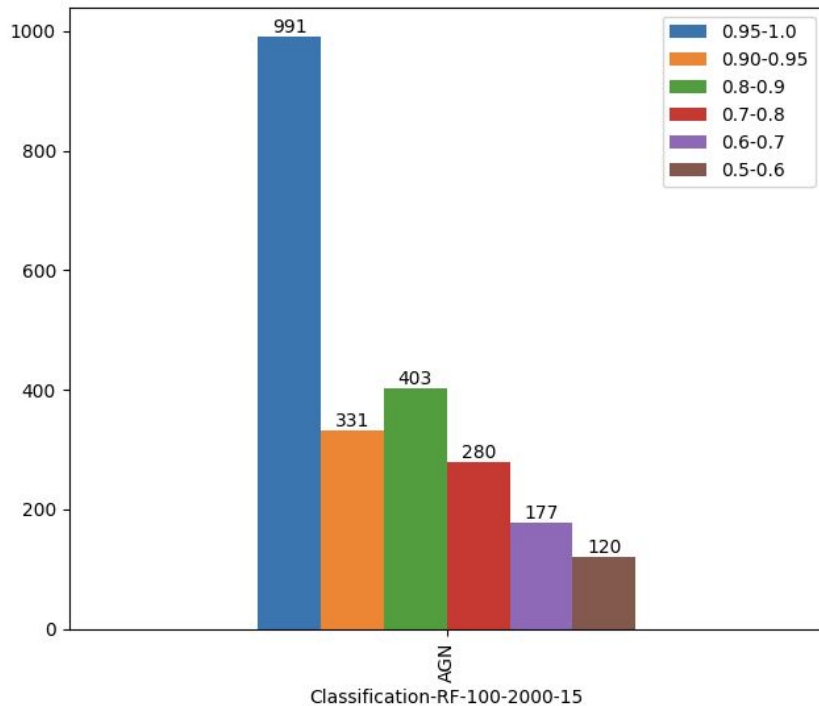


Random Forests

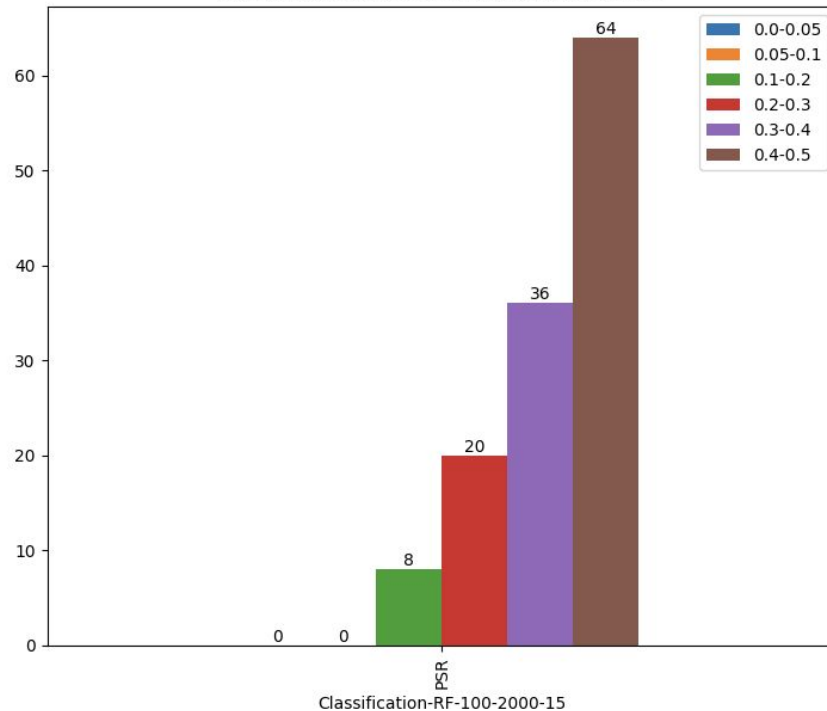
- #trees: 100, # samples: 2000, depth: 15

Total: 2430 unassociated sources

2302 unassociated sources AGN classification



128 unassociated sources PSR classification

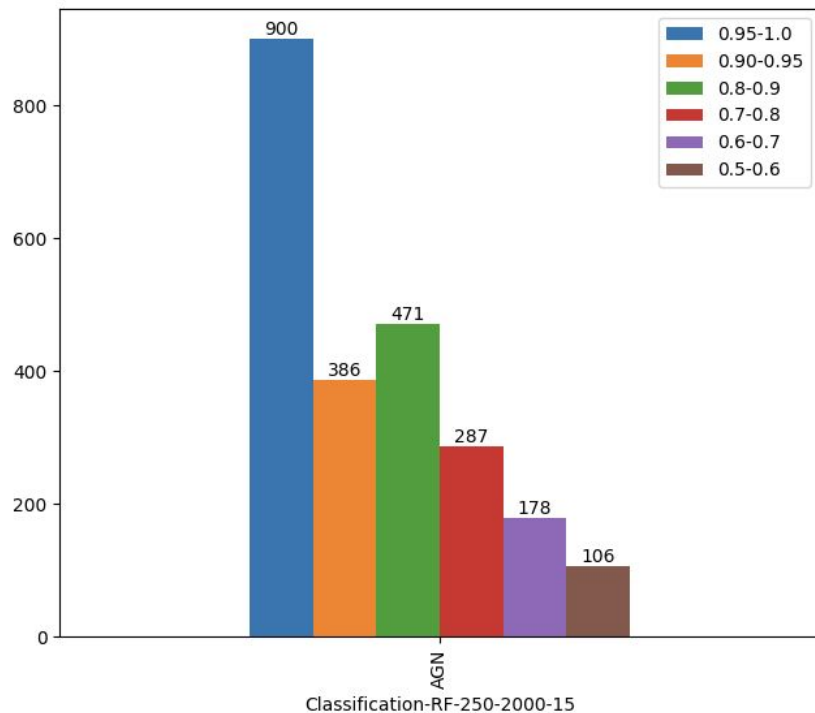


Random Forests

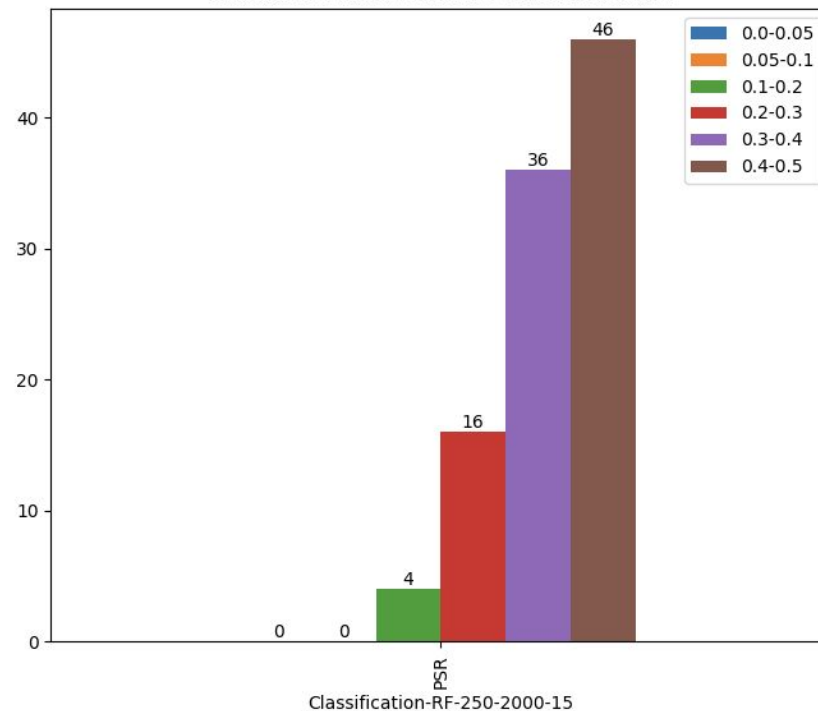
- #trees: 250, # samples: 2000, depth: 15

Total: 2430 unassociated sources

2328 unassociated sources AGN classification



102 unassociated sources PSR classification

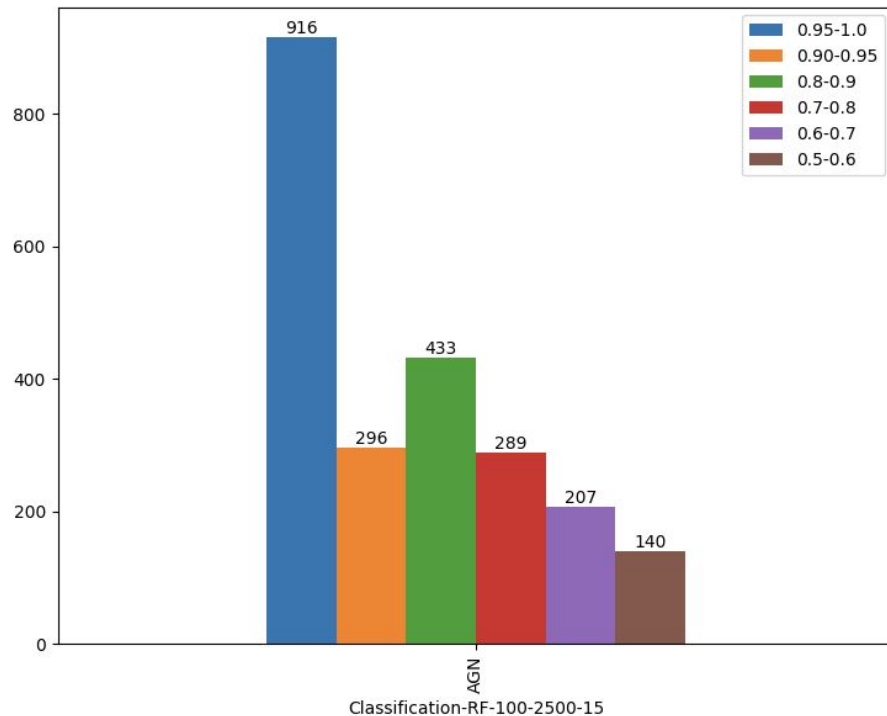


Random Forests

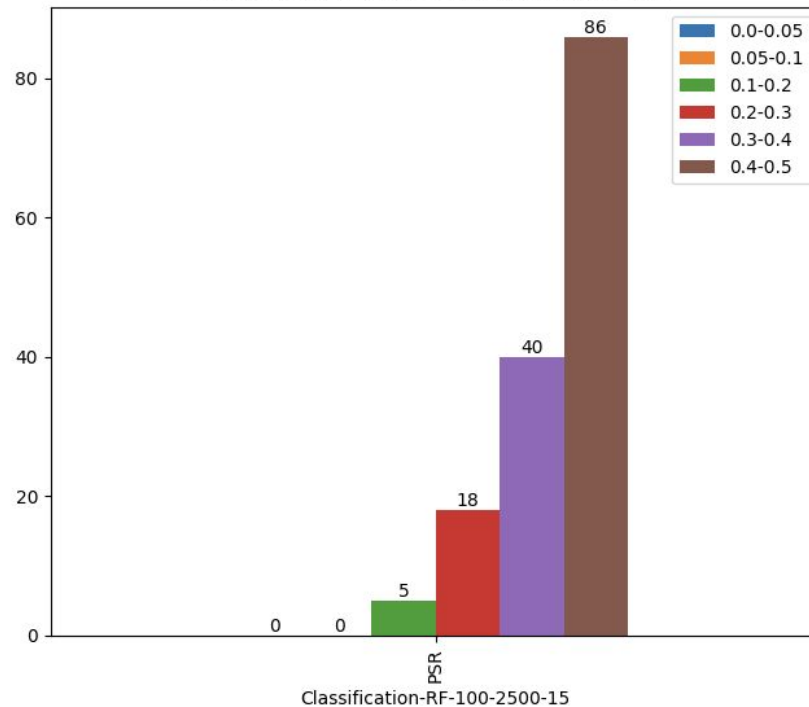
- #trees: 100, # samples: 2500, depth: 15

Total: 2430 unassociated sources

2281 unassociated sources AGN classification



149 unassociated sources PSR classification

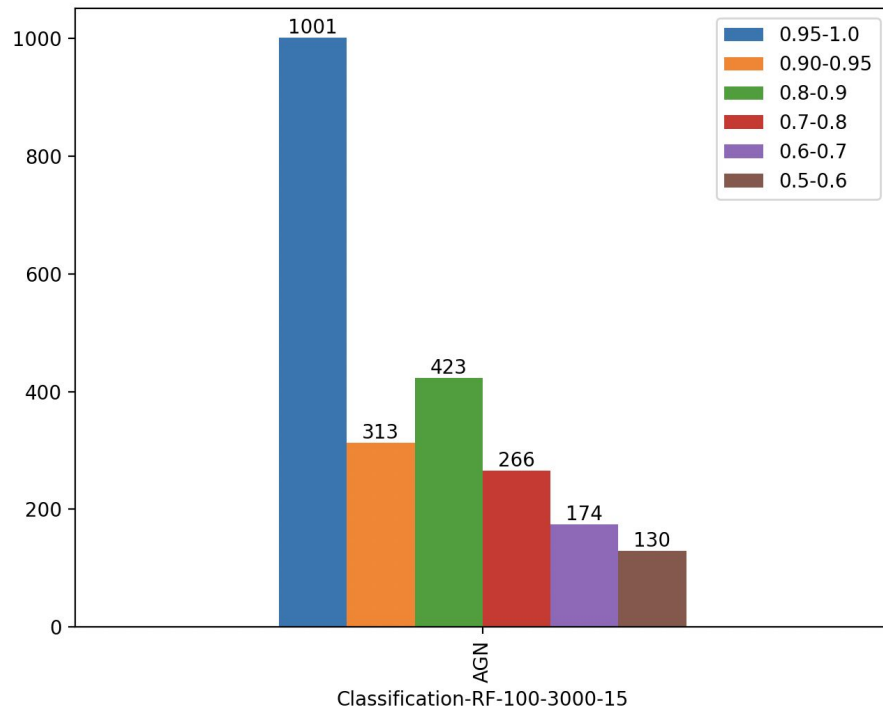


Random Forests

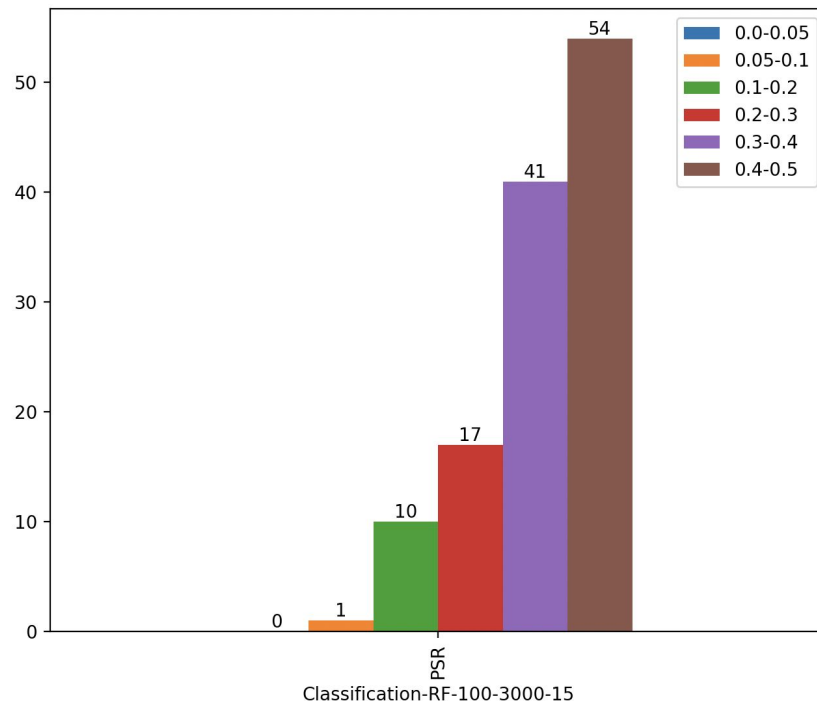
- #trees: 100, # samples: 3000, depth: 15

Total: 2430 unassociated sources

2307 unassociated sources AGN classification



123 unassociated sources PSR classification



Results

- LR:
 - Firmly predicted 1433-1465 AGNs in [0.95-1.0] range
 - Firmly predicted 58-65 PSRs in [0.0-0.05] range until 200 iterations.
- Neural Networks
 - Firmly predicted 1465-1543 AGNs [0.95-1.0] range
 - Firmly predicted 47-119 PSRs [0.0-0.05] range
- Deep Neural Networks
 - Firmly predicted 1464-1927 AGNs [0.95-1.0] range
 - Firmly predicted 37-254 PSRs [0.0-0.05] range
 - More iterations predicted more AGNs in [0.95-1.0] range and more PSRs in [0.0-0.05] range
- Random Forests
 - Firmly predicted 927-992 AGNs [0.95-1.0] range
 - Firmly predicted 0 PSRs in the [0.0-0.05] range
 - Predicted 0-4 PSRs in the [0.5-0.10] range
 - Weakly predicted 45-65 PSRs in [0.45-0.5] range

Update: 03/22/2024

Progress

- Solved the **unbalanced dataset problem** by multiplying total count of Pulsar rows
- So to multiply by 10, created 9 more rows for each Pular row

Code Changes

```
unit.append(float(source[wanted_feature_names[9]]))
unit.append(float(source[wanted_feature_names[10]]))
unit.append(float(source[wanted_feature_names[11]]))
unit.append(float(source[wanted_feature_names[12]]))
# Calculate hardness ratios (also considered in the paper)
for index in range(1, len(flux_list)):
    unit.append((flux_list[index] - flux_list[index-1] + 0.0) /
                (flux_list[index] + flux_list[index-1] + 0.0))
xdata.append(unit)
if source[wanted_type_col] in ("PSR", "psr", "MSP", "msp"):
    ydata.append([0]) # 0 is for pulsar
else:
    ydata.append([1]) # 1 is for AGN
# Multiply Pulsar rows by 10. Create 9 more rows for each Pulsar row
if source[wanted_type_col] in ("PSR", "psr", "MSP", "msp"):
    for i in range(9):
        xdata.append(unit)
        ydata.append([0]) # 0 is for pulsar
self.x = torch.from_numpy(np.array(xdata, float))
self.y = torch.from_numpy(np.array(ydata, float))
self.size = len(xdata)
```

Multiply Pulsar rows
by creating 9 more
rows for each Pulsar
row to get 10x rows

Code Changes

- Create **3 more PSR rows** for each PSR row

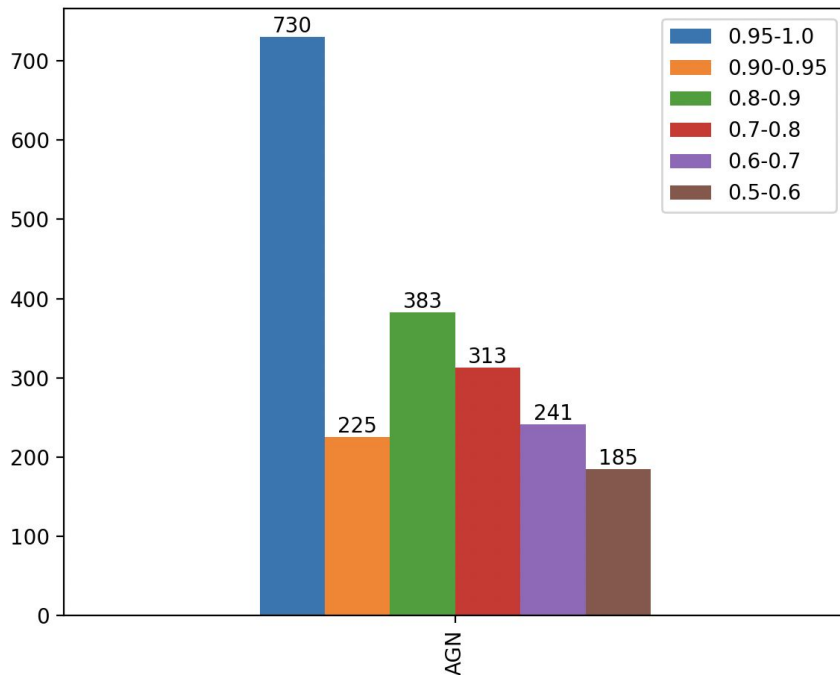
```
correct = 1527
test_size = 1587
correct_agm = 1170
total_agm = 1187 ←
correct_psr = 357
total_psr = 400 ←
sensitivity = 96.21928166351607
agm_sensitivity = 98.56781802864364
psr_sensitivity = 89.25
```

Code Changes

- #trees: 20, # samples: 500, depth: 15

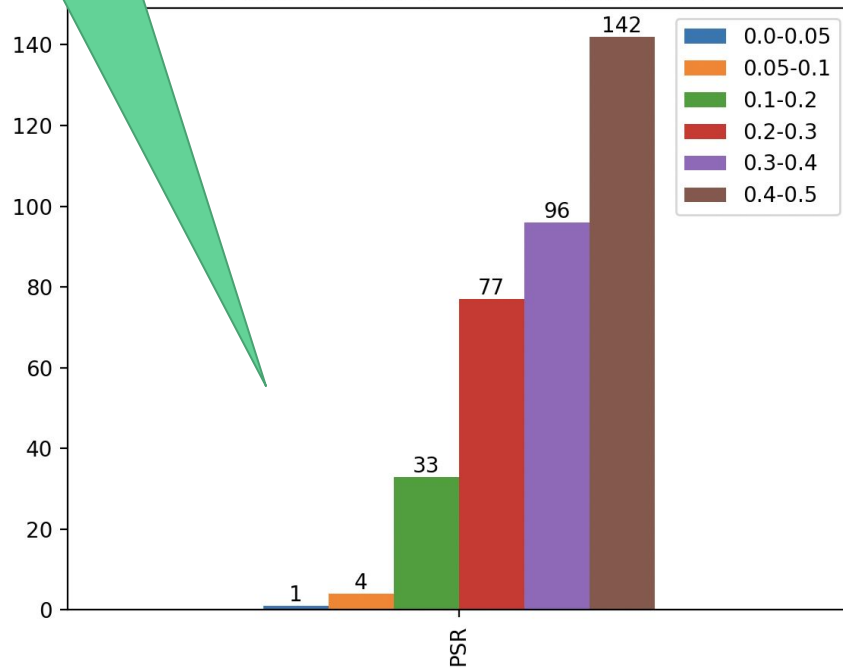
Now PSRs have
been strongly
identified

2077 unassociated sources AGN classification



Classification-RF-20-500-15

353 unassociated sources PSR classification



Classification-RF-20-500-15

Code Changes

- Create **4 more PSR rows** for each PSR row

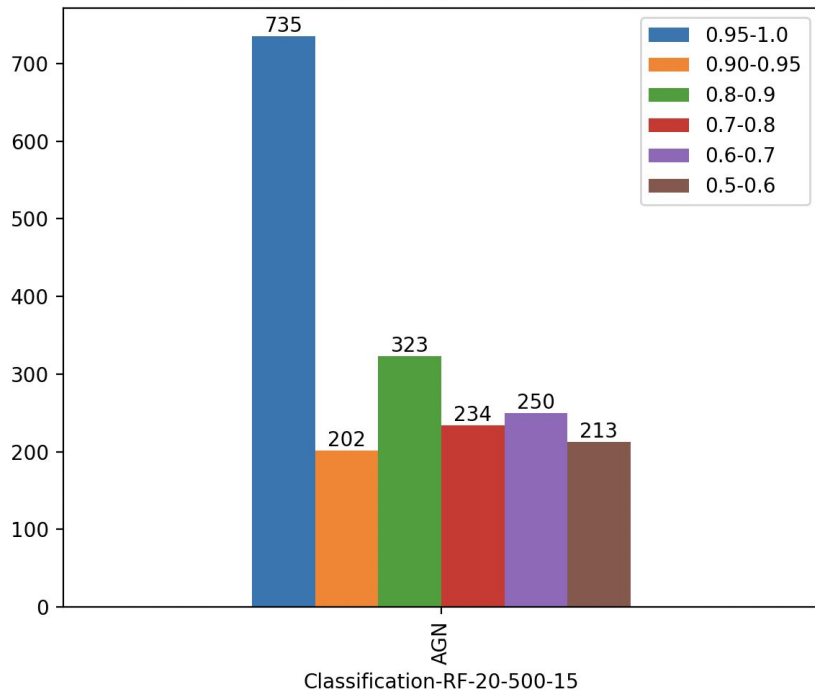
```
correct = 1649
test_size = 1683
correct_agm = 1171
total_agm = 1190 ←
correct_psr = 478
total_psr = 493 ←
sensitivity = 97.97979797979798
agm_sensitivity = 98.40336134453781
psr_sensitivity = 96.95740365111561
```

Code Changes

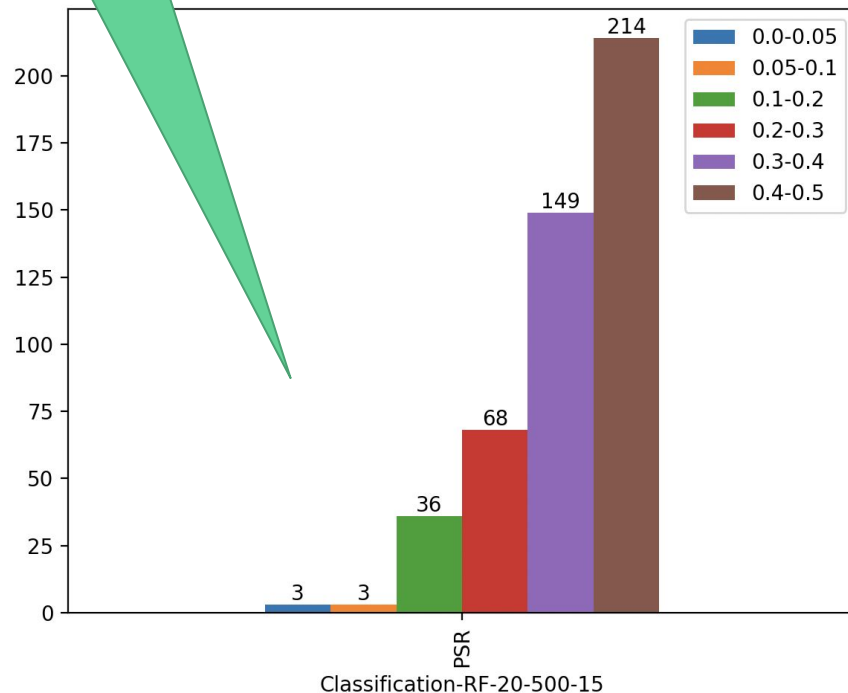
- #trees: 20, # samples: 500, depth: 15

Now PSRs have
been strongly
identified

1957 unassociated sources AGN classification



473 unassociated sources PSR classification



Code Changes

- Create **9 more PSR rows** for each PSR row

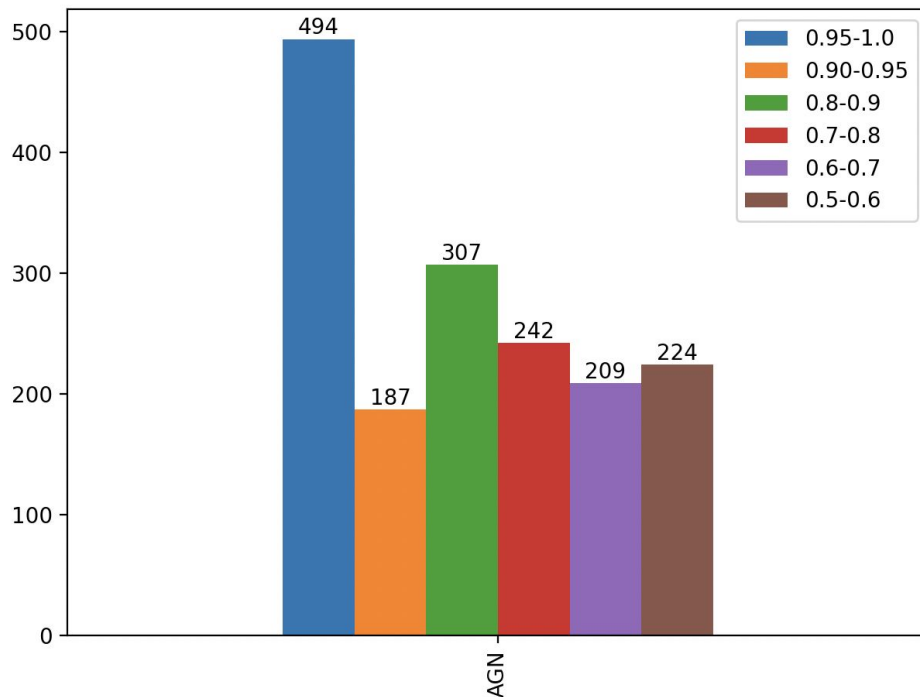
```
correct = 2112
test_size = 2163
correct_agm = 1169
total_agm = 1196 ←
correct_psr = 943
total_psr = 967 ←
sensitivity = 97.64216366158114
agm_sensitivity = 97.74247491638796
psr_sensitivity = 97.51809720785936
```

Code Changes

- #trees: 20, # samples: 500, depth: 15

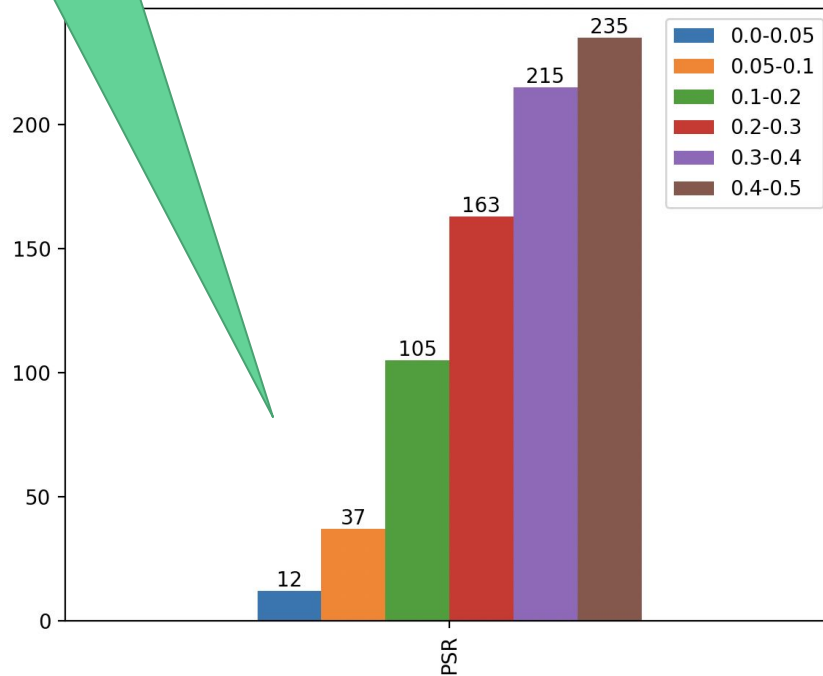
Now PSRs have
been strongly
identified

1663 unassociated sources AGN classification



Classification-RF-20-500-15

767 unassociated sources PSR classification



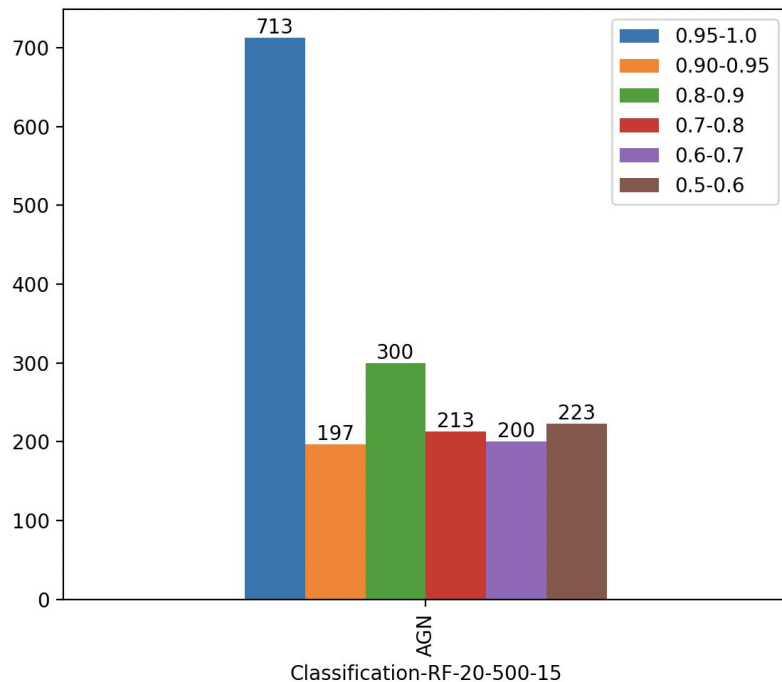
Classification-RF-20-500-15

Code Changes

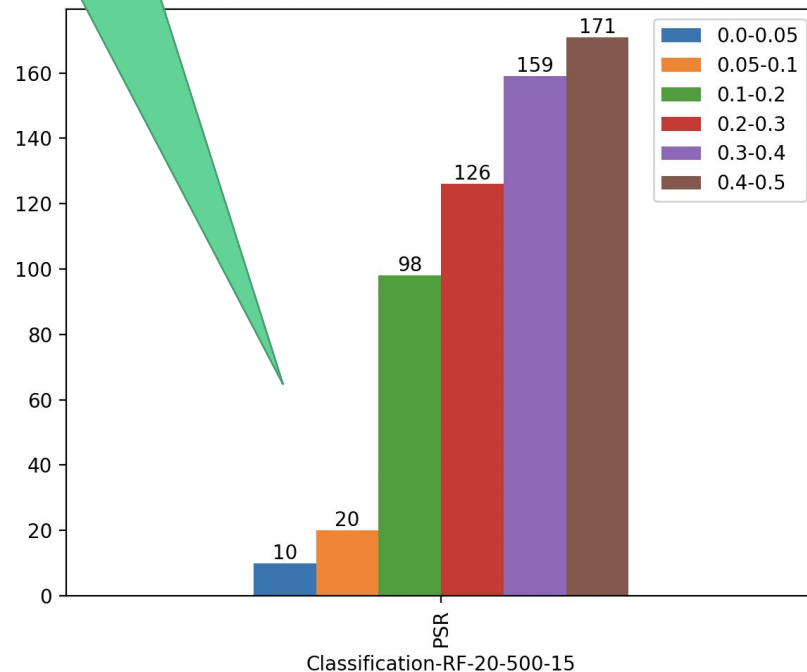
- #trees: 20, # samples: 500, depth: 15

Now PSRs have
been strongly
identified

1846 unassociated sources AGN classification



584 unassociated sources PSR classification



Code Changes

- Create **11 more PSR rows** for each PSR row

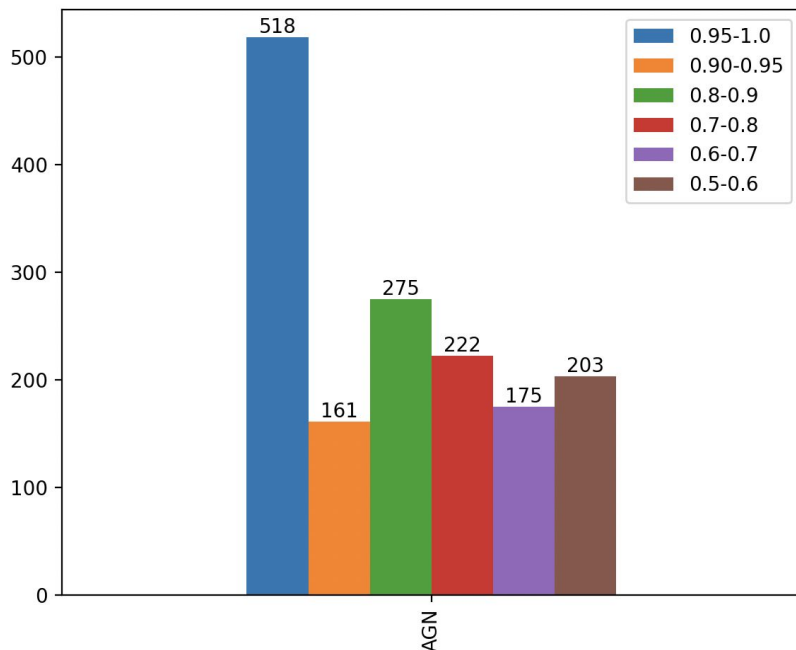
```
correct = 2298
test_size = 2355
correct_agm = 1180
total_agm = 1223 ←
correct_psr = 1118
total_psr = 1132 ←
sensitivity = 97.57961783439491
agm_sensitivity = 96.4840556009812
psr_sensitivity = 98.76325088339223
```

Code Changes

- #trees: 20, # samples: 500, depth: 15

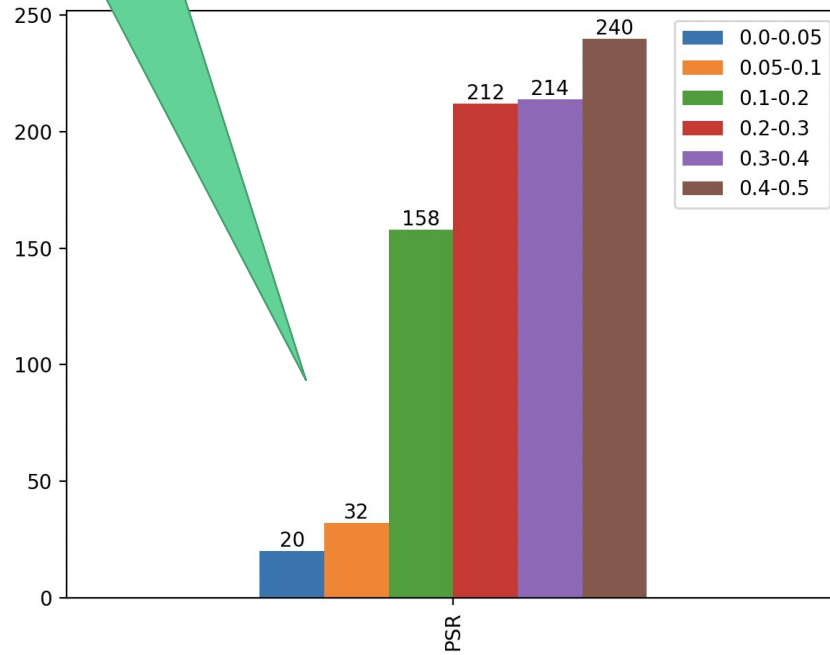
Now PSRs have
been strongly
identified

1554 unassociated sources AGN classification



Classification-RF-20-500-15

876 unassociated sources PSR classification



Classification-RF-20-500-15

Results

- As more rows were duplicated for Pulsars bringing them closer to the AGN rows, the PSRs were more strongly identified for unassociated sources.
- Discuss measure of closeness
 - Sum of predictions of individual trees / number of trees

Update: 04/12/2024

Sort Fields

Catalog	Sort Fields
BSL	sqrt_TS sqrt_TS23 sqrt_TS35
1FGL	Signif_avg Signif_peak sqrt_TS30_100 sqrt_TS100_300 sqrt_TS300_1000 sqrt_TS1000_3000 sqrt_TS3000_10000 sqrt_TS10000_100000

Catalog	Sort Fields
2FGL	Signif_avg Signif_peak Signif_curve sqrt_TS30_100 sqrt_TS100_300 sqrt_TS300_1000 sqrt_TS1000_3000 sqrt_TS3000_10000 sqrt_TS10000_100000
3FGL	Signif_avg Signif_peak Signif_curve sqrt_TS30_100 sqrt_TS100_300 sqrt_TS300_1000 sqrt_TS1000_3000 sqrt_TS3000_10000 sqrt_TS10000_100000

Catalog	Sort Fields
4FGL-DR1	Signif_avg Signif_peak Signif2_peak
4FGL-DR1	Signif_avg Signif_peak
4FGL-DR2	Signif_avg Signif_peak
4FGL-DR3	Signif_avg Signif_peak
4FGL-DR4	Signif_avg Signif_peak

Process

- Opened the FITS files in TopCat
- Found unassociated sources (class1 is null) in catalogs like BSL, 1/2/3FGL, 4FGL DR1/2/3/4
- If exists in 3FGL/4FGL, looked prior associations in columns shown below

Table Browser for 9: gll_psc_v16_3FGL.fit

	Flux_History	Unc_Flux_History	OFGL_Name	1FGL_Name	2FGL_Name
2740	(2.6007621E-8, 2.6720793E-8, 2.856525E-8,...	((-2.0991994E-8, 2.2942334E-8), (-2.364542...			
2742	(2.2480405E-8, 1.0676823E-8, 1.2815452E-...	((-5.183114E-9, 5.8511316E-9), (-3.7521533...		1FGL J2112.5-3044	2FGL J2112.5-3042
2744	(1.5292892E-8, 3.6051482E-8, 2.5758847E-...	((-1.2139872E-8, 1.3199144E-8), (-1.382965...			

Table Browser for 27: gll_psc_v27_4FGL_DR2.fit

	Unc_Flux_History	Sqrt_TS_History	ASSOC_FGL
1806	((-2.2582012E-9, 2.3462454E-9), (-2.349689...	(13.781338, 22.222631, 22.628527, 24.8065...	3FGL J0850.0+4855
251	((-1.2570351E-9, 1.3408505E-9), (-1.214068...	(14.391105, 17.910778, 18.003508, 18.3883...	3FGL J0106.5+4855
2302	((-3.7645798E-9, 3.7645798E-9), (-2.491857...	(26.477457, 13.359099, 16.868233, 15.9703...	3FGL J1103.9-5357
5258	((-8.396821E-10, 8.396821E-10), (-8.894406...	(16.858236, 19.474714, 16.138826, 15.2607...	3FGL J2112.5-3044
4825	((-1.4758287E-9, 1.5485185E-9), (-1.319461...	(15.454493, 13.886233, 15.747384, 18.8581...	3FGL J1918.2-4110

Prior Association

Table Browser for 41: gll_psc_v33_4FGL_DR4.fit

	Sqrt_TS_History	ASSOC_4FGL	ASSOC_FGL
3499	(12.577743, 85.35981, 17.117466, 5.910452...	4FGL J1312.6+4828	3FGL J1312.7+4828
2942	(17.42409, 16.94912, 17.928568, 18.0316, 1...	4FGL J1120.0-2204	3FGL J1119.9-2204
4384	(27.16054, 28.981466, 14.323773, 2.578804, ...	4FGL J1603.8-4903	3FGL J1603.9-4903
6518	(17.27783, 19.530153, 16.588514, 15.34258...	4FGL J2112.5-3043	3FGL J2112.5-3044

Examples

- Vela, Crab, Greninga
- Unassociated source #1 → PSR in 3FGL
- Unassociated source #2 → PSR in 3FGL
- Unassociated source #3: spp → HMB
- Unassociated source #4 → Blazar
- Unassociated source #5 (Eta Carinae) → Binary
- Unassociated source #6 → PSR in 4FGL-DR1
- Unassociated source #7 thru 4FGL-DR4
- Unassociated source #8

Vela, Crab, Greminga

Table Browser for 1: gll_psc3month_BSL_v2.fit

	Source_Name	RA	DEC	GLON	GLAT	Sqrt_TS	CLASS1	ASSOC1	ASSOC2	Conf_95_Radius	Flux100_1000	Unc_Flux100_1000	Flux1000_1
179	0FGL J2021.5+4026	305.398	40.439	78.23	2.07	69.6	PSR	LAT PSR J2021+4026		0.053	1.23600E-6	5.10000E-8	1.0600
24	0FGL J0238.6+1636	39.663	16.613	156.775	-39.112	85.7	bzb	CGRaBS J0238+1636	AO 0235+164	0.052	6.07000E-7	2.10000E-8	6.8100
136	0FGL J1709.7+4428	257.427	-44.475	343.106	-2.679	85.8	PSR	PSR J1709-4429		0.048	1.15300E-6	5.00000E-8	1.5820
114	0FGL J1504.4+1030	226.115	10.505	11.409	54.577	88.2	bzq	CGRaBS J1504+1029	PKS 1502+106	0.054	6.34000E-7	2.10000E-8	5.8600
158	0FGL J1836.2+5924	279.056	59.406	88.855	24.996	98.	PSR	LAT PSR J1836+5925		0.053	4.44000E-7	1.70000E-8	8.3600
41	0FGL J0534.6+2201	83.653	22.022	184.562	-5.764	139.2	PSR	PSR J0534+2200	Crab	0.046	2.04000E-6	9.20000E-8	1.5400
200	0FGL J2254.0+1609	343.502	16.151	86.125	-38.187	149.1	bzq	CGRaBS J2253+1608	3C 454.3	0.051	2.11700E-6	4.30000E-8	9.8300
49	0FGL J0634.0+1745	98.503	17.76	195.155	4.285	283.2	PSR	PSR J0633+1746	Geminga	0.043	2.86200E-6	3.80000E-8	6.1610
63	0FGL J0835.4-4510	128.865	-45.17	263.56	-2.767	374.2	PSR	PSR J0835-4510	Vela	0.042	8.03100E-6	5.70000E-8	1.1200

Table Browser for 2: gll_psc_v03_1FGL.fit

	Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Flux_Density	Spectral_Index	Conf_68_SemiMajor	Conf
862	1FGL J1512.8-0906	228.207	-9.1025	351.284	40.1392	BZQ		PKS 1510-08		101.448	61.8159	411.604	5.93235E-10	2.40756	0.009578	
257	1FGL J0457.0-2325	74.2595	-23.4228	223.719	-34.9024	bzq		PKS 0454-234		115.436	74.3592	517.004	1.64470E-10	2.20649	0.010875	
10	1FGL J0007.0+7303	1.7565	73.0523	119.66	10.4634	PSR		LAT PSR J0007+7...		119.65	60.5047	868.207	7.02733E-11	1.96724		
712	1FGL J1256.2-0547	194.053	-5.7911	305.117	57.0603	BZQ		3C 279		121.1	80.8158	489.11	2.28447E-10	2.32061	0.01013	
1395	1FGL J2253.9+1608	343.495	16.1495	86.1169	-38.1851	BZQ		3C 454.3		154.136	109.331	382.865	7.31831E-10	2.46619	0.009924	
991	1FGL J1709.7+4429	257.428	-44.4856	343.098	-2.68594	PSR		PSR J1709-4429		168.269	95.0664	744.267	3.18954E-10	2.09819		
848	1FGL J1504.4+1029	226.102	10.4927	11.3786	54.581	BZQ		PKS 1502+106		188.71	125.043	488.027	1.98856E-10	2.22063	0.007615	
1129	1FGL J1836.2+5925	279.058	59.4251	88.8758	24.999	PSR		LAT PSR J1836+5...		191.177	120.676	680.539	1.93301E-10	1.98779		
298	1FGL J0534.5+2200	83.6332	22.0145	184.558	-5.78403	PSR	PWN	PSR J0534+2200	Crab	277.644	174.338	494.555	1.00005E-9	2.24752		
357	1FGL J0633.9+1746	98.4756	17.7703	195.134	4.26605	PSR		PSR J0633+1746	Geminga	49.801	352.127	663.604	1.25520E-9	1.92058		
460	1FGL J0835.3-4510	128.836	-45.1763	263.552	-2.78715	PSR	PWN	PSR J0835-4510	Vela	640.132	398.091	582.982	3.66952E-9	2.0244		

Table Browser for 5: gll_psc_v09_2FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Pivot_Energy	Sqrt_TS300_1000	Spectral_Index	ASSOC_TEV
776	2FGL J1104.4+3812	166.12	38.2134	179.818	65.0348	bzb		Mkn 421		131.491	1162.36	58.993	1.77058	Markarian 421
422	2FGL J0617.2+2234e	94.31	22.58	189.048	3.0337	SNR		SNR G189.1-03...	IC 443	132.926	640.492	75.9253	1.7912	IC443
762	2FGL J1057.9-5226	164.496	-52.4485	285.985	6.64999	PSR		PSR J1057-5226		133.667	652.629	81.5958	1.01356	
181	2FGL J0240.5+6113	40.1298	61.2249	135.677	1.08177	HMB		LS I+61 303		133.955	505.083	98.3982	2.18321	LSI +61 303
886	2FGL J1229.1+0202	187.277	2.04167	289.954	64.3492	BZQ		3C 273		140.633	231.151	90.3276	2.4524	
917	2FGL J1256.1-0547	194.042	-5.79369	305.097	57.0581	BZQ		3C 279		147.848	294.806	95.6947	2.22124	3C279
381	2FGL J0538.8-4405	84.706	-44.0838	250.08	-31.0919	BZB		PKS 0537-441		164.992	371.966	98.0431	2.01159	
878	2FGL J1224.9+2122	186.226	21.38	255.07	81.6598	BZQ		4C +21.35		182.403	317.119	115.726	2.12155	4C+21.35
7	2FGL J0007.0+7303	1.77352	73.0545	119.665	10.4647	PSR		LAT PSR J0007+...		189.52	727.398	95.97	1.44987	
1107	2FGL J1512.8-0906	228.207	-9.10336	351.283	40.1383	BZQ		PKS 1510-08		202.414	259.642	128.682	2.28839	PKS 1510-089
1276	2FGL J1709.7+4429	257.433	-44.4852	343.101	-2.68855	PSR		PSR J1709-4429		256.76	607.611	147.288	1.62488	HESS J1708-443
1470	2FGL J1836.2+5926	279.065	59.4339	88.8862	24.9969	PSR		LAT PSR J1836+...		303.698	550.467	190.746	1.23136	
1800	2FGL J2253.9+1609	343.497	16.1526	86.1217	-38.1839	BZQ		3C 454.3		339.435	297.596	255.301	2.22639	
373	2FGL J0534.5+2201	83.6315	22.0196	184.553	-5.78261	PSR		PSR J0534+2200	Crab	376.634	485.021	240.037	2.14454	Crab
439	2FGL J0633.9+1746	98.4785	17.7739	195.132	4.27019	PSR		PSR J0633+1746	Geminga	791.02	510.79	486.28	1.22089	Geminga
570	2FGL J0835.3-4510	128.839	-45.1792	263.556	-2.78707	PSR		PSR J0835-4510	Vela	1027.8	433.286	654.836	1.52334	Vela X

Look here!

Vela, Crab, Greminga

Table Browser for 9: gl_l_psc_v16_3FGL.fit

	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Spectral_Index	CLASS1	ASSOC1	ASSOC2	ASSOC_TEV	Extended_Source_Name
2013	78.2317	2.08315	655.304	3.80554E-10	237.049	134.378	58.7847	1.56906	PSR	LAI PSR J2021+4026			
1483	255.059	81.6615	306.157	5.16984E-10	219.674	137.609	4.99403	2.18542	FSRQ	4C +21.35		4C +21.35	
20	119.662	10.4606	913.709	7.55546E-11	288.746	145.293	22.6506	1.20806	PSR	LAT PSR J0007+7303		CTA 1	
622	184.552	-5.77575	635.591	5.54130E-10	30.6699	177.273	15.8003	2.04501	PSR	PSR J0534+2200	Crab	Crab Pulsar	
2114	343.098	-2.68919	849.288	3.10332E-10	360.818	188.988	28.4751	1.03076	PSR	PSR J1709-4429		HESS J1708-443	
2391	88.8821	24.9983	747.86	2.01251E-10	438.122	260.318	71.7768	0.886055	PSR	LAT PSR J1836+5925			
2913	86.1214	-38.1918	412.747	1.33123E-9	480.741	308.06	18.2082	1.63463	FSRQ	3C 454.3			
961	263.55463	-2.78727	832.557	2.33308E-9	1048.96	528.538	53.9787	1.00295	PSR	PSR J0835-4510	Vela	Vela Pulsar	
726	195.133	4.27009	927.902	9.37508E-10	1012.14	568.663	84.973	0.916183	PSR	PSR J0633+1746	Geminga	Geminga	
620	184.553	-5.78121	1000.	1.83367E-11	—∞		0.	1.64	PWN	Crab		Crab	
956	263.333	-3.10593	1000.	2.59146E-11	—∞		0.	2.41006	PWN	Vela X		Vela X	Vela X
1848	320.27	-1.27147	1000.	1.78847E-12	—∞		0.	1.57134	PWN	MSH 15-52		MSH 15-52	MSH 15-52
2613	78.2413	2.19658	1000.	4.00366E-13	—∞		0.	1.53	SNR	Gamma Cygni		VER J2019+407	gamma Cygni
2639	79.6012	1.39613	1000.	6.84591E-11	—∞		0.	2.175	SFR	Cygnus Cocoon		MGRO J2031+41	Cygnus Cocoon

Table Browser for 19: gl_l_psc_v22_4FGL_DR1.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_Source_Name	Pivot_Energy	Signif_Peak	LP_SigCuv
2792	4FGL J1427.9-4206	216.987	-42.106	321.449	17.2695	258.031	FSRQ		PKS 1424-41				677.451	216.453	17.1327
1309	4FGL J0721.9+7120	110.488	71.3485	143.985	28.0224	260.283	BLL		S5 0716+71		S5 0716+714		717.339	141.324	9.47317
2048	4FGL J1104.4+3812	166.119	38.207	179.833	65.0357	294.278	BLL		Mkn 421		Markarian 421		1286.47	146.43	3.21282
25	4FGL J0007.0+7303	1.768	73.0512	119.663	10.4617	329.094	PSR		PSR J0007+7303		CTA 1		1452.3		62.4081
3441	4FGL J1709.7-4429	257.43	-44.4854	343.099	-2.68665	374.423	PSR		PSR J1709-4429				1411.45		74.0677
4881	4FGL J2253.9+1609	343.496	16.1506	86.1185	-38.1849	443.845	FSRQ		3C 454.3				517.462	355.842	38.1457
4002	4FGL J1836.2+5925	279.056	59.4279	88.8783	25.0006	466.08	PSR		PSR J1836+5925				1361.54		111.226
1560	4FGL J0835.3-4510	128.837	-45.1781	263.554	-2.78773	496.139	PSR		PSR J0835-4510	Vela X field	Vela Pulsar		1746.46		196.328
1178	4FGL J0633.9+1746	98.483	17.7723	195.135	4.27298	792.691	PSR		PSR J0633+1746		Geminga		1585.48		207.659
994	4FGL J0534.5+2201i	83.633	22.02	184.553	-5.78148		PWN		Crab Nebula		Crab	Crab IC	10000.		0.

Table Browser for 27: gl_l_psc_v27_4FGL_DR2.fit

	Source_Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_Source_Name
698	4FGL J0319.8+4130	1	49.9575	41.5121	150.58	-13.2581	235.649	952.578	RDG		NGC 1275		NGC 1275	
569	4FGL J0240.5+6113	1	40.1426	61.2287	135.68	1.08765	236.903	1185.92	HMB		LSI +61 303		LSI +61 303	
3321	4FGL J1512.8-0906	1	228.215	-9.1064	351.288	40.1312	238.707	723.638	FSRQ		PKS 1510-089		PKS 1510-0...	
904	4FGL J0428.6-3756	1	67.173	-37.9403	240.71	-43.6147	249.909	808.988	bll		PKS 0426-380			
2273	4FGL J1057.9-5227	1	164.498	-52.4528	285.987	6.64635	269.544	1030.07	PSR		PSR J1057-52...			
2756	4FGL J1256.1-0547	1	194.842	-5.7887	305.095	57.0632	291.633	466.046	FSRQ		3C 279		3C 279	
3144	4FGL J1427.9-4206	1	216.987	-42.106	321.449	17.2695	293.386	645.788	FSRQ		PKS 1424-41			
1463	4FGL J0721.9+7120	1	110.488	71.3485	143.985	28.0224	298.165	734.586	BLL		S5 0716+71		S5 0716+714	
2306	4FGL J1104.4+3812	1	166.119	38.207	179.833	65.0357	325.523	1263.75	BLL		Mkn 421		Markarian 4...	
5504	4FGL J2232.6+1143	1	338.152	11.7306	77.4385	-38.5832	349.938	531.25	FSRQ		CTA 102			
30	4FGL J0007.0+7303	1	1.768	73.0512	119.663	10.4617	351.673	1472.99	PSR		PSR J0007+7...		CTA 1	
3893	4FGL J1709.7-4429	1	257.43	-44.4854	343.099	-2.68665	408.691	1462.71	PSR		PSR J1709-44...			
5577	4FGL J2253.9+1609	1	343.496	16.1506	86.1185	-38.1849	465.154	232.352	FSRQ		3C 454.3			
4559	4FGL J1836.2+5925	1	279.056	59.4279	88.8783	25.0006	523.634	1024.48	PSR		PSR J1836+5...			
1750	4FGL J0835.3-4510	1	128.837	-45.1781	263.554	-2.78773	548.109	1760.38	PSR		PSR J0835-45...	Vela X field	Vela Pulsar	
1323	4FGL J0633.9+1746	1	98.483	17.7723	195.135	4.27298	840.475	1328.4	PSR		PSR J0633+1...		Geminga	
1118	4FGL J0534.5+2201i	1	83.633	22.02	184.553	-5.78148		10000.	PWN		Crab Nebula		Crab	Crab IC

Vela, Crab, Gremlinga

Table Browser for 34: gll_psc_v31_4FGL_DR3.fit

Source Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Signif_Avg	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_Source_Name
653	4FGLJ0240.5+6113	1	40.1426	61.2287	135.68	1.08765	1260.67	249.812	HMB	LSI +61 303		LSI +61 3...	
1036	4FGLJ0428.6-3756	1	67.173	-37.9403	240.71	-43.6147	850.508	256.684	bll	PKS 0426-380			
2616	4FGLJ1057.9-5227	1	164.498	-52.4528	285.987	6.64635	1408.27	277.55	PSR	PSR J1057-5226			
3598	4FGLJ1427.9-4206	1	216.987	-42.106	321.449	17.2695	1367.49	297.359	FSRQ	PKS 1424-41			
3153	4FGLJ1256.1-0547	1	194.042	-5.7887	305.095	57.0632	954.15	302.454	FSRQ	3C 279		3C 279	
1694	4FGLJ0721.9+7120	1	110.488	71.3405	143.985	28.0224	1510.63	321.002	BLL	SS 0716+71		SS 0716+...	
2650	4FGLJ1104.4+3812	1	166.119	38.207	179.833	65.0357	1258.26	343.769	BLL	Mkn 421		Markarian ...	
6329	4FGLJ2232.6+1143	1	338.152	11.7306	77.4385	-38.5832	1082.42	345.948	FSRQ	CTA 102			
37	4FGLJ0007.0+7303	1	1.768	73.0512	119.663	10.4617	2213.57	370.992	PSR	PSR J0007+7303		CTA 1	
4471	4FGLJ1709.7-4429	1	257.43	-44.4854	343.099	-2.68665	2078.25	413.705	PSR	PSR J1709-4429		HESS PSR ...	
6412	4FGLJ2253.9+1609	1	343.496	16.1506	86.1185	-38.1849	891.907	454.905	FSRQ	3C 454.3			
5226	4FGLJ1836.2+5925	1	279.056	59.4279	88.8783	25.0006	1428.31	555.913	PSR	PSR J1836+5925			
2014	4FGLJ0835.3-4510	1	128.837	-45.1781	263.554	-2.78773	1914.57	665.559	PSR	PSR J0835-4510	Vela X fi...	Vela Pulsar	
1534	4FGLJ0633.9+1746	1	98.483	17.7723	195.135	4.27298	1670.18	884.643	PSR	PSR J0633+1746		Geminga P...	
1294	4FGLJ0534.5+2201i	1	83.633	22.02	184.553	-5.78148	10000.		PWN	Crab Nebula		Crab	Crab IC

Table Browser for 41: gll_psc_v33_4FGL_DR4.fit

Source Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Extended_Source_Name	Signif_Avg	Signif_Peak	Pivot_Energy	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV
1216	4FGLJ0457.0-2324	1	74.2608	-23.4149	223.71	-34.8991	247.863	106.397	692.189	FSRQ		PKS 0454-234		
6706	4FGLJ2158.8-3013	1	329.714	-30.2251	17.7308	-52.2434	248.351	100.358	1168.97	bll		PKS 2155-304		PKS 2155-304
1411	4FGLJ0538.8-4405	1	84.7089	-44.0862	250.083	-31.0904	249.871	153.837	653.401	BLL		PKS 0537-441		
850	4FGLJ0319.8+4130	1	49.9575	41.5121	150.58	-13.2581	251.542	138.776	974.9	RDG		NGC 1275		NGC 1275
1110	4FGLJ0428.6-3756	1	67.173	-37.9403	240.71	-43.6147	258.51	112.784	873.599	bll		PKS 0426-380		
697	4FGLJ0240.5+6113	1	40.1426	61.2287	135.68	1.08765	261.254	109.965	1517.29	HMB		LSI +61 303		LSI +61 303
2828	4FGLJ1057.9-5227	1	164.498	-52.4528	285.987	6.64635	297.348	→	1445.88	PSR		PSR J1057-52...		
3891	4FGLJ1427.9-4206	1	216.987	-42.106	321.449	17.2695	305.414	216.333	1550.06	FSRQ		PKS 1424-41		
1840	4FGLJ0721.9+7120	1	110.488	71.3405	143.985	28.0224	318.785	130.895	1606.17	BLL		SS 0716+71		SS 0716+714
3412	4FGLJ1256.1-0547	1	194.042	-5.7887	305.095	57.0632	319.282	303.431	908.925	FSRQ		3C 279		3C 279
6837	4FGLJ2232.6+1143	1	338.152	11.7306	77.4385	-38.5832	340.599	400.519	1098.69	FSRQ		CTA 102		
6722	4FGLJ2202.7+4216	1	330.695	42.2821	92.6007	-10.4441	343.185	280.56	992.378	BLL		BL Lac		BL Lacertae
2866	4FGLJ1104.4+3812	1	166.119	38.207	179.833	65.0357	363.7	147.932	1328.17	BLL		Mkn 421		Markarian 421
40	4FGLJ0007.0+7303	1	1.768	73.0512	119.663	10.4617	401.705	→	2247.63	PSR		PSR J0007+73...		CTA 1
4830	4FGLJ1709.7-4429	1	257.43	-44.4854	343.099	-2.68665	431.37	→	2136.25	PSR		PSR J1709-44...		HESS PSR B17...
6924	4FGLJ2253.9+1609	1	343.496	16.1506	86.1185	-38.1849	448.024	365.922	908.003	FSRQ		3C 454.3		
5643	4FGLJ1836.2+5925	1	279.056	59.4279	88.8783	25.0006	587.57	→	1496.23	PSR		PSR J1836+59...		
2175	4FGLJ0835.3-4510	1	128.837	-45.1781	263.554	-2.78773	689.218	→	1976.37	PSR		PSR J0835-45...	Vela X field	Vela Pulsar
1660	4FGLJ0633.9+1746	1	98.483	17.7723	195.135	4.27298	897.975	→	1705.33	PSR		PSR J0633+17...		Geminga Pulsar
1390	4FGLJ0534.5+2201i	1	83.633	22.02	184.553	-5.78148		→	10000.	PWN		Crab Nebula		Crab

Unassociated Source #1 → PSR in 3FGL

Table Browser for 2: gl_lpsc_v03_1FGL.fit

Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
1523 1FGL J1214.7-3358	321.183	-33.9791	10.9255	-45.4378	PSR		PSR J1214-3358		26.0896	11.915	991.297
1248 1FGL J2015.7+3708	303.929	37.1399	74.8571	1.16046					26.1037	13.8629	1050.77
306 1FGL J0538.9-6914	84.7396	-69.2398	279.625	-31.6338	gal		LMC field		26.1676	14.6717	791.00
880 1FGL J1536.5-4949	234.146	-49.818	328.223	4.76784					26.3268	9.86395	951.00
452 1FGL J0825.8-2230	126.468	-22.5104	243.973	8.89926	bzb		PKS 0823-223		26.5001	13.3148	891.00
1064 1FGL J1751.5+0937	267.89	9.6275	34.8997	17.6326	bzb		4C +09.57		26.5094	15.0708	771.00
910 1FGL J1603.8-4903	240.972	-49.0662	332.158	2.57019	agu		PMN J1603-4904		26.8036	9.10865	1453.07
1354 1FGL J2203.5+1726	330.882	17.4387	75.6957	-29.6413	bzq		PKS 2201+171		26.8739	15.9251	574.137
1268 1FGL J2032.2+4127	308.054	41.4567	80.224	1.02802	PSR		LAT PSR J2032+4...		26.9134	7.49773	1696.68
1261 1FGL J2027.6+3335	306.915	33.592	73.3268	-2.86108					26.9593	18.1954	702.555
669 1FGL J1217.7+3007	184.446	30.1206	188.893	82.0351	bzb		B2 1215+30		27.3569	8.64911	1325.08
53 1FGL J0050.6-0928	12.6646	-9.4832	122.299	-72.3539	bzb		B15 0048-09		27.4003	16.1701	764.423
74 1FGL J0112.0+2247	18.0179	22.7945	129.129	-39.8288	bzb		CGRaBS J0112+2...		27.5516	14.694	717.733
740 1FGL J1311.7-3429	197.947	-34.4944	307.689	28.1893					27.7337	14.9313	881.639
165 1FGL J0303.5-2406	45.889	-24.1124	214.619	-60.15	bzb		PKS 0301-243		27.8553	14.2392	1086.94

Unassociated
Class1 == null

Table Browser for 5: gl_lpsc_v09_2FGL.fit

Source_Name	RA	DEC	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
1724 2FGL J2139.3-4236	324.841	-42.608	358.291	-48.3171	bzb			MH 2136-428		39.1281	21.3043	805.105	2.09528
86 2FGL J0112.8+3208	18.2069	32.1387	128.188	-30.5146	bzq			4C 31.03		39.4861	23.6512	561.489	2.31038
1130 2FGL J1536.4-4949	234.124	-49.8294	328.205	4.76692						40.3338	16.5487	865.906	1.94332
95 2FGL J0118.8-2142	19.7243	-21.7062	173.461	-81.7284	bzq			PKS 0116-219		40.3492	26.6873	439.865	2.12153
1597 2FGL J2000.0+6509	300.02	65.1572	98.016	17.6657	bzb			1F5 1959+650		40.4049	17.3951	1402.26	1.93676
873 2FGL J1221.4+2814	185.374	28.2391	201.691	83.2804	bzb			W Comae		40.5612	19.8744	1106.2	2.01857
150 2FGL J0217.9+0143	34.4833	1.73026	162.199	-54.4104	bzq			PKS 0215+015		42.2751	23.2729	724.696	2.15159
1387 2FGL J1800.5+7829	270.147	78.4834	110.057	29.0718	bzb			SS 1803+784		42.5332	22.2863	665.466	2.23857
764 2FGL J1058.6+5628	164.666	56.4861	149.575	54.4212	bzb			TXS 1055+567		42.8403	19.4494	1236.75	1.92735
947 2FGL J1311.7-3429	197.947	-34.4944	307.689	28.1893						43.0656	26.4636	706.553	2.01692
1355 2FGL J1745.6-2858	266.423	-28.9786	359.973	-0.036044	spp					43.1212	23.2346	1618.9	2.33888

Associated to
PSR in 3FGL

Table Browser for 9: gl_lpsc_v16_3FGL.fit

Table browser for 3fgl_psc_v10_3fglcm													
	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Spectral_Index	CLASS1	ASSOC1
1372	3FGL J1136.1+7523	174.044	75.3885	129.104	40.8251	1514.61	9.41672E-14	4.3202	0.850643	0.726054	2.30112	PSR	PSR J1137+7523
1385	3FGL J1142.9+0120	175.742	1.3493	267.576	59.4367	1830.62	3.34363E-13	13.7362	3.78123	3.98751	1.18626	PSR	PSR J1142+0120
1496	3FGL J1231.2-1411	187.801	-14.1981	295.539	48.383	917.884	2.16312E-11	105.655	51.8699	23.9051	1.1804	PSR	PSR J1231-1411
1565	3FGL J1301.6+0832	195.418	8.5398	310.829	71.2544	765.506	2.14075E-12	13.6457	8.33066	2.59564	2.37894	PSR	PSR J1301+0832
1566	3FGL J1302.3-3259	195.595	-32.9879	305.577	29.8276	1440.63	1.12547E-12	18.7593	6.91077	6.14404	1.12307	PSR	PSR J1302-3259
1595	3FGL J1311.8-3430	197.959	-34.5021	307.699	28.1807	658.035	2.41142E-11	62.3248	35.3774	9.05289	1.90862	PSR	LAT PSR J1311.8-3430
1600	3FGL J1312.7+0051	198.195	0.861	314.85	63.2377	837.744	4.26514E-12	20.6608	7.59733	7.17686	1.52349	PSR	PSR J1312+0051

Prior
associations

1FGL_Name	2FGL_Name	1FHL_Name	ASSOC_GAM1	ASSOC_GAM2	ASSOC_GAM3
1FGL J1142.7+0127	2FGL J1142.9+0121				
1FGL J1231.1-1410	2FGL J1231.2-1411	1FHL J1231.2-1414			EGR J1
1FGL J1301.8+0837	2FGL J1301.5+0835				
1FGL J1302.3-3255	2FGL J1302.4-3257				
1FGL J1311.7-3429	2FGL J1311.7-3429			3EG J1314-3431	
1FGL J1312.6+0048	2FGL J1312.7+0051				

Unassociated Source #2 → PSR in 3FGL

Table Browser for 2: gll_psc_v03_1FGL.fit

	Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
91	1FGL J0136.5+3905	24.1305	39.0856	132.416	-22.9545	bzb		B3 0133+388		25.6661	6.22098	2336.53
781	1FGL J1344.2-1723	206.071	-17.3925	320.482	43.671	bzq		CGRaBS J1344-1...		25.7139	13.8585	918.528
81	1FGL J0118.7-2137	19.6825	-21.6276	172.877	-81.7038	bzq		PKS 0116-219		25.7165	16.3633	533.797
501	1FGL J0909.0+0126	137.273	1.4439	228.85	30.9519	bzq		PKS 0906+01		25.7542	16.6642	385.549
217	1FGL J0403.9-3603	60.9774	-36.0636	237.713	-48.4798	bzq		PKS 0402-362		25.8157	17.0821	452.558
1323	1FGL J2124.7-3358	321.183	-33.9791	10.9255	-45.4378	PSR		PSR J2124-3358		26.0896	11.915	991.297
1248	1FGL J2015.7+3708	303.929	37.1399	74.8571	1.16046					26.1037	13.8629	1050.77
306	1FGL J0538.9-6914	84.7396	-69.2398	279.625	-31.6338	gal		LMC field		26.1676	14.6717	793.557
880	1FGL J1536.5-4949	234.146	-49.818	328.223	4.76784					26.3268	9.86395	952.782
452	1FGL J0825.8-2230	126.468	-22.5104	243.973	8.89926	bzb		PKS 0823-223		26.5001	13.3148	890.931

Unassociated
Class1 == null

Table Browser for 5: gll_psc_v09_2FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
239	2FGL J0334.2-4008	53.5596	-40.1484	244.777	-54.0715	bzb		PKS 0332-403		38.3389	23.6149	696.373	2.18878
566	2FGL J0831.9+0429	127.987	4.48609	220.719	24.3573	bzb		PKS 0829+046		38.3503	22.1135	527.964	2.04695
1778	2FGL J2236.4+2828	339.102	28.4755	90.1205	-25.6591	bzb		B2 2234+28A		38.7141	22.8297	460.793	2.05423
350	2FGL J0521.7+2113	80.449	21.2204	183.603	-8.69891	bzb		VER J0521+211		38.9956	13.0665	1874.43	1.92896
1724	2FGL J2139.3-4236	324.841	-42.608	358.291	-48.3171	bzb		MH 2136-428		39.1281	21.3043	805.105	2.09528
86	2FGL J0112.8+3208	18.2069	32.1387	128.188	-30.5146	bzq		4C 31.03		39.4861	23.6512	561.489	2.31038
1130	2FGL J1536.4-4949	234.124	-49.8294	328.205	4.76692					40.3338	16.5487	865.906	1.94332
95	2FGL J0118.8-2142	19.7243	-21.7062	173.461	-81.7284	bzq		PKS 0116-219		40.3492	26.6873	439.865	2.12153
1597	2FGL J2000.0+6509	300.02	65.1572	98.016	17.6657	bzb		1ES 1959+650		40.4049	17.3951	1402.26	1.93676

Unassociated Source #2 → PSR in 3FGL

Gets associated in 3FGL and appears in 4FGL also

Table Browser for 9: gll_psc_v16_3FGL.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Spectral_Index	CLASS1	ASSOC1
1349	3FGL J1125.6-6012	171.403	-60.2093	292.46	0.905667	2.24512E-13	5.64072	3.52324	2.56519	2.37394	psr	PSR J1125-6014
1427	3FGL J1203.9-6243	180.996	-62.7268	297.527	-0.354157	2.34999E-12	14.8116	9.35934	5.75902	2.5673	psr	PSR J1203-63
1434	3FGL J1207.2-5052	181.824	-50.87	295.854	11.3941	3.26278E-13	7.73423	3.00148	3.92012	2.14214	psr	PSR J1207-5050
1489	3FGL J1227.9-4854	186.976	-48.9012	298.954	13.7886	3.73728E-11	38.6305	22.9382	4.63259	2.23497	psr	PSR J1227-4853
1704	3FGL J1400.5-1437	210.145	-14.617	326.929	45.0087	1.13245E-12	10.1603	5.40907	3.0576	2.29915	psr	PSR J1400-1438
1844	3FGL J1513.4-2549	228.368	-25.8271	338.848	26.9653	1.23075E-12	11.0081	5.65408	4.87454	2.03904	psr	PSR J1513-25
1889	3FGL J1536.3-4949	234.099	-49.8225	328.196	4.7821	1.21817E-11	60.0751	23.3951	8.53276	1.86792	psr	PSR J1536-4948
1984	3FGL J1622.9-0312	245.741	-3.2039	10.758	30.7141	6.82104E-13	9.47165	5.1476	2.29817	2.23295	psr	PSR J1622-03
2002	3FGL J1628.0-3203	247.016	-32.0608	347.447	11.5151	2.71948E-12	11.8772	6.40167	5.45218	2.26867	psr	PSR J1628-3205

Table Browser for 19: gll_psc_v22_4FGL_DR1.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC3	Pivot_Energy	Signif_Peak	LP_SigC
4483	4FGL J2032.2+4127	308.063	41.459	80.2294	1.02414	86.9727	PSR	PSR J2032-4127			2235.47		22.
1474	4FGL J0809.8+5218	122.462	52.3143	166.248	32.9144	87.0045	BLL	1ES 0806+524			1297.	44.2755	3.
4695	4FGL J2143.5+1743	325.894	17.7306	72.1129	-26.0814	87.5121	fsrq	OX 169			394.828	36.8899	4.
2733	4FGL J1413.5-6205	213.382	-62.0903	312.378	-0.733427	87.7349	PSR	PSR J1413-6205			2051.79		22.
2021	4FGL J1058.4+0133	164.624	1.5641	251.514	52.7728	87.9966	BLL	4C +01.28			777.085	40.389	4.
329	4FGL J0144.6+2705	26.1502	27.0899	137.276	-34.2967	88.0051	bll	TXS 0141+268			767.909	40.1945	7.
2189	4FGL J1146.9+3958	176.74	39.9775	164.951	71.4682	91.0036	fsrq	S4 1144+40			540.053	57.648	5.
3038	4FGL J1536.4-4948	234.1	-49.8164	328.2	4.78665	91.2146	PSR	PSR J1536-49			1475.37		18.
3572	4FGL J1732.5-3131	263.145	-31.5174	356.315	1.00632	91.5022	PSR	PSR J1732-3131			2426.08		32.
403	4FGL J0210.7-5101	32.6946	-51.0218	276.106	-61.7737	91.9811	FSRQ	PKS 0208-512			458.644	52.4501	7.
4113	4FGL J1855.9+0121e	283.99	1.355	34.6534	-0.388004	92.3315	SNR	W 44		W 44	2448.21		28.

Associated to
PSR in 3FGL

Table Browser for 27: gll_psc_v27_4FGL_DR2.fit

Source_Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC3	ASSOC4	ASSOC5
5370	4FGL J2147.3-7536	1	326.827	-75.6024	315.801	-36.5338	97.4077	FSRQ		PKS 2142-75				
606	4FGL J0252.8-2219	1	43.2007	-22.3203	209.742	-62.0938	97.4665	fsrq		PKS 0250-225				
3080	4FGL J1413.5-6205	1	213.382	-62.0903	312.378	-0.733427	98.4685	PSR		PSR J1413-62...				
2467	4FGL J1146.9+3958	1	176.74	39.9775	164.951	71.4682	99.3588	fsrq		S4 1144+40				
761	4FGL J0339.5-0146	1	54.8771	-1.7769	188.	-42.4563	99.6133	fsrq		PKS 0336-01				
1825	4FGL J0854.8+2006	1	133.707	20.1159	206.805	35.8264	100.244	BLL		OJ 287				
5520	4FGL J2236.5-1433	1	339.144	-14.5557	47.8811	-56.2157	100.773	BLL		PKS 2233-148				
370	4FGL J0144.6+2705	1	26.1502	27.0899	137.276	-34.2967	100.824	bll		TXS 0141+268				
3421	4FGL J1536.4-4948	1	234.1	-49.8164	328.2	4.78665	101.719	PSR		PSR J1536-49				
1069	4FGL J0522.9-3628	1	80.737	-36.4686	240.618	-32.7216	101.878	AGN		PKS 0521-36				

Unassociated Source #3: spp → HMB

spp in 1FGL to HMB in 2FGL

Table Browser for 2: gll_psc_v03_1FGL.fit

Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
975 1FGLJ1653.9+3945	253.49	39.7527	63.5928	38.8413	bzb		Mkn 501		40.2217	15.5982	1476.79
546 1FGLJ1015.1+4927	153.788	49.4526	165.496	52.7181	bzb		IES 1011+496		40.3215	15.9248	1122.12
31 1FGLJ0030.4+0451	7.6143	4.861	113.142	-57.6114	PSR		PSR J0030+0451		40.6079	23.8943	831.998
1217 1FGLJ1952.9+3252	298.242	32.8779	68.7655	2.82324	PSR		PSR J1952+3252		40.7442	22.6044	824.035
449 1FGLJ0818.2+4222	124.552	42.3816	178.224	33.392	bzb		B3 0814+425		40.8058	22.0282	758.552
1124 1FGLJ1833.6-2103	278.406	-21.0637	12.1595	-5.7045	bzq		PKS 1830-21		41.0232	27.4163	481.52
70 1FGLJ0108.6+0135	17.1741	1.5879	131.852	-60.9845	bzq		PKS 0108+013		42.1262	27.3067	434.662
1170 1FGLJ1907.9+0602	286.978	6.0378	40.1822	-0.894193	PSR		PSR J1907+0...		42.4456	24.3516	1008.68
1188 1FGLJ1923.5-2104	290.885	-21.0771	17.178	-16.2559	bzq		OV -235		42.5887	24.6697	751.494
549 1FGLJ1018.6-5856	154.673	-58.9388	284.32278	-1.70069	spp				42.9696	28.2287	766.352
1110 1FGLJ1826.1-1256	276.535	-12.9434	18.5552	-0.377947	PSR		LAT PSR J1826-1...		43.0842	20.5847	980.67

spp

Table Browser for 5: gll_psc_v09_2FGL.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
992 2FGLJ1344.2-1723	206.068	-17.3956	320.477	43.6688	bzq		PMN J1344-1723		30.7749	16.3747	641.316	1.95405
1336 2FGLJ1740.2+5212	265.06	52.2012	79.5656	31.8057	bzq				30.7753	19.8098	481.282	2.50226
1622 2FGLJ2017.3+0603	304.35	6.05396	48.6261	-16.0298	PSR				31.2384	8.00843	1608.34	1.04592
763 2FGLJ1058.4+0133	164.615	1.56646	251.501	52.7681	bzb				31.4827	16.1794	991.442	2.21744
1779 2FGLJ2236.5-1431	339.133	-14.5331	47.9086	-56.1957	BZB		PKS 2233-148		31.5358	18.6361	673.396	2.23955
1527 2FGLJ1911.1-2005	287.791	-20.097	16.8922	-13.2106	bzq		PKS B1908-201		31.851	19.5889	437.277	2.20548
708 2FGLJ1019.0-5856	154.759	-58.9416	284.362	-1.67841	HMB		1FGLJ1018.6-5856		31.8745	17.4002	1263.62	2.59492
1852 2FGLJ2339.6-0532	354.909	-5.54204	81.3579	-62.4673					32.2053	11.3601	973.939	1.66875
1580 2FGLJ1108.6-1845	200.554	-18.7623	65.0764	0.352095	PSR		LAT PSR J1108-1845		32.2511	16.717	2108.44	1.24055

HMB

Table Browser for 9: gll_psc_v16_3FGL.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Spectral_Index	CLASS1	ASSOC1
2913 3FGLJ2254.0+1608	343.502	16.1459	86.1214	-38.1918	412.747	1.33123E-9	480.741	308.06	18.2082	1.63463	FSRQ	3C 454.3
2940 3FGLJ2311.0+3425	347.764	34.4231	100.411	-24.021	368.599	7.98852E-11	66.8727	40.6246	5.56969	2.19157	FSRQ	
3003 3FGLJ2345.2-1554	356.307	-15.9131	65.6972	-70.9857	825.789	2.02701E-11	82.3238	40.1362	6.27866	2.01772	FSRQ	
123 3FGLJ0059.0-7242e	14.75	-72.7	302.145	-44.4167	665.532	1.05933E-11	27.9457	18.7552	2.50357	2.22098	GAL	SMC
603 3FGLJ0526.6-6825e	81.65	-68.42	278.843	-32.8502	396.322	1.83021E-10	83.8399	51.3288	6.56124	2.01708	GAL	MC
320 3FGLJ0240.5+6113	40.1379	61.2281	135.679	1.08628	547.579	2.79382E-10	196.606	127.731	16.7763	2.18968	HMB	LS I+61 303
1173 3FGLJ1018.9-5856	154.73	-58.9456	284.351	-1.69011	1350.21	2.15732E-11	52.2459	22.9137	21.1823	2.54647	HMB	1FGLJ1018.6-5856
2359 3FGLJ1826.2-1450	276.568	-14.8473	16.8856	-1.2938	280.328	7.86167E-10	35.0597	24.3102	5.33324	2.37947	HMB	LS 5039
994 3FGLJ1040.0-5108	120.487	-51.1445	167.603	20.1703	706.662	5.31324E-12	22.3503	18.0837	2.75833	2.28245	MCV1	CRC 0846+513

HMB

Unassociated Source #4 → Blazar

Table Browser for 2: gll_psc_v03_1FGL.fit

	Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
155	1FGL J0252.8-2219	43.2096	-22.323	209.753	-62.0864	bzq		PKS 0250-225		34.455	22.8489	513.47
675	1FGL J1221.5+2814	185.39	28.2467	201.611	83.2931	bzb		W Com		34.724	17.1719	988.293
547	1FGL J1016.1+0514	154.032	5.2351	236.514	47.0585	bzq		CRATES J1016+0...		34.7958	20.7301	692.145
962	1FGL J1642.5+3947	250.6398	39.7941	63.4273	41.0283					34.8074	21.9664	548.416
392	1FGL J0714.0+1935	108.515	19.5866	197.659	13.6204	bzq		CLASS J0713+19...		35.1604	22.1187	558.232

Table Browser for 5: gll_psc_v09_2FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
170	2FGL J0229.3-3644	43.2096	-36.7336	243.952	-67.1972	bzq				16.1813	10.452	475.342
1243	2FGL J1647.5+4950	251.881	49.84	76.6545	40.0848	agn				16.1842	9.63322	554.614
1515	2FGL J1901.1+0427	285.298	4.45755	38.0103	-0.132545					16.2615	12.2794	913.074
249	2FGL J0340.5+5307	55.1495	53.1314	146.755	-1.73135					16.2887	10.1242	250.669
1186	2FGL J1618.2-7718	244.562	-77.3007	313.438	-18.8793	bzq		PKS 1610-77		16.3579	10.4627	503.719
48	2FGL J0043.7+3426	10.9409	34.4394	121.134	-28.4055	bzq		GB6 J0043+3426		16.498	8.31669	1071.46
1237	2FGL J1642.9+3949	250.748	39.8487	63.4801	40.9473	BZQ		3C 345		16.5409	7.8223	972.728
1144	2FGL J1549.5+0237	237.388	2.62742	10.8693	40.8972	bzq		PKS 1546+027		16.6181	8.75505	572.201

Unassociated
Class1 == null

Table Browser for 9: gll_psc_v16_3FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Spectral_Index	CLASS1	ASSOC1
1804	3FGL J1451.4-5539	224.555	-55.6501	329.000	40.550	410.111	39.2553	39.2553	3.17304	2.10093	FSRQ	PKS 1451-554
1817	3FGL J1504.4+1029	226.105	10.4936	11.382	54.5796	345.173	189.471	118.838	9.19739	2.07509	FSRQ	PKS 1502+106
1841	3FGL J1512.8-0906	228.213	-9.1032	351.289	40.1348	445.227	207.272	131.263	7.7404	2.30471	FSRQ	PKS 1510-08
1995	3FGL J1626.0-2951	246.518	-29.8624	348.809	13.3169	938.06	5.04271E-12	20.9228	14.1516	4.38472	FSRQ	PKS 1622-29
2021	3FGL J1635.2+3809	248.812	38.1525	61.1099	42.3395	701.191	4.47701E-11	86.9665	53.7671	9.3616	FSRQ	4C +38.41
2036	3FGL J1640.6+3945	250.159	39.7553	63.3465	41.3952	1222.25	3.50900E-12	28.2796	13.7391	4.73747	FSRQ	NRAO 512
2043	3FGL J1642.9+3950	250.748	39.8487	63.5063	40.9489	1156.74	2.54784E-12	20.5269	9.98583	0.273285	FSRQ	3C 345
2179	3FGL J1733.0-1305	263.256	-13.0858	12.0253	10.8129	516.34	3.74577E-11	36.8071	21.9722	3.76453	FSRQ	PKS 1730-13

BZQ

Table Browser for 19: gll_psc_v22_4FGL_DR1.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_So...	Pivot_Energy	Signif_Peak	LP_Sig
3344	4FGL J1647.5+4950	251.892	49.8336	76.6451	40.0778	37.1897	bl						808.93	18.7059	1
2487	4FGL J1306.3-6043	196.587	-60.7199	304.755	2.09904	37.2479						FGES J1036...	2121.84		1
1947	4FGL J1036.3-5833e	159.094	-58.563	286.081	-0.181491	37.2569							5171.9		1
1494	4FGL J0816.3+5739	124.1	57.6616	159.851	33.8992	37.2642	bl						1483.76	15.9236	1
4278	4FGL J1932.3+1916	293.076	19.2685	54.65	0.083537	37.3203	FSRQ		PSR J1932+1916				1537.65		1
4705	4FGL J2147.1+0931	326.783	9.5181	65.808	-32.241	37.3583	FSRQ		PKS 2144+092				467.316	36.2753	1
2993	4FGL J1522.6-2730	230.664	-27.5059	339.588	24.4001	37.3679	bl		PKS 1519-273				1203.56	22.2033	1
3316	4FGL J1642.9+3948	250.734	39.8164	63.4624	40.9576	37.381	FSRQ		3C 345				1408.69	20.9635	1
2491	4FGL J1307.6-4259	196.91	-42.995	306.079	19.7777	37.7179	bl		1RXS J130737.8...				2525.42		1
2553	4FGL J1323.0+2941	200.762	29.6871	54.9945	82.5956	38.0299	bcu		4C +29.48				1142.81		1
4096	4FGL J1852.4+4856	283.116	48.935	78.5979	19.9487	38.0499	fsrq		S4 1851+48				766.87	33.2299	1

FSRQ

FSRQ

Unassociated Source #5 (Eta Carinae) → Binary

Table Browser for 1: gll_psc3month_BSL_v2.fit

	Source_Name	RA	DEC	GLON	GLAT	Sqrt_TS	CLASS1	ASSOC1	95_Radius	Flux1
4	0FGLJ0030.3+0450	7.6	4.848	113.111	-57.622	18.7	PSR	PSR J0030+0451	0.138	8.
14	0FGLJ0137.1+4751	24.286	47.854	130.818	-14.317	18.8	bzq	CGRaBS J0136+4751 DA 55	0.12	1.
135	0FGLJ1653.9+3946	253.492	39.767	63.612	38.841	19.	bzb	CGRaBS J1653+3945 Mrk 501	0.068	2.
22	0FGLJ0229.5-3640	37.375	-36.681	243.801	-67.189	19.2	bzq	BZQ J0229-3643 PKS 0227-369	0.138	1.
147	0FGLJ1805.3-2138	271.329	-21.649	8.536	-0.165	19.2			0.186	7.
77	0FGLJ1045.6-5937	161.409	-59.631	287.637	-0.548	19.5			0.123	3.
189	0FGLJ2147.1+0931	326.777	9.519	65.805	-32.236	19.9	bzq	CGRaBS J2147+0925 PKS 2144+092	0.137	1.

Unassociated
Class1 == null

Table Browser for 2: gll_psc_v03_1FGL.fit

	Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
547	1FGLJ1016.1+0514	154.032	5.2351	236.514	47.0585	bzq		CRATES J1016+0...		34.7958	20.7301	692.145
962	1FGLJ1642.5+3947	250.6398	39.7941	63.4273	41.0283					34.8074	21.9664	548.416
392	1FGLJ0714.0+1935	108.515	19.5866	197.659	13.6204	bzq		CLASS J0713+19...		35.1604	22.1187	558.232
1333	1FGLJ2139.3-4235	324.849	-42.588	358.319	-48.3256	bzb		CRATES J2139-4...		35.2554	18.8526	856.989
585	1FGLJ1058.4+0134	164.62	1.5815	251.49	52.7827	bzq		PKS 1055+01		35.5381	20.491	660.283
572	1FGLJ1045.2-5942	161.305	-59.7057	287.625	-0.63878					35.6078	19.7799	1191.51
1071	1FGLJ1801.3-2322c	270.331	-23.383	6.57232	-0.21566	sdp				35.6355	18.8774	1052.3

Table Browser for 5: gll_psc_v09_2FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
1732	2FGLJ2147.3+0930	326.835	9.51476	65.8455	-32.2825	bzq		PKS 2144+092		35.8819	23.1634	455.413	2.514
1812	2FGLJ2302.7+4443	345.698	44.7202	103.403	-13.9924	PSR		PSR J2302+4442		35.9022	14.5797	1156.86	1.02671
1784	2FGLJ2241.7-5236	340.448	-52.6161	337.436	-54.935	PSR		PSR J2241-5236		36.0254	15.4306	924.773	1.35051
744	2FGLJ1045.0-5941	161.25116	-59.692	287.595	-0.639366			Eta Carinae		36.0695	16.2337	1847.65	2.32184
1817	2FGLJ2311.0+3425	347.771	34.4298	100.42	-24.0173	bzq		B2 2308+34		36.0738	21.088	421.488	2.16433
907	2FGLJ1248.2+5820	192.063	58.3499	123.738	58.7735	bzb		PG 1246+586		36.1449	17.2007	1329.24	1.94943

Unassociated Source #5 (Eta Carinae) → Binary

Table Browser for 9: gll_psc_v16_3FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Signif_Index	CLASS1	ASSOC1
249	3FGLJ0210.7-5101	32.6972	-51.0282	276.112	-61.7677	385.06	5.92996E-11	62.9476	38.0937	4.74627	2.3901	BCU	PKS 0208-512
591	3FGLJ0522.9-3628	80.7416	-36.4715	240.623	-32.7183	321.474	1.14793E-10	70.4297	44.9991	2.77891	2.3901	BCU	PKS 0521-36
612	3FGLJ0532.0-4827	83.0004	-48.4585	255.024	-32.8055	791.888	1.58149E-11	73.8326	35.9432	4.12102	2.0315	BCU	PMN J0531-4827
707	3FGLJ0622.9+3326	95.7346	33.4357	179.941	9.15656	507.631	3.72538E-11	56.1443	30.2938	2.97232	2.09513	BCU	B2 0619+33
1737	3FGLJ1418.5+3543	214.65	35.7186	63.1902	69.5785	1220.98	1.77572E-12	36.3181	15.6653	2.26862	2.05842	BCU	NVSS J141828+354250
1243	3FGLJ1045.1-5941	161.277	-59.6916	287.606	-0.633075	3281.82	1.54551E-12	33.0856	8.16258	8.88966	2.3901	BIN	Eta Carinae
112	3FGLJ0050.6-0929	12.6744	-9.4849	122.331	-72.3557	1023.06	3.79958E-12	41.9779	21.0804	1.25283	2.09308	BLL	PKS 0048-09

Associated to
BIN

Table Browser for 19: gll_psc_v22_4FGL_DR1.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_So...	Pivot_Energy	Signif_Peak
4905	4FGLJ2302.7+4443	345.692	44.7182	103.398	-13.9925	81.5316	PSR		PSR J2302+4442				1285.56	
979	4FGLJ0532.0-4827	83.0022	-48.4607	255.026	-32.8046		BLL		PMN J0531-4827				910.36	78.8813
4646	4FGLJ2124.7-3358	321.192	-33.9786	10.9268	-45.4451		PSR		PSR J2124-3358				1142.05	
4918	4FGLJ2311.0+3425	347.768	34.4223	100.414	-24.0231		FSRQ		B2 2308+34				499.986	66.5599
2973	4FGLJ1517.7-2422	229.425	-24.373	340.681	27.5766	83.55	bll		AP Librae		AP Lib		824.093	35.5893
308	4FGLJ0136.5+3906	24.1423	39.1	132.422	-22.9387	84.7523	bll		B3 0133+388		RGB J0136+391		1922.08	39.5364
4247	4FGLJ1923.2+1408e	290.818	14.145	49.1158	-0.461602	85.2124	SNR		W 51C		W 51C		2529.51	
1974	4FGLJ1045.1-5940	161.277	-59.6822	287.601	-0.624566	85.7256	BIN		Eta Carinae	FGES J1036.3-5833 field	Eta Carinae		2014.83	33.5116
4483	4FGLJ2032.2+4127	308.063	41.459	80.2294	1.02414	86.9727	PSR		PSR J2032+4127	Cygnus Cocoon field	PSR J2032+4127		2235.47	
1474	4FGLJ1040.8+5218	122.462	52.3143	166.248	32.9144	87.0045	RII		1FS 0806+524		1FS 0806+524		1297.	44.2755

BIN

BIN

Table Browser for 27: gll_psc_v27_4FGL_DR2.fit

	Source_Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_Source_Name	Signif_Peak
4844	4FGLJ1923.2+1408e	1	290.818	14.145	49.1158	-0.461602	93.9032	2456.97	SNR		W 51C		W 51C	W 51C	29.6422
2221	4FGLJ1045.1-5940	1	161.277	-59.6822	287.601	-0.624566	94.0432	2109.55	BIN		Eta Carinae	FGES J1036....	Eta Carinae		33.4744
1643	4FGLJ0808.2-0751	1	122.065	-7.8556	229.044	13.1616	94.4208	620.534	fsrq		PKS 0805-07				61.7413
2276	4FGLJ1058.4+0133	1	164.624	1.5641	251.514	52.7728	94.7605	802.762	BLL		4C +01.28				39.9912
3351	4FGLJ1517.7-2422	1	229.425	-24.373	340.681	27.5766	84.057	911.196	bll		AP Librae		AP Lib		35.7046

Unassociated Source #6 → PSR in 4FGL-DR1

Table Browser for 2: gll_psc_v03_1FGL.fit

Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	Sqrt_TS300_1000	Pivot_Energy	Flux_Density	Spectral_Index
1339	1FGL J2146.6-1345	326.672	-13.7535	40.654	-44.982			2.03775	3173.25	1.41837E-13	1.82421
982	1FGL J1702.4-4147c	255.604	-41.7859	344.45	0.003416			5.71774	1192.53	7.48946E-12	2.39258
93	1FGL J0137.5-2428	24.3927	-24.4815	201.197	-79.2893	bzq	PKS 0135-247	9.80455	6.51846	4.47998E-12	2.44899
1140	1FGL J1844.3-0309c	281.09	-3.157	29.3163	0.132846			9.8154	0.	1.07250E-12	2.15436
565	1FGL J1036.2-6719	159.0655	-67.3229	290.432	-7.79229			9.81386	3.55857	3.80255E-12	2.34984
1334	1FGL J2139.9+4715	324.987	47.2579	92.627	-4.03097			9.82447	4.17288	6.14558E-12	2.36444

Unassociated
Class1 == null

Table Browser for 5: gll_psc_v09_2FGL.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
1292	2FGL J1719.3+1744	259.83	17.7425	39.5258	28.0706	bzb	PKS 1717+177		17.6009	4.67481	2048.13	1.84168
1657	2FGL J2039.8-5620	309.965	-56.3406	341.192	-37.1778				17.6212	9.81252	834.837	2.19114
390	2FGL J0543.9-5532	85.9921	-55.5337	263.513	-31.4719	bzb	1RXS J054357.3-553...		17.7402	5.0146	3006.36	1.74014
957	2FGL J1315.9-3339	198.992	-33.6632	308.763	28.9306	bzq	PKS 1313-333		17.7481	10.6962	905.7	2.30522
734	2FGL J1036.1-6722	159.039	-67.3716	290.448	-7.83956				17.7547	6.97096	1096.66	1.88395
1509	2FGL J1857.6+0211	284.419	2.18709	35.5898	-0.389722				17.7549	10.269	1475.28	2.56639

Table Browser for 9: gll_psc_v16_3FGL.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Spectral_Index	CLASS1	ASSOC1
1212	3FGL J1033.0-5945	158.266	-59.7513	286.308	-1.4227	1555.61	1.13956E-12	11.3052	5.49087	1.70744	2.37901	
1214	3FGL J1033.4-5035	158.374	-50.5946	281.724	6.50911	2603.06	1.33345E-13	10.4081	1.91365	0.694635	2.01941	
1216	3FGL J1034.0-5731c	158.516	-57.5174	285.295	0.572408	1999.22	4.86822E-13	7.54238	5.12755	2.93315	2.28352	
1218	3FGL J1035.7-6720	158.926	-67.3335	290.391	-7.82838	778.904	8.65322E-12	30.4761	15.5561	8.39136	1.95648	
1220	3FGL J1036.4-4348	159.113	-43.812	278.678	12.6117	1252.55	2.79663E-13	4.61889	1.20748	1.22018	2.27681	

Table Browser for 19: gll_psc_v22_4FGL_DR1.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extend...	Pivot_Energy
4680	4FGL J2140.0+4716	325.001	47.2686	92.6408	-4.0291	PSR	PSR J2139+4716				941.996
407	4FGL J0212.1+5321	33.0445	53.3507	134.926	-7.63041						1033.57
1944	4FGL J1035.4-6720	158.865	-67.338	290.372	-7.84417	PSR	PSR J1035-6720				1640.89

Associated to
PSR

Table Browser for 27: gll_psc_v27_4FGL_DR2.fit

Source_Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	CLASS1	CLASS2	ASSOC1	ASSOC2
5163	4FGL J2045.2+5026e	1	311.32	50.44	88.7529	4.67043	48.3538	SNR		HB 21	
5538	4FGL J2241.7-5236	1	340.438	-52.6084	337.7452	-54.9341	48.4299	PSR		PSR J2241-5236	
2185	4FGL J1035.4-6720	1	158.865	-67.338	290.372	-7.84417	48.495	PSR		PSR J1035-6720	
3959	4FGL J1719.2+1745	1	259.806	17.7533	39.5272	28.0959	48.5274	bll		PKS 1717+177	

Unassociated source #7, not present in BSL

Table Browser for 2: gll_psc_v03_1FGL.fit

Source_Name	RA	DEC	GLON	GLAT	Signif_Avg	Pivot_Energy	Flux1000	Unc_Flux1000	Energy_Flux	Unc_Energy_Flux	Sqrt_TS300_1000	ASSOC1	CLASS1
1090	1FGL J1813.4+3141	273.355	31.694	58.6128	21.383	12.08	1209.52	2.61684E-9	1.81393E-10	1.91990E-11	4.08612E-12	4.94638	B2 1811...
959	1FGL J1639.5-5152	249.896	-51.8772	334.19	-3.42661	12.7029	722.549	3.94941E-9	7.04672E-10	4.70342E-11	9.30263E-12	8.11435	
1385	1FGL J2243.1-2541	340.788	-25.685	28.4482	-61.3159	12.7208	736.055	1.83617E-9	3.31700E-10	1.67093E-11	3.12147E-12	6.89736	PKS 224...
1315	1FGL J2112.5-3044	318.137	-30.7357	14.8952	-42.4425	12.7843	1107.29	2.87275E-9	4.10476E-10	1.78860E-11	3.47843E-12	5.55027	
54	1FGL J0051.1-0649	12.7826	-6.8184	122.713	-69.69	12.821	781.285	1.86046E-9	3.30053E-10	1.98122E-11	3.86006E-12	8.70655	PKS 004...
174	1FGL J0312.6+0131	48.1538	1.5333	178.507	-45.5564	12.8365	812.796	1.82553E-9	3.30055E-10	1.85719E-11	3.52029E-12	7.09747	PKS 031...
934	1FGL J1624.0-4041	246.016	-40.6897	340.546	6.16336	12.8394	808.767	4.90696E-9	6.89157E-10	3.45534E-11	6.07750E-12	7.66264	

Total: 1,451 Visible: 1,451 Selected: 1

Table Browser for 5: gll_psc_v09_2FGL.fit

Source_Name	RAJ2000	DECJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	CLASS1	CLASS2	ASSOC1
370	2FGL J0555.0+4623	85.2000	40.3936	102.131	8.2057	19.5040	672.070	2.07784E-9	2.42030E-10	3.51523E-11	2.31038E-12	bzq	1A5 0529+4...
1536	2FGL J1917.6-1921	289.423	-19.3503	18.2433	-14.3047	19.5752	2263.19	3.23330E-9	2.66117E-10	3.72461E-11	3.44170E-12	bzb	1H 1914-194
797	2FGL J1118.8-6128	169.703	-61.4735	292.107	-0.563835	19.5814	946.988	9.02700E-9	6.01848E-10	7.91638E-11	5.75340E-12	PSR	PSR J1119-6...
970	2FGL J1326.7-5254	201.7	-52.9149	308.325	9.58611	19.6766	997.444	3.08156E-9	2.96810E-10	4.50878E-11	3.00896E-12	bzb	PMN J1326-5...
1700	2FGL J2112.5-3042	318.149	-30.7106	14.9311	-42.4479	19.6973	1152.89	3.39351E-9	2.89512E-10	1.89001E-11	1.98985E-12		
409	2FGL J0608.0-1521	92.0034	-15.3667	222.112	-16.4071	19.7407	387.331	2.34312E-9	2.40045E-10	2.85808E-11	1.95358E-12	bzq	PMN J0608-1...
760	2FGL J1057.0-8004	164.267	-80.0826	297.951	-18.3356	19.7782	544.676	2.75386E-9	2.41736E-10	2.48207E-11	1.96758E-12	bzb	PKS 1057-79

Total: 1,873 Visible: 1,873 Selected: 3

Table Browser for 9: gll_psc_v16_3FGL.fit

Source_Name	RAJ2000	DECJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	Flux_Density	Unc_Flux_Density	CLASS1
758	3FGL J0650.7+2503	102.697	25.0632	190.272	11.0038	29.4866	3626.36	2.16351E-9	1.42736E-10	2.86996E-11	2.26063E-12	1.76277E-13	1.16144E-14 bl
1115	3FGL J0953.0-0839	148.255	-8.6635	246.175	33.8993	29.8033	2221.09	2.31321E-9	1.44562E-10	2.84761E-11	2.18646E-12	4.52124E-13	2.75543E-14 bl
2157	3FGL J1727.1+4531	261.8	45.5272	71.4457	33.3279	29.9227	333.074	1.40029E-9	1.05124E-10	1.99875E-11	9.45562E-13	3.84612E-14	1.87911E-12 fsrq
539	3FGL J0455.7-4617	73.9447	-46.2845	251.992	-38.8259	30.0136	451.944	1.46289E-9	1.07174E-10	2.28016E-11	1.00137E-12	1.70886E-11	7.42053E-13 fsrq
1055	3FGL J0915.8+2933	138.965	29.5604	196.644	42.9327	30.0152	1948.02	2.00434E-9	1.24068E-10	2.34761E-11	1.71877E-12	5.12150E-13	3.04401E-14 bl
1472	3FGL J1222.4+0414	185.611	4.234	284.851	66.0826	30.0163	263.937	1.19957E-9	1.28931E-10	2.86599E-11	1.41116E-12	1.11763E-10	5.68161E-12 fsrq
2742	3FGL J2112.5-3044	318.145	-30.7344	14.8984	-42.4487	30.1418	1133.51	3.25896E-9	1.89762E-10	1.90079E-11	1.38509E-12	2.86716E-12	1.77038E-13
1333	3FGL J1119.9-2204	169.984	-22.0673	276.47	36.0588	30.1602	791.708	2.69949E-9	1.66004E-10	1.68464E-11	1.03209E-12	6.46429E-12	3.78659E-13

Unassociated source #7, not present in BSL

Table Browser for 14: gll_psc_v22_4FGL_DR1.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	CLASS1	CLASS2	ASSOC1
3927	4FGLJ1824.1+5651	276.039	56.8585	85.7488	26.0775	49.7303	611.672	1.93342E-9	7.58437E-11	2.22824E-11	8.64008E-13	bl	4C+56.27
2379	4FGLJ1231.7+2847	187.935	28.7917	180.861	85.3436	50.1482	1580.74	2.03677E-9	8.01939E-11	2.28336E-11	9.66106E-13	bl	B2 1229+29
298	4FGLJ0132.7-1654	23.176	-16.9103	168.1	-76.0186	50.2573	406.555	1.77349E-9	7.98956E-11	2.17270E-11	7.64664E-13	fsrq	PKS 0130-17
472	4FGLJ0228.3-5547	37.0854	-55.7931	278.263	-56.4474	50.2735	1146.11	2.73168E-9	9.43999E-11	2.62402E-11	1.62447E-12	fsrq	PKS 0226-559
4612	4FGLJ2112.5-3043	318.14	-30.7293	14.9039	-42.4435	50.3083	1343.1	3.40176E-9	1.24364E-10	1.84531E-11	8.13833E-13	bl	
1797	4FGLJ0953.0-0840	148.263	-8.6695	246.186	33.9809	50.2735	1946.86	2.40997E-9	9.38699E-11	2.78344E-11	1.23263E-12	bl	PMN J0953-0...
3414	4FGLJ1703.6-6213	255.912	-62.1111	132.441	-12.7269	50.3604	764.101	3.13613E-9	1.09085E-10	3.25072E-11	1.27231E-12	fsrq	MRC 1659-6...
202	4FGLJ0101.1-6422	15.200	-64.2222	15.200	-64.2222	50.4295	937.511	2.10791E-9	8.44320E-11	1.29872E-11	5.42054E-13	PSR	PSR J0101-6...
3231	4FGLJ1625.7-2527	245.917	-25.2727	162.527	-25.2727	50.7237	1179.44	5.98051E-9	1.83377E-10	5.29311E-11	2.35345E-12	fsrq	PKS 1622-253
1690	4FGLJ0915.9+2933	135.917	29.3333	91.517	29.3333	50.7639	1345.56	2.24586E-9	9.22299E-11	2.80234E-11	5.97505E-13	bl	Ton 0396
1475	4FGLJ0811.0-7529	122.717	-75.2929	122.717	-75.2929	50.8021	1639.68	2.56166E-9	7.70019E-11	2.41555E-11	5.97505E-13	bl	PMN J0810-7...
3547	4FGLJ1728.3+5013	262.078	50.1333	172.833	50.1333	50.9726	3005.02	1.75913E-9	7.00500E-11	2.20234E-11	5.97505E-13	bl	

Total: 5,065 Visible: 5,065 Selected: 1

Table Browser for 22: gll_psc_v27_4FGL_DR2.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	CLASS1	ASSOC1	ASSOC_FGL
5250	4FGLJ2110.7+3129	317.880	31.4958	92.217	18.0378	18.0378	1561.46	1.80167E-9	7.68640E-10	1.44101E-11	1.11478E-11	3.48697E-12	
5257	4FGLJ2111.4+4606	317.863	46.1048	88.533	18.0378	18.0378	1561.46	1.80167E-9	7.68640E-10	1.44101E-11	1.11478E-11	3.48697E-12	
5258	4FGLJ2112.5-3043	318.14	-30.7293	14.9039	-42.4435	56.3453	1373.16	3.47246E-9	1.12888E-11	1.82917E-11	7.04271E-13		
5259	4FGLJ2112.7+0819	318.185	8.3195	58.6974	-26.3095	55.8537	3971.92	bl	1RXS ...	3FGLJ2112.7+08...	4.78079E-10	5.51715E-12	5.97505E-13
5260	4FGLJ2114.3+5023	318.59	50.3993	91.7634	1.14875	11.9353	1651.51	1.31164E-9	1.30023E-10	1.81205E-11	2.25549E-12		
5261	4FGLJ2114.7+3130	318.691	31.5038	77.9907	-11.8843	11.8848	1276.32	4.33096E-9	5.08231E-11	1.22239E-11	7.22339E-13		
5262	4FGLJ2114.8+2831	318.716	28.5222	75.7417	-13.9054	56.5697	713.699	fsrq	B2 21...	2.43509E-10	4.56981E-11		
5263	4FGLJ2114.8+2026	318.717	20.4418	69.3912	-19.2232	33.9928	2989.52	agn	TXS 2...	1.59863E-10	3.74911E-11		
5264	4FGLJ2114.9-3326	318.73	-33.4353	11.3563	-43.3489	9.01754	2775.03	3.25338E-10	4.51391E-11	1.22239E-11	7.22339E-13		

Total: 5,788 Visible: 5,788 Selected: 1

Table Browser for 29: gll_psc_v31_4FGL_DR3.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	CLASS1	ASSOC1	ASSOC_FGL
1127	4FGLJ0510.0+1800	18.0181	18.0181	184.735	-12.7821	60.2364	2982.84	1.62214E-9	5.43588E-11	1.67130E-11	6.27078E-13		
4620	4FGLJ1728.3+5013	262.078	50.2267	77.0763	33.5371	60.2364	2982.84	1.62214E-9	5.43588E-11	1.67130E-11	6.27078E-13		
6037	4FGLJ2112.5-3043	318.14	-30.7293	14.9039	-42.4435	60.4539	1431.25	3.48063E-9	1.04856E-10	1.80194E-11	6.44335E-13		
280	4FGLJ0106.4+4855	16.6094	48.9269	125.469	-13.871	60.4727	1577.5	PSR	PSR J0106+4855	3.46684E-9	1.01081E-11	1.92126E-11	7.22339E-13
4877	4FGLJ1754.2+3212	268.553	32.2007	57.7348	25.4093	60.6251	1455.82	bl	RX J1754.1+32...	2.32678E-9	7.58325E-11	2.03398E-11	9.03015E-13
256	4FGLJ0101.1-6422	15.2997	-64.3683	301.189	-52.7269	61.0144	903.834	MSP	PSR J0101-6422	2.08845E-9	6.83483E-11	1.30303E-11	4.41994E-13
4388	4FGLJ1657.7+4808	254.438	48.1368	74.3614	38.4824	61.2709	576.543	fsrq	4C+48.41	1.78467E-9	5.84558E-11	2.05647E-11	6.28426E-13
6500	4FGLJ2321.9+3204	350.478	32.0737	181.686	-27.0778	61.2169	845.670	fsrq	B2 2310+31	2.85501E-9	8.77644E-11	2.55721E-11	9.80112E-13

Total: 6,659 Visible: 6,659 Selected: 1

Table Browser for 36: gll_psc_v33_4FGL_DR4.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	CLASS1	ASSOC1	ASSOC_FGL
3499	4FGLJ1112.6+4828	118.169	48.4701	113.396	68.263	64.3245	827.214	1.71635E-9	5.43802E-11	1.67130E-11	6.27078E-13		
2942	4FGLJ1120.0-2204	178.002	-22.0779	36.0559	64.3332	847.502		2.90012E-9	8.73870E-11	1.77850E-11	5.77320E-13		
4384	4FGLJ1603.8-4903	240.967	-49.0617	332.158	2.57594	64.5449	3359.9	6.00857E-9	1.93476E-10	5.94111E-11	2.46915E-12		
6518	4FGLJ2112.5-3043	318.14	-30.7293	14.9039	-42.4435	64.5728	1385.6	3.48812E-9	9.84653E-11	1.84894E-11	6.17647E-13		
54	4FGLJ0009.3+5030	2.3466	50.5115	116.124	-11.8036	64.8969	1330.3	bl	NVSS J000922+50...	2.37991E-9	6.97634E-11	2.20884E-11	8.34268E-13
4986	4FGLJ1728.3+5013	262.078	50.2267	77.0763	33.5371	64.9075	1891.18	bl	I Zw 187	1.62234E-9	5.32592E-11	1.93064E-11	7.61677E-13
4648	4FGLJ1642.9+3948	258.734	39.8164	63.4624	40.9576	64.956	1250.67	FSRQ	3C 345	3.11057E-9	8.07993E-11	3.64885E-11	1.41579E-12

Unassociated source #8

Table Browser for 1: gl_l_psc3month_BSL_v2.fit

Source_Name	RA	DEC	GLON	GLAT	Sqrt_TS	Flux100_1000	Unc_Flux100_1000	CLASS1	CLASS2	ASSOC1	ASSOC2	Flux1000_100000	Unc_Flux1000_1000
5 OFGLJ0033.6-1921	8.401	-19.36	94.215	-81.22	10.7	1.40000E-8	0.	bzb		BZB J0033-1921	KUV 00311-1938	3.60000E-9	7.00000E
33 OFGLJ0412.9-5341	63.23	-53.686	263.001	-44.716	10.7	5.70000E-8	1.30000E-8	bzb		CRATES J0413-5332		2.90000E-9	7.00000E
121 OFGLJ1536.7-4947	234.197	-49.798	328.261	4.764	10.7	1.90000E-7	0.					7.40000E-9	1.30000E
185 OFGLJ2110.8+4608	317.702	46.137	88.261	-1.351	10.7	1.05000E-7	3.30000E-8					6.40000E-9	1.20000E
86 OFGLJ1106.4-6055	166.605	-60.919	290.516	-0.604	10.8	1.69000E-7	5.20000E-8	x				1.43000E-8	2.20000E
116 OFGLJ1511.2-0536	227.814	-5.613	354.099	42.948	10.8	8.30000E-8	1.70000E-8	bzq		PKS 1508-05	BZQJ1510-0543	3.40000E-9	8.00000E
23 OFGLJ0238.4+2855	39.601	28.923	149.522	-28.368	10.9	8.30000E-8	1.60000E-8	bzq		CGRaBS J0237+2848	B2 0234+28	3.40000E-9	8.00000E

Total: 205 Visible: 205 Selected: 1

Unassociated
Class1 == null

Table Browser for 2: gl_l_psc_v03_1FGL.fit

Source_Name	RA	DEC	GLON	GLAT	Sqrt_TS	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	Sqrt_TS300_1000	ASSOC1	CLASS1
8 1FGLJ0005.7+3815	1.4364	38.2594	113.151	-23.7429	8.36664	363.651	6.24795E-10	2.92986E-10	1.35817E-11	3.12512E-12	4.37164	B2 0003... bzb
1014 1FGLJ1726.2-0724	261.561	-7.4051	16.1325	15.1768	8.38771	837.232	1.80879E-9	4.75535E-10	2.62828E-11	5.05270E-12	4.03250	
882 1FGLJ1539.0-3328	234.763	-33.4684	338.682	17.5233	8.38784	1804.2	3.56342E-9	5.06702E-10	1.27525E-11	3.67577E-12	0.	
427 1FGLJ0752.8+5353	118.202	53.894	164.184	30.4758	8.40213	2316.11	1.22187E-9	2.74070E-10	1.08243E-11	4.10910E-12	1.9177	4C +54.... bzb
277 1FGLJ0515.9+1528	78.9777	15.4668	187.702	-13.0416	8.41787	3037.3	1.17092E-9	3.65462E-10	1.70621E-11	4.35959E-12	1.77922	
1114 1FGLJ1829.6-1006	277.408	-10.1134	21.4582	0.181272	8.42706	1428.7	6.94492E-9	1.13077E-9	4.64454E-11	1.36119E-11	2.56718	
940 1FGLJ1626.2-2956	246.566	-29.941	348.779	13.2325	8.43641	809.279	2.23321E-9	4.73436E-10	1.85789E-11	4.70111E-12	5.16958	PKS 162... bza

Total: 1,451 Visible: 1,451 Selected: 1

Table Browser for 5: gl_l_psc_v09_2FGL.fit

Source_Name	RA	DEC	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	CLASS1	CLASS2	ASSOC1
159 2FGLJ0222.0-1615	35.5177	-16.2644		190.233	-66.3674	6.88071	663.928	5.70891E-10	1.17699E-10	8.09901E-12	1.26008E-12	bzb		PKS 0219-164
501 2FGLJ0737.1-3235	114.294	-32.5874		246.859	-5.56216	6.88755	869.793	1.33420E-9	2.34937E-10	1.66216E-11	2.48982E-12			
1350 2FGLJ1745.1-1729	266.277	-17.4977		9.73587	6.03521	6.90026	2154.19	1.72167E-9	2.80750E-10	1.91245E-11	3.11557E-12			1RXS J17445...
1134 2FGLJ1539.3-4636	234.835	-46.611		330.511	7.07367	6.91418	546.297	1.06522E-9	2.15605E-10	1.78825E-11	2.65789E-12			
1710 2FGLJ2125.0-4632	321.265	-46.5465		352.994	-45.338	6.92033	670.536	6.54296E-10	1.34271E-10	9.02194E-12	1.39882E-12			
1278 2FGLJ1710.0-0323	257.523	-3.39311		17.5944	20.6829	6.92346	522.71	9.96225E-10	2.00121E-10	1.59000E-11	2.33016E-12			
1574 2FGLJ1949.7+2405	297.437	24.0857		60.8408	-1.05511	6.92868	686.437	1.78901E-9	3.91542E-10	1.85934E-11	2.66922E-12			

Total: 1,873 Visible: 1,873 Selected: 1

TOPCAT(9): Table Browser

Table Browser for 9: gl_l_psc_v16_3FGL.fit

Source_Name	RA	DEC	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	Flux_Density	Unc_Flux_Density	CLASS1
1278 3FGLJ1058.1+7010	164.536	70.1755		135.858	43.9999	12.4335	472.177	4.49052E-10	6.23365E-11	8.72761E-12	7.81379E-13	5.81277E-12	5.13055E-13	fsrq
2871 3FGLJ2230.5-7817	337.641	-78.2979		311.292	-36.4397	12.4368	616.943	7.12232E-10	9.27920E-11	1.05900E-11	9.90376E-13	3.60838E-12	3.36987E-13	bcu
400 3FGLJ0326.2+0225	51.5554	2.4178		180.748	-42.4352	12.4886	2272.86	8.05968E-10	1.04334E-10	9.61144E-12	1.19800E-12	1.68025E-13	2.02919E-12	bpl
1892 3FGLJ1538.4-4656	234.603	-46.9354		330.187	6.90878	12.4962	320.779	8.57160E-10	1.36290E-10	2.10750E-11	1.76297E-12	4.04517E-11	3.38476E-12	
2715 3FGLJ2056.7+4938	314.179	49.6415		89.3014	2.74449	12.5043	7206.76	1.30116E-9	1.88963E-10	1.63057E-11	2.08529E-12	3.09427E-14	3.95719E-15	bcu
1744 3FGLJ1419.8+3819	214.956	38.3248		69.6926	68.3859	12.5054	793.405	5.98921E-10	7.67526E-11	7.65662E-12	7.72506E-13	1.41483E-12	1.42545E-13	fsrq
432 3FGLJ0343.2-2534	55.8245	-25.5675		220.629	-51.6542	12.5067	577.415	5.92093E-10	7.74420E-11	9.39635E-12	8.82862E-13	3.76872E-12	3.52574E-13	fsrq

Unassociated source #8

Table Browser for 14: gl_l_psc_v22_4FGL_DR1.fit

Source Name	RAJ2000	DEC2000	GLON	GLAT	Signif_Avg	Pivot_Energy	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	CLASS1	CLASS2	ASSOC1
2200 4FGL J1150.4+2418	177.603	24.3016	221.157	75.9766	30.2533	1095.55	1.43838E-9	7.52204E-11	1.65645E-11	7.72806E-13	bll		OM 280
637 4FGL J0325.6-1646	51.4179	-16.781	204.879	-52.9356	30.419	2795.32	1.06938E-9	6.63612E-11	1.28569E-11	9.01493E-13	bll		RBS 0421
2340 4FGL J1222.5+0414	185.627	4.2389	284.885	66.0918	30.4311	357.188	1.04697E-9	6.85125E-11	2.26787E-11	1.03355E-12	fsrq		4C +04.42
1258 4FGL J0701.5-4634	105.391	-46.5729	256.802	-17.7064	30.4616	635.254	1.28761E-9	7.36302E-11	1.48404E-11	8.23983E-13	fsrq		PKS 0700-465
4075 4FGL J1848.5+3243	282.146	32.7309	62.4593	14.8255	30.47	1258.6	2.26347E-9	1.11396E-10	2.85471E-11	1.73588E-12	fsrq		B2 1846+328
3048 4FGL J1539.4-3323	234.851	-33.3987	338.788	17.5321	30.5315	1890.84	2.35742E-9	1.20536E-10	1.14377E-11	6.81213E-11			
4 4FGL J0001.5+2113	0.3815	21.2183	107.649	-40.1677	30.666	376.396	9.67927E-10	6.66420E-11	1.92604E-11	8.02958E-13	fsrq		TXS 2358+2...
4159 4FGL J1902.9-6218	285.743	-62.8068	327.76	-26.0474	30.6809	657.545	1.11970E-9	6.49767E-11	1.21305E-11	7.54722E-13	fsrq		PMN J1903-6...
1015 4FGL J0539.9-2839	84.9952	-28.6585	232.938	-27.2745	30.7296	453.68	7.80914E-10	5.57520E-11	1.62593E-11	7.97316E-13	fsrq		PKS 0537-286
1259 4FGL J0702.7-1951	105.678	-19.8517	231.923	-6.52771	30.7468	1820.6	2.06814E-9	1.04087E-10	2.31446E-11	1.14384E-12	bll		TXS 0700-197
330 4FGL J0145.0-2732	26.2585	-27.5363	217.074	78.0057	30.7855	317.527	7.46509E-10	5.54529E-11	1.20727E-11	5.63070E-13	fsrq		PKS 0142-208
4058 4FGL J1836.5-0651e	279.143	-6.866	25.1304	0.158304	36.9105	6098.15	1.92883E-8	6.99271E-10	1.24528E-10	7.55094E-12			

Total: 5,065 Visible: 5,065 Selected: 1

Unassociated
Class1 == null

Table Browser for 22: gl_l_psc_v27_4FGL_DR2.fit

Source Name	RAJ2000	DEC2000	GLON	GLAT	Signif_Avg	Pivot_Energy	CLASS1	ASSOC1	ASSOC_FGL	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100
3426 4FGL J1537.9-1344	234.487	-13.7335	352.884	52.5461	5.41502	7787.48	bcl	1RXS ...		1.31060E-10	3.6121...	1.75157E-12	4.36716E-13
3427 4FGL J1538.0-6638	234.524	-46.6422	330.319	7.17651	10.5876	921.131			3FGL J1538.4-4656	6.12367E-10	7.7140...	2.7170E-12	1.34249E-12
3428 4FGL J1538.2-6201	234.554	-62.0307	321.188	-5.22992	4.08019	942.535				2.46432E-10	5.7958...	4.42070E-12	1.07448E-12
3429 4FGL J1538.9+0425	234.734	4.4321	10.869	44.1024	5.87586	1507.02				1.97518E-10	3.8064...	2.40515E-12	4.67267E-13
3430 4FGL J1539.1+1008	234.793	10.1345	17.8462	47.0532	5.87807	667.225				1.56618E-10	3.4452...	2.74301E-12	4.90575E-13
3431 4FGL J1539.4-3323	234.851	-33.3987	338.788	17.5321	34.4732	1836.43			3FGL J1539.2-3324	2.43457E-9	1.0933...	1.18449E-11	6.23559E-13
3432 4FGL J1539.6+2743	234.902	27.7277	43.849	52.8012	23.0003	941.229	fsrq	MG2 J...	3FGL J1539.5+27...	6.58902E-10	4.6706...	8.18170E-12	4.78219E-13
3433 4FGL J1539.7-1127	234.928	-11.4636	355.171	33.840	11.4067	4683.57	bll	PMN J...	3FGL J1539.8-1128	3.75552E-10	5.2378...	4.48757E-12	6.03110E-13
3434 4FGL J1539.9+4220	234.977	42.3381	68.1079	52.6638	6.56511	1170.54	bll	87GB ...		1.60692E-10	2.9133...	1.99152E-12	3.37194E-13

Total: 5,788 Visible: 5,788 Selected: 1

Table Browser for 29: gl_l_psc_v31_4FGL_DR3.fit

Source Name	RAJ2000	DEC2000	GLON	GLAT	Signif_Avg	Pivot_Energy	CLASS1	ASSOC1	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	Cor
3331 4FGL J1333.7+5056	203.439	50.9366	107.12	64.8726	36.0717	770.618	fsrq	CLASS J1333+50...	8.97635E-10	4.44843E-11	9.05566E-12	5.10891E-13	
875 4FGL J0340.5+2118	55.1477	-21.3158	213.779	-51.1783	36.6753	783.645	bll	PKS 0338-214	1.07773E-9	5.54493E-11	1.07860E-11	6.29518E-13	
2848 4FGL J1150.4+2418	177.603	24.3016	221.157	75.9766	36.6833	983.149	bll	OM 280	1.52235E-9	6.78834E-11	5.29699E-11	7.82401E-13	
3934 4FGL J1539.4-3323	234.851	-33.3987	338.788	17.5321	36.7929	1845.26			2.44897E-9	1.03235E-10	1.20685E-11	6.12293E-13	
3750 4FGL J1501.0+2238	225.257	22.6364	31.4478	60.3515	36.8093	1298.57	bll	MS 1458.8+2249	1.23754E-9	6.11580E-11	1.18614E-11	7.32432E-13	
214 4FGL J0050.0-5736	12.5197	-57.6164	303.291	-59.5109	36.8886	821.795	fsrq	PKS 0047-579	8.38575E-10	4.25134E-11	1.15611E-11	4.62011E-13	
5229 4FGL J1836.5-0651e	279.143	-6.866	25.1304	0.158304	36.9105	6098.15	pwn	HESS J1837-069	1.92883E-8	6.99271E-10	1.24528E-10	7.55094E-12	

Total: 6,659 Visible: 6,659 Selected: 1

Table Browser for 36: gl_l_psc_v33_4FGL_DR4.fit

Source Name	RAJ2000	DEC2000	GLON	GLAT	Signif_Avg	Pivot_Energy	CLASS1	ASSOC1	Flux1000	Unc_Flux1000	Energy_Flux100	Unc_Energy_Flux100	C
4332 4FGL J1553.6-2422	238.403	-24.3687	347.584	22.3119	39.3382	1125.27	fsrq	PKS 1550-242	1.90632E-9	7.94698E-11	1.86400E-11	1.00863E-12	
2680 4FGL J1027.0-8542	156.764	-85.7163	300.179	-23.6403	39.3804	2446.56	bll	PKS 1029-85	1.08203E-9	5.31208E-11	1.09449E-11	6.71541E-13	
7044 4FGL J2324.7-4041	351.182	-40.6834	350.193	-67.58	39.5812	2942.49	bll	1ES 2322-409	9.98137E-10	5.45467E-11	1.15224E-11	7.48148E-13	
3087 4FGL J1151.5+5859	177.88	58.9894	136.888	56.4858	39.5817	2157.89		TXS 1148+592	5.58035E-10	4.19115E-11	7.98655E-12	5.31482E-13	
4247 4FGL J1539.4-3323	234.851	-33.3987	338.788	17.5321	39.6321	1904.32			2.47999E-9	9.81446E-11	1.21582E-11	5.63199E-13	
3942 4FGL J1438.9+3710	219.74	37.1752	63.6093	65.2363	39.6601	609.166	fsrq	B2 1436+378	8.83737E-10	4.22492E-11	9.65365E-12	4.58701E-13	
4463 4FGL J11617.9-7718	244.481	-77.304	313.422	-18.8684	39.6781	603.177	fsrq	PKS 1610-77	1.25073E-9	5.28172E-11	1.98890E-11	7.86662E-13	

Update: 05/24/2024

Summary

Catalog	Total	AGNs	Pulsars	UNK	Nulls	Others
4FGL-DR4	7195	4008	320	147	2430	
4FGL-DR3	6659	3809	292	134	2157	
4FGL-DR2	5788	3503	271	115	1667	
4FGL-DR1	5065	3202	239	92	1336	
3FGL	3034	1744	167	0	1010	
2FGL	1873	25	108	0	652	211 agu 342 bzb 413 bzq 17 BZQ
1FGL	1451	28	63	0	630	92 agu 274 bzb 295 bzq 4 BZQ
BSL	205	2	29	0	39	62 bzb 46 bzq 11 bzu

Progress

- Created reporting script:
 - /Python/Reporting/4FGL_Reporting.py
 - /Python/Utils/FGL4Dataset_Reporting.py
- This enabled numbers to be reported from the different catalogs

Progress

4FGL_Reporting.py

```
# wanted_type_col
# 70 (4FGL_DR4, 4FGL_DR3), 65 (4FGL_DR2), 75 (4FGL_DR1)
# 73 (3FGL), 63 (2FGL), 54 (1FGL), 17 (BSL)
```

```
wanted_type_col = 17
```

```
# KrishivB: Modified path to 4FGL file
```

```
path = os.path.join(os.getcwd(), "../FITS/gll_psc3month_BSL_v2.fit")
```

```
# path = os.path.join(os.getcwd(), "../FITS/gll_psc_v03_1FGL.fit")
```

```
# path = os.path.join(os.getcwd(), "../FITS/gll_psc_v09_2FGL.fit")
```

```
# path = os.path.join(os.getcwd(), "../FITS/gll_psc_v16.fit")
```

```
# path = os.path.join(os.getcwd(), "../FITS/gll_psc_v22_4FGL_DR1.fit")
```

```
# path = os.path.join(os.getcwd(), "../FITS/gll_psc_v27_4FGL_DR2.fit")
```

```
# path = os.path.join(os.getcwd(), "../FITS/gll_psc_v31_4FGL_DR3.fit")
```

```
# path = os.path.join(os.getcwd(), "../FITS/gll_psc_v33_4FGL_DR4.fit")
```

set wanted_type_col

set path of FITS file

```
# Just call its constructor here
```

```
dataset = FGL4Dataset_Reporting(point_source_catalogue, wanted_type_col, all_types, psr_types, agn_types)
```

Call constructor of reporting file

4FGL_Dataset_Reporting.py

Declare count variables

```
all_cnt = null_cnt = agn_cnt = AGN_cnt = psr_cnt = PSR_cnt = 0
msp_cnt = MSP_cnt = fsrq_cnt = FSRQ_cnt = bll_cnt = BLL_cnt = bcu_cnt = BCU_cnt = 0
RDG_cnt = rdg_cnt = NLSY1_cnt = nlsy1_cnt = YNG_cnt = yng_cnt = ssrq_cnt = sey_cnt = 0
unk_cnt = bin_cnt = glc_cnt = sbg_cnt = clust_cnt = spp_cnt = snr_cnt = SNR_cnt = 0
gal_cnt = HMB_cnt = pwn_cnt = GAL_cnt = PWN_cnt = BIN_cnt = css_cnt = SFR_cnt = NOV_cnt = 0
bzb_cnt = bzq_cnt = agn_cnt = BZQ_cnt = BZB_cnt = SEY_cnt = HXB_cnt = MQO_cnt = 0
glb_cnt = hxb_cnt = bzu_cnt = 0
```

Count types based on source[wanted_type_col]

```
if source[wanted_type_col] == '' or source[wanted_type_col] is None:
    null_cnt += 1
elif source[wanted_type_col] == 'AGN':
    AGN_cnt += 1
elif source[wanted_type_col] == 'agn':
    agn_cnt += 1
elif source[wanted_type_col] == 'FSRQ':
    FSRQ_cnt += 1
elif source[wanted_type_col] == 'fsrq':
    fsrq_cnt += 1
elif source[wanted_type_col] == 'BLL':
    BLL_cnt += 1
elif source[wanted_type_col] == 'bll':
    bll_cnt += 1
elif source[wanted_type_col] == 'BCU':
    BCU_cnt += 1
elif source[wanted_type_col] == 'bcu':
    bcu_cnt += 1
elif source[wanted_type_col] == 'RDG':
    RDG_cnt += 1
elif source[wanted_type_col] == 'rdg':
    rdg_cnt += 1
```

Prepare data frames for reporting

```
df1 = pd.DataFrame([
    ['AGN', AGN_cnt, agn_cnt, FSRQ_cnt, fsrq_cnt, BLL_cnt, bll_cnt, BCU_cnt, bcu_cnt, RDG_cnt, rdg_cnt, NLSY1_cnt, nlsy1_cnt, YNG_cnt, yng_cnt, ssrq_cnt, sey_cnt],
    columns=[
        'AGN count', 'AGN', 'agn', 'FSRQ', 'fsrq', 'BLL', 'bll', 'BCU', 'bcu', 'RDG', 'rdg', 'NLSY1', 'nlsy1', 'YNG', 'yng', 'ssrq', 'sey'
    ])

df2 = pd.DataFrame([
    ['PSR', PSR_cnt, psr_cnt, MSP_cnt, msp_cnt]],
    columns=[
        'PSR count', 'PSR', 'psr', 'MSP', 'msp'])

df3 = pd.DataFrame([
    ['Total', all_cnt, null_cnt, psr_cnt+msp_cnt+PSR_cnt+MSP_cnt,
    AGN_cnt+agn_cnt+FSRQ_cnt+fsrq_cnt+BLL_cnt+bll_cnt+BCU_cnt+bcu_cnt+RDG_cnt+rdg_cnt+NLSY1_cnt+nlsy1_cnt+YNG_cnt+yng_cnt+ssrq_cnt+sey_cnt,
    unk_cnt, bin_cnt, glc_cnt, sbg_cnt, clust_cnt, spp_cnt, snr_cnt, SNR_cnt, gal_cnt, GAL_cnt, HMB_cnt,
    pwn_cnt, PWN_cnt, BIN_cnt, css_cnt, SFR_cnt, NOV_cnt, bzq_cnt, bzb_cnt, agn_cnt, BZQ_cnt, HXB_cnt, MQO_cnt,
    glb_cnt, bzu_cnt, hxb_cnt,
    ]],
    columns=[
        'Total', 'All', 'Null count', 'PSR', 'AGN', 'UNK', 'BIN', 'GLC', 'SGB', 'CLUST', 'spp', 'snr', 'SNR',
        'gal', 'GAL', 'HMB', 'pwn', 'PWN', 'BIN', 'css', 'SFR', 'NOV', 'bzb', 'bzq', 'agu', 'BZQ', 'HXB', 'MQO',
        'glb', 'bzu', 'hxb'])
```

Create plots

```
# plot grouped bar chart
ax1 = df1.plot(
    x='AGN count',
    kind='bar',
    stacked=False,
    title='AGN count')
for container in ax1.containers:
    ax1.bar_label(container)

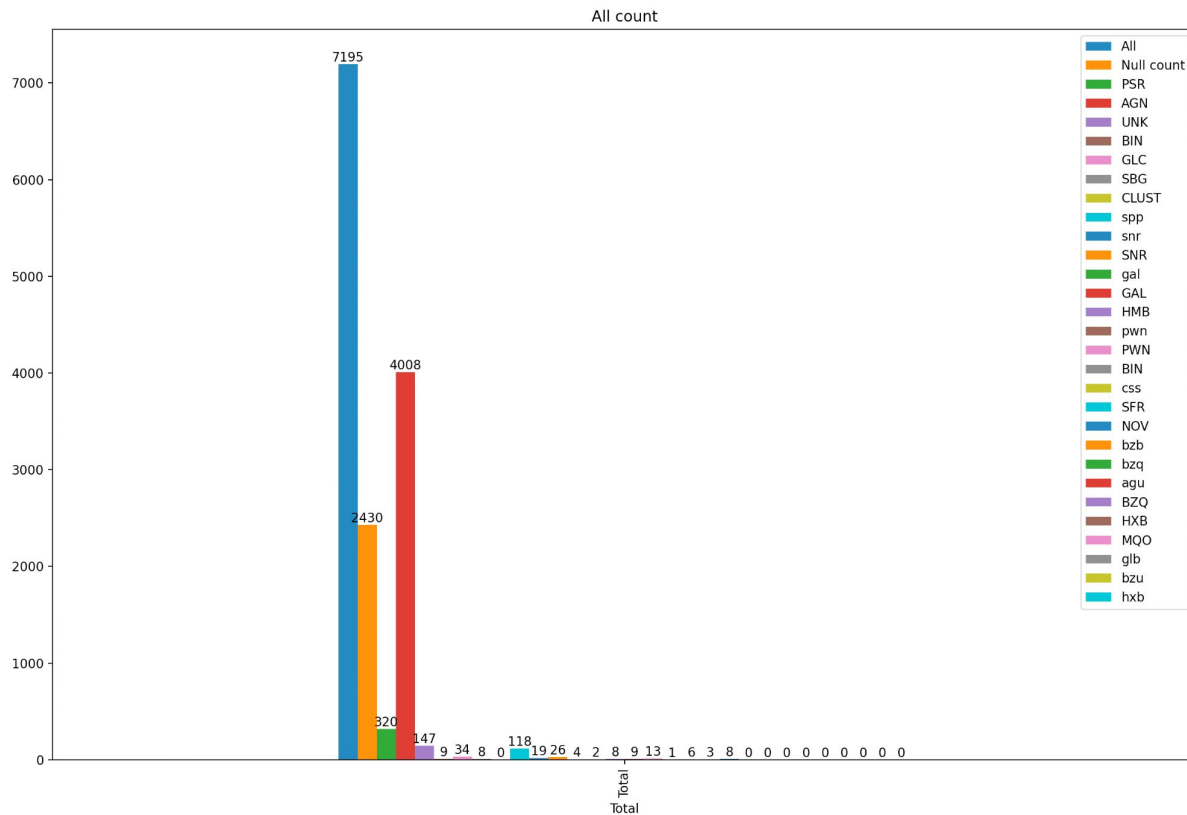
ax2 = df2.plot(
    x='PSR count',
    kind='bar',
    stacked=False,
    title='PSR count')
for container in ax2.containers:
    ax2.bar_label(container)

ax3 = df3.plot(
    x='Total',
    kind='bar',
    stacked=False,
    title='All count')
for container in ax3.containers:
    ax3.bar_label(container)

plt.show()
```

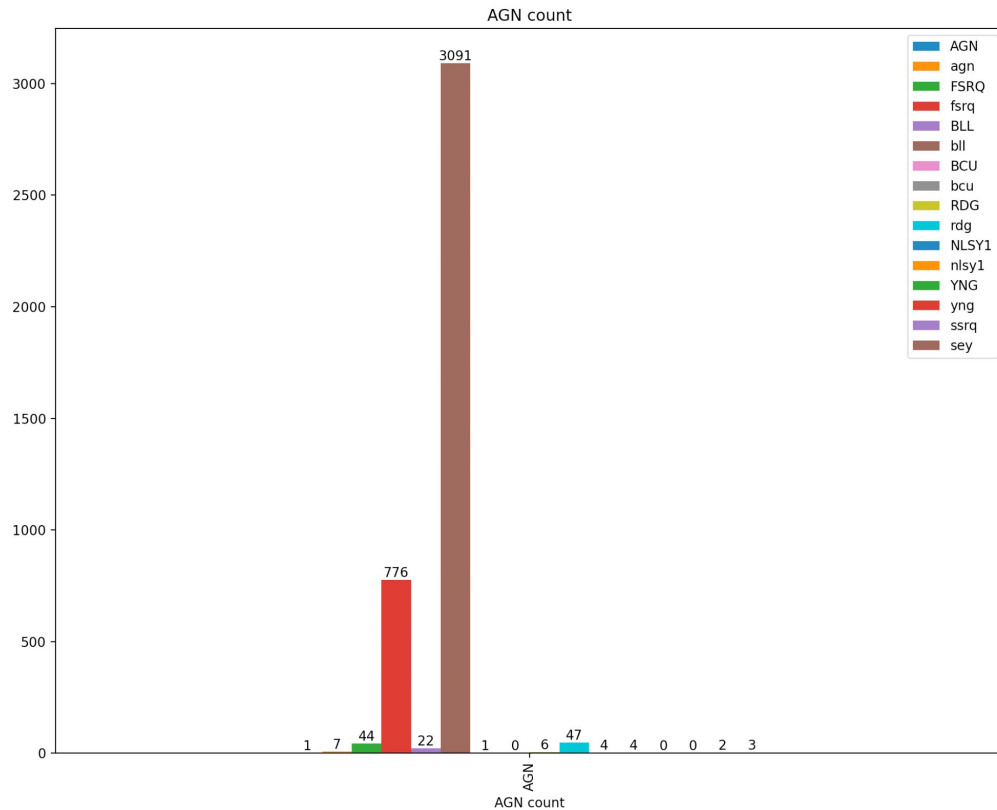
4FGL-DR4

4FGL-DR4 Catalog

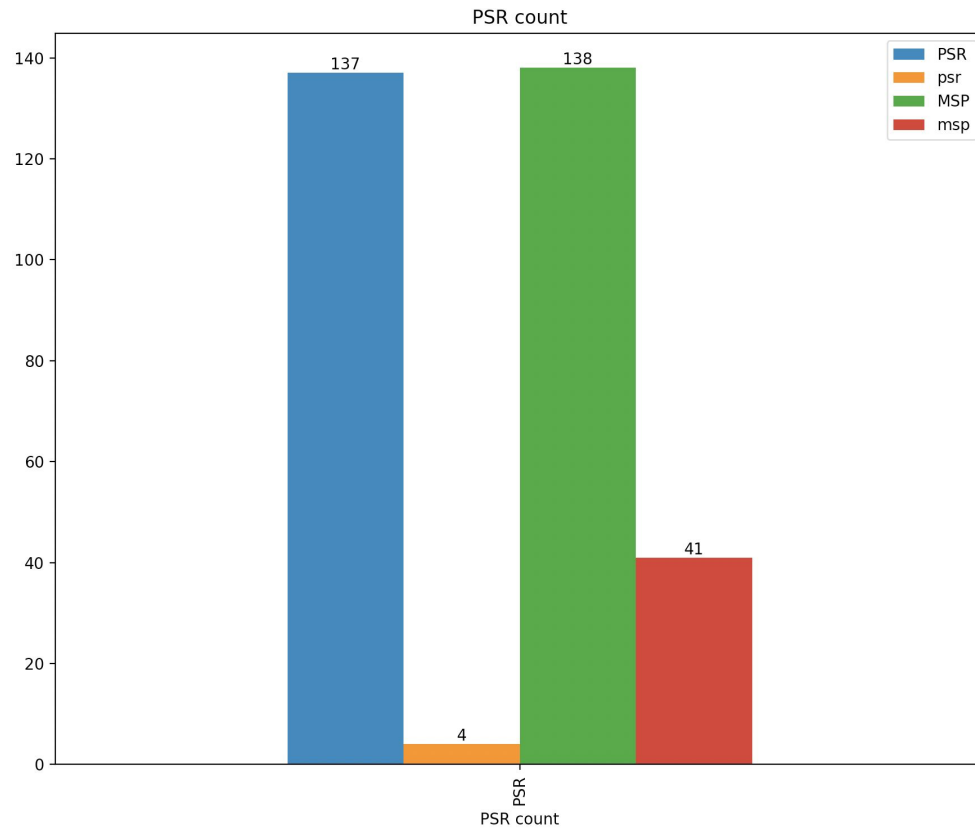


- 7195 Total
- 4008 AGNs
- 320 Pulsars
- 147 UNK
- 2430 Nulls

4FGL-DR4 Catalog

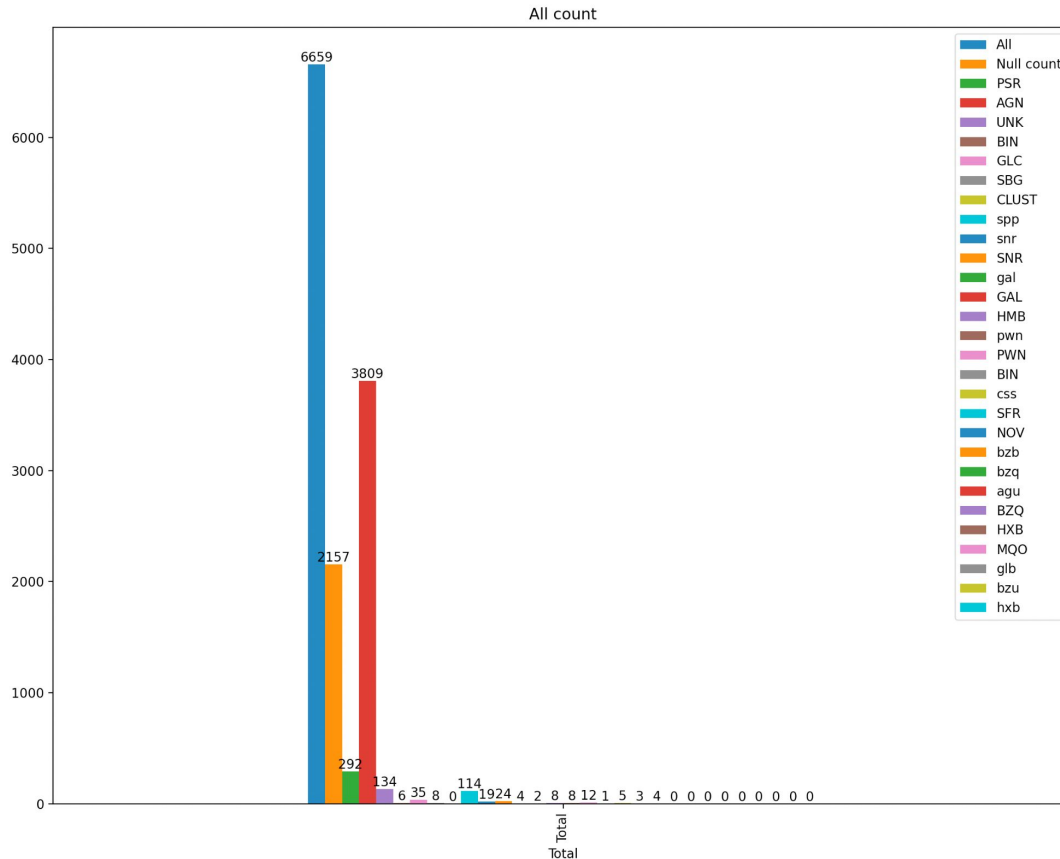


4FGL-DR4 Catalog



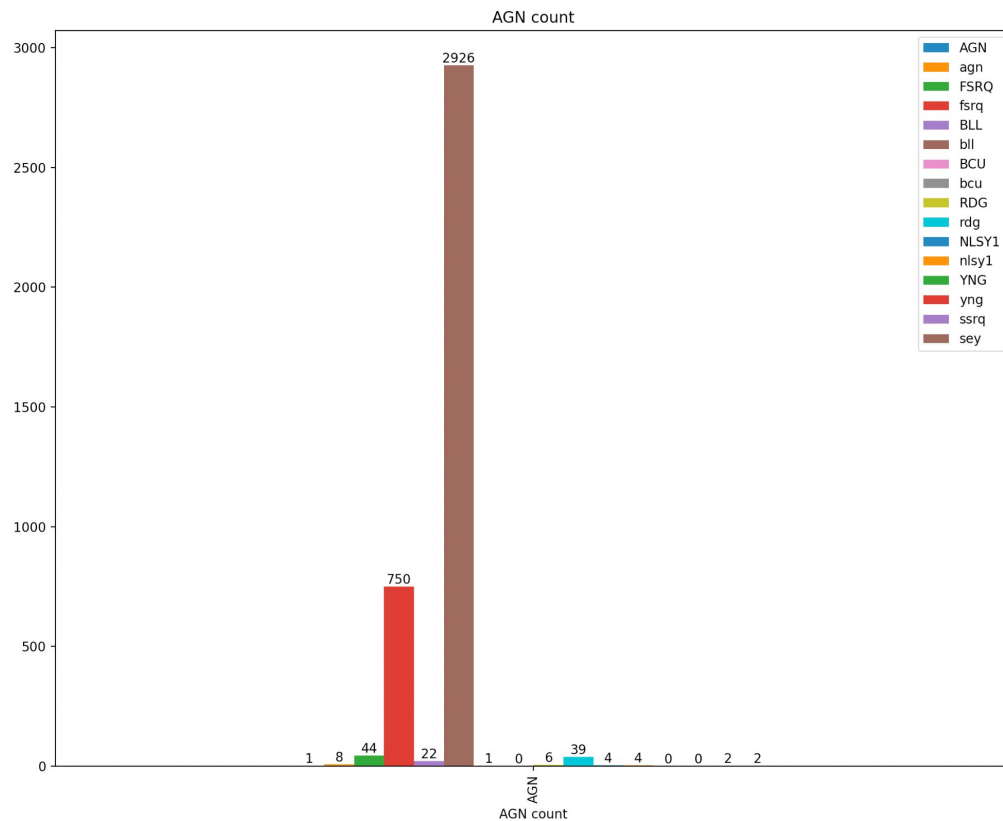
4FGL-DR3

4FGL-DR3 Catalog

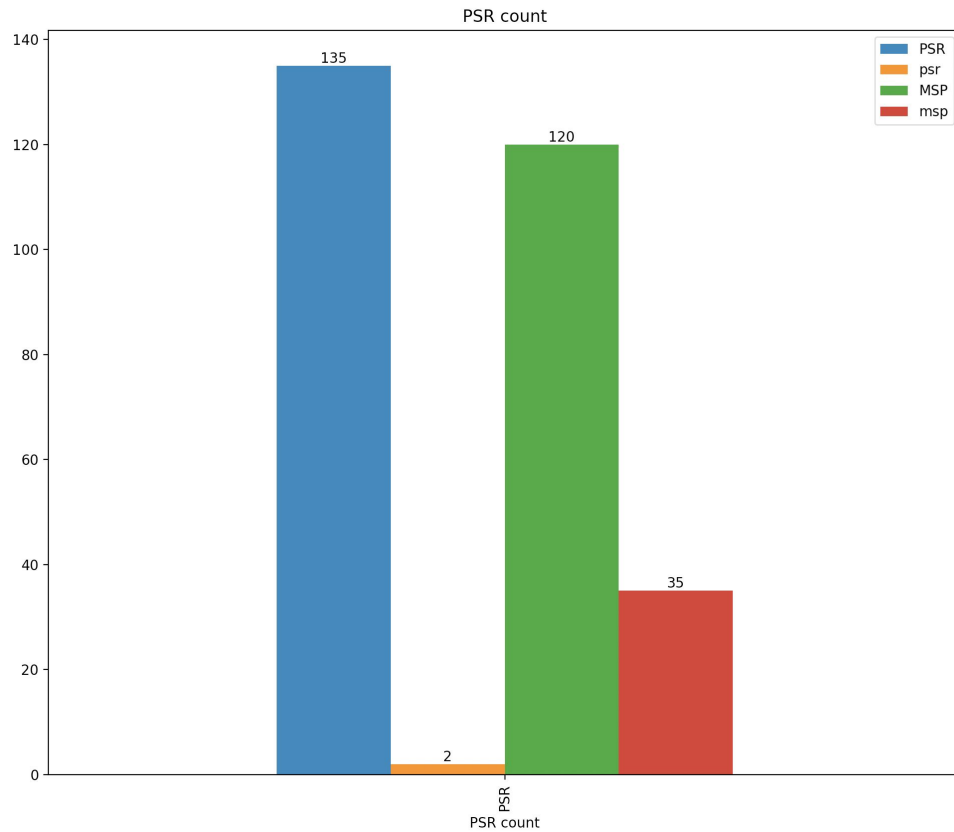


- 6659 Total
- 3809 AGNs
- 292 Pulsars
- 134 UNK
- 2157 Nulls

4FGL-DR3 Catalog

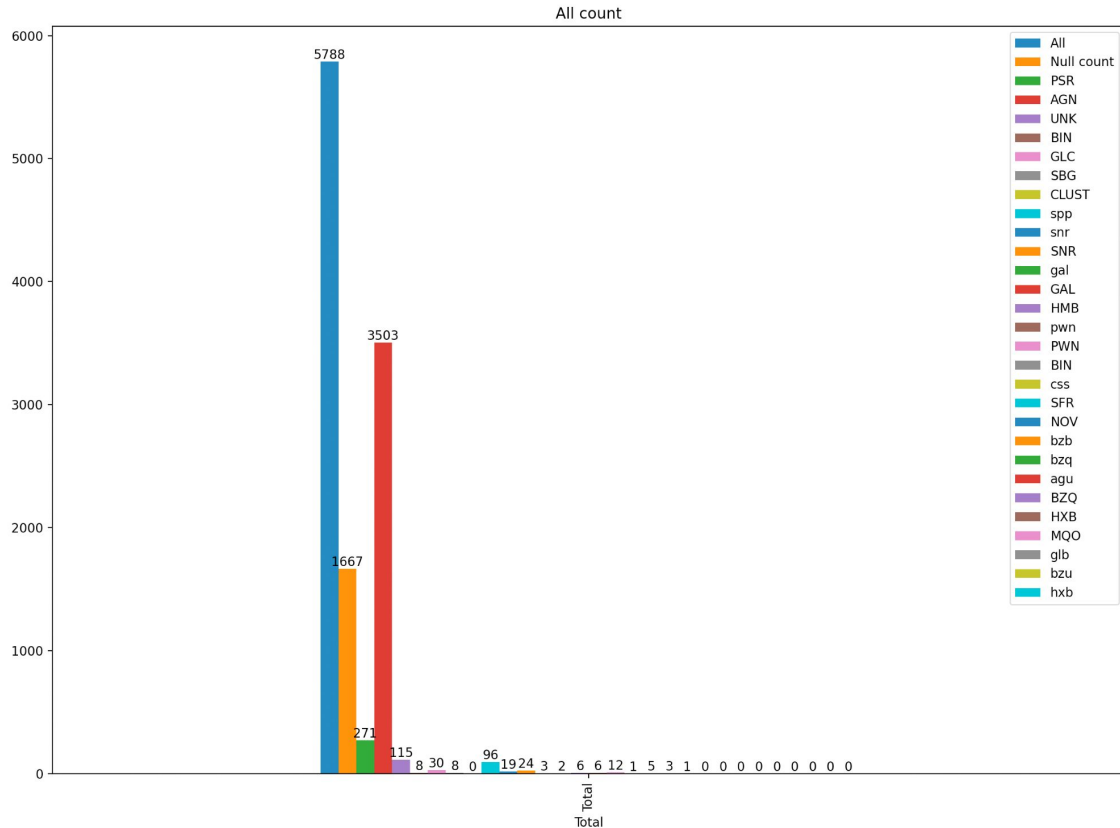


4FGL-DR3 Catalog



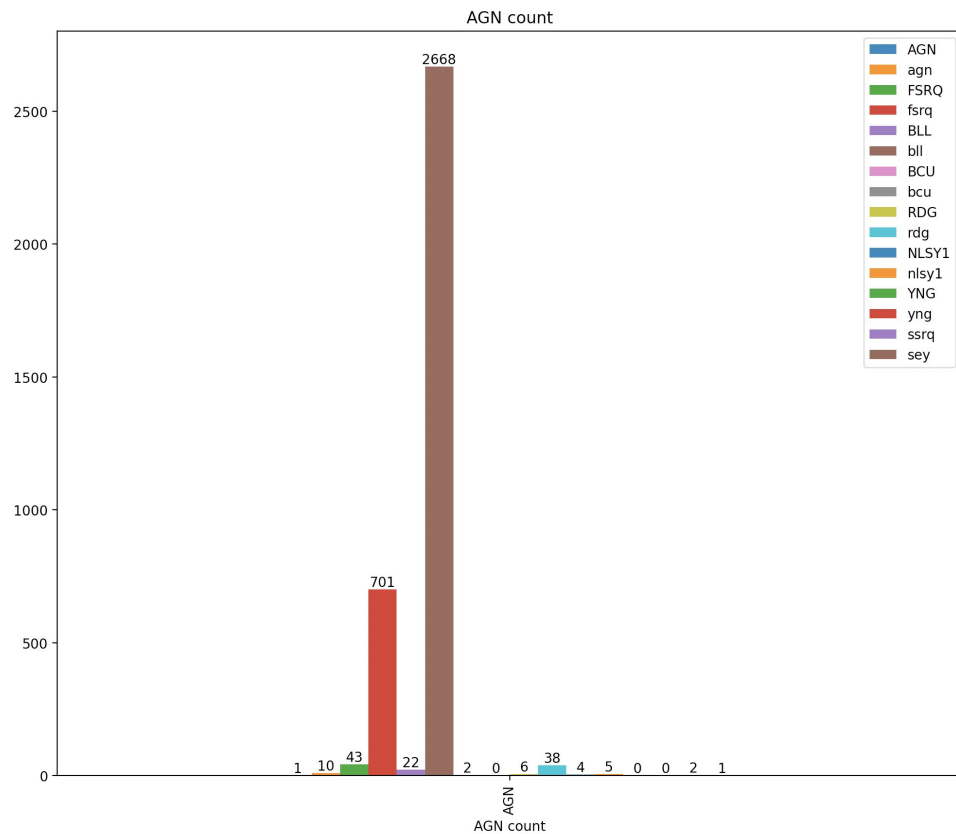
4FGL-DR2

4FGL-DR2 Catalog

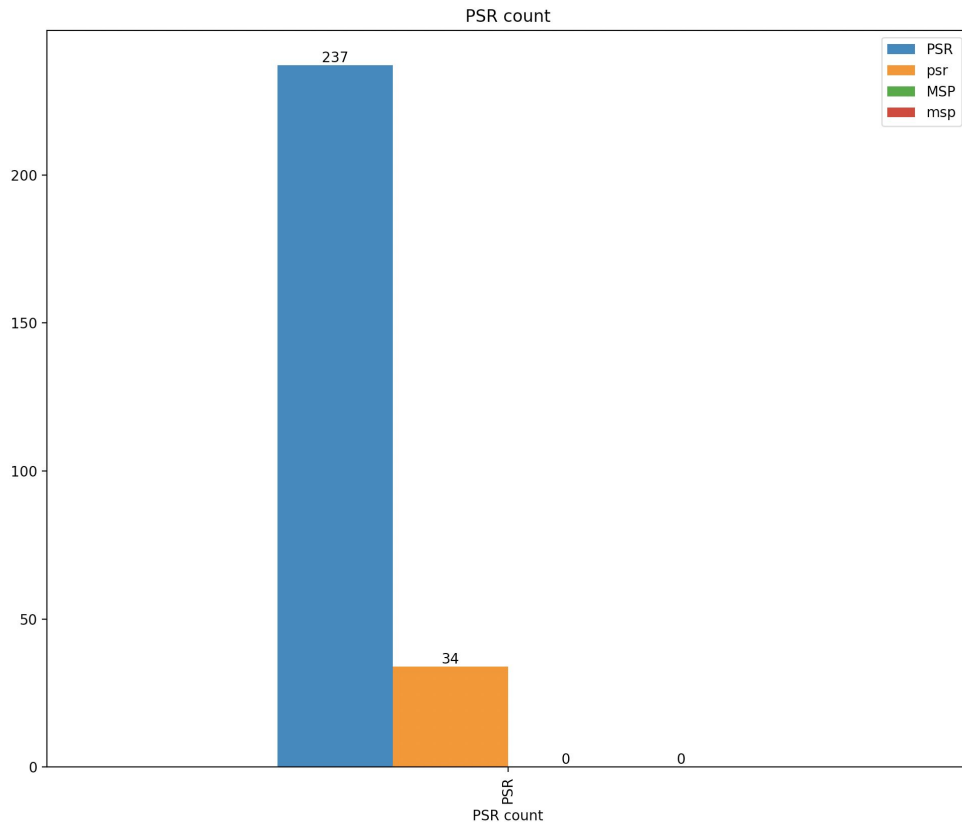


- 5788 Total
- 3503 AGNs
- 271 Pulsars
- 115 UNK
- 1667 Nulls

4FGL-DR2 Catalog

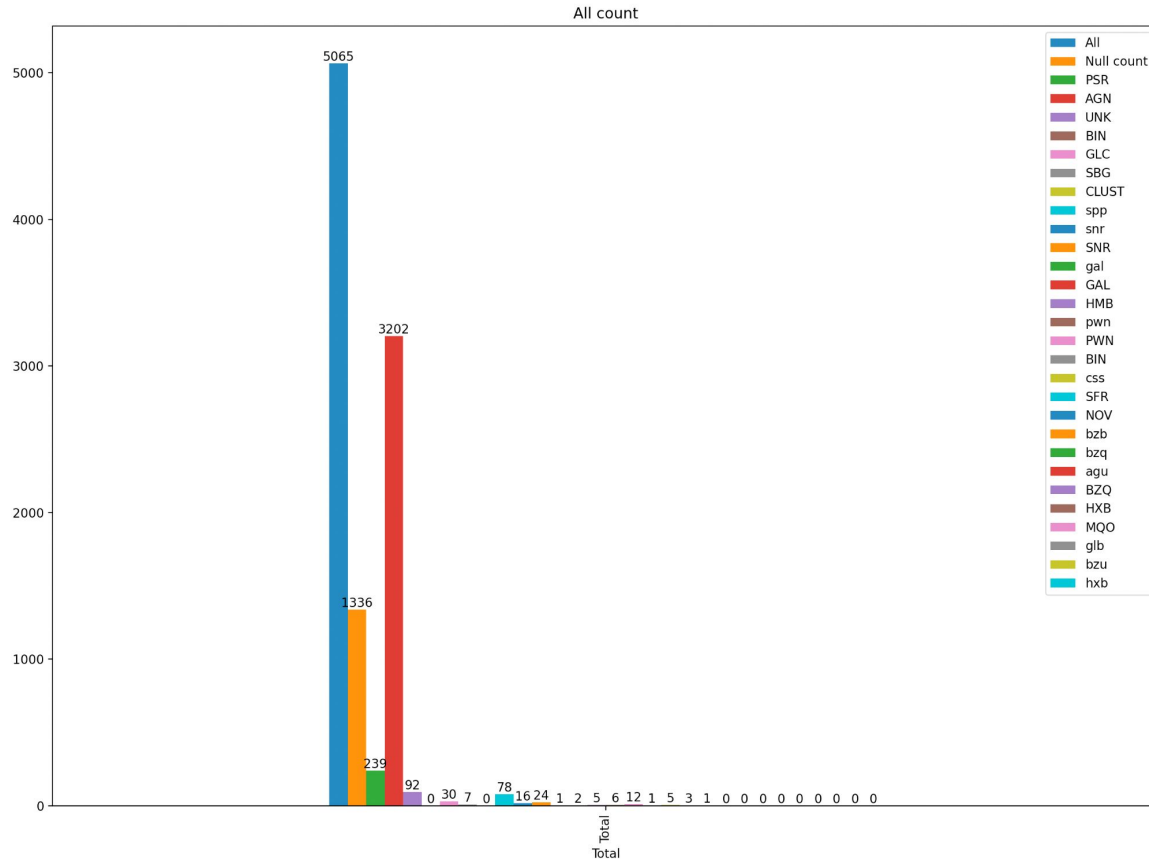


4FGL-DR2 Catalog



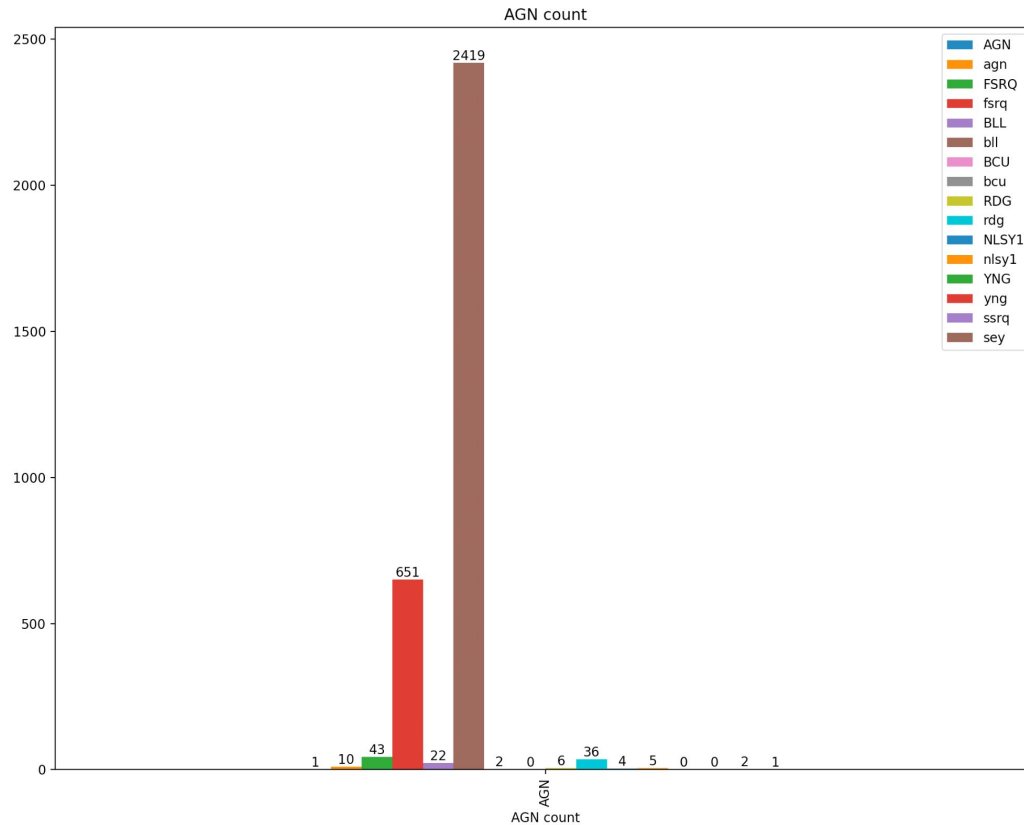
4FGL-DR1

4FGL-DR1 Catalog

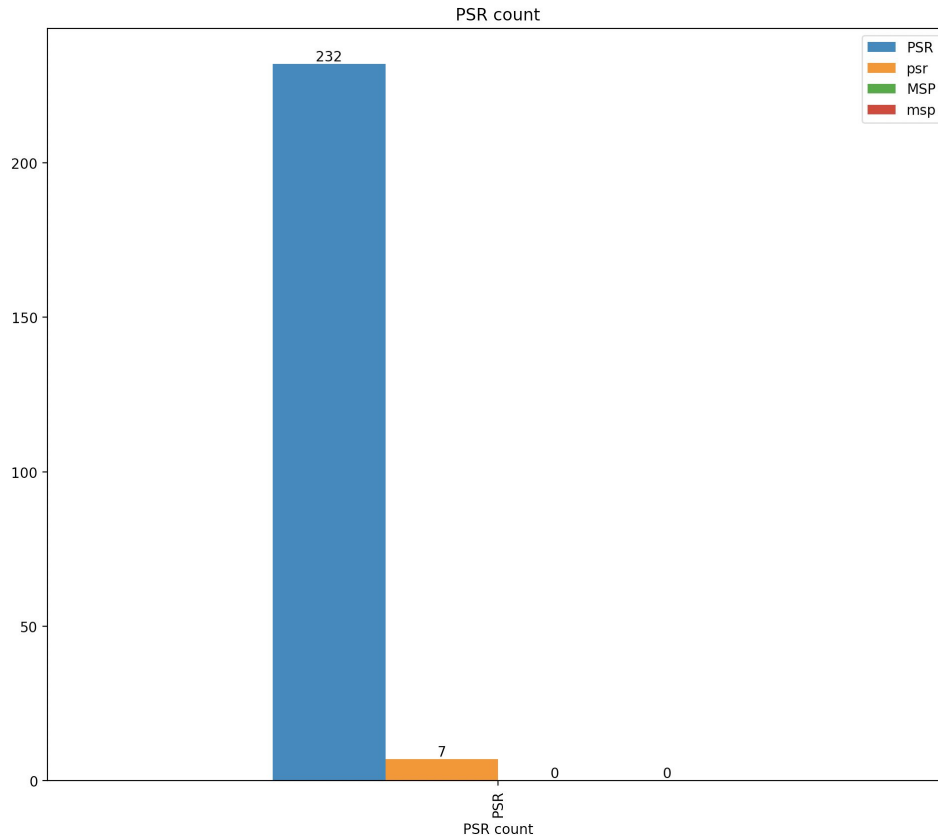


- 5065 Total
- 3202 AGNs
- 239 Pulsars
- 92 UNK
- 1336 Nulls

4FGL-DR1 Catalog

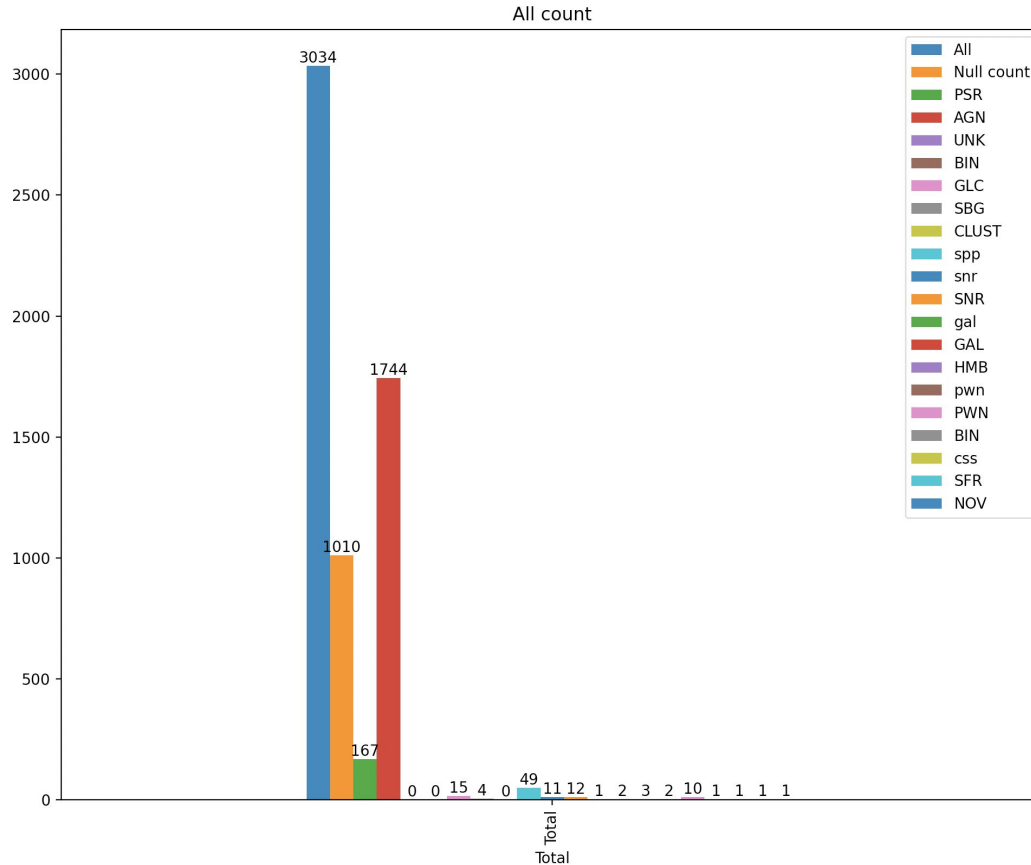


4FGL-DR1 Catalog



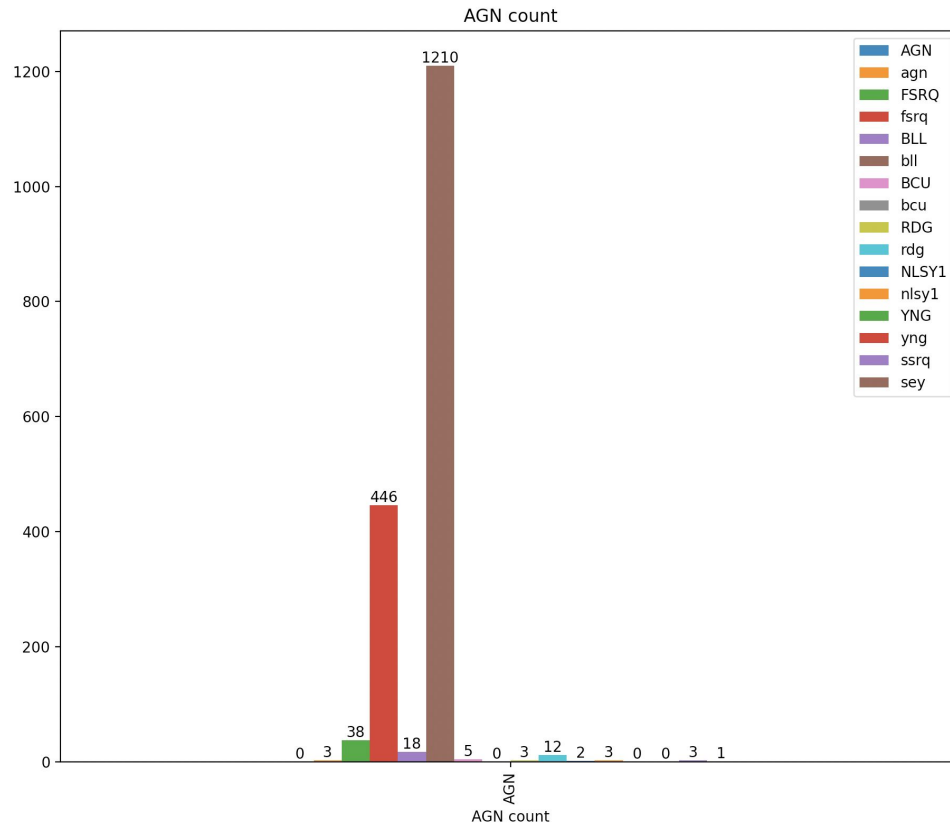
3FGL

3FGL Catalog

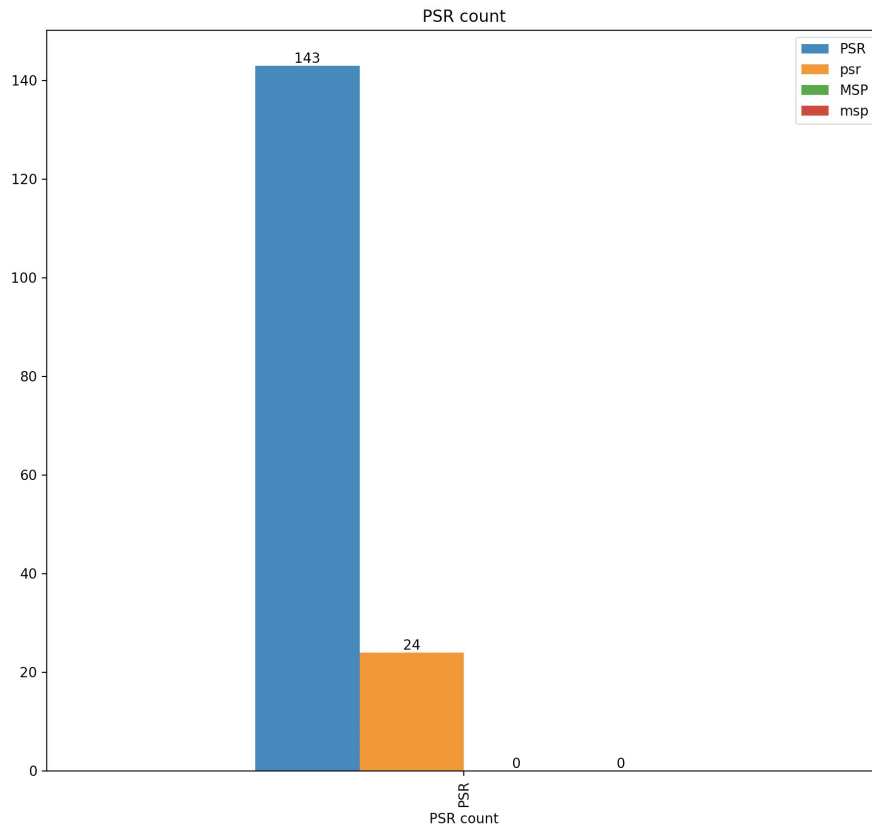


- 3034 Total
- 1744 AGNs
- 167 Pulsars
- 0 UNK
- 1010 Nulls

3FGL Catalog

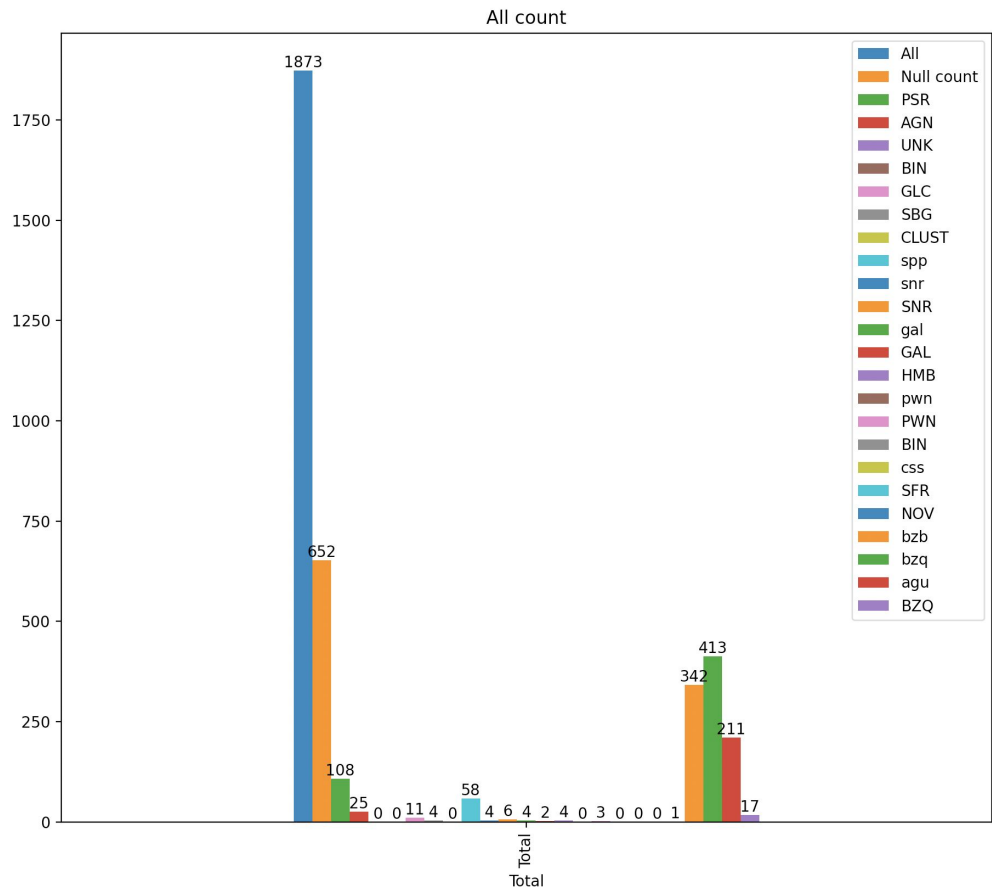


3FGL Catalog



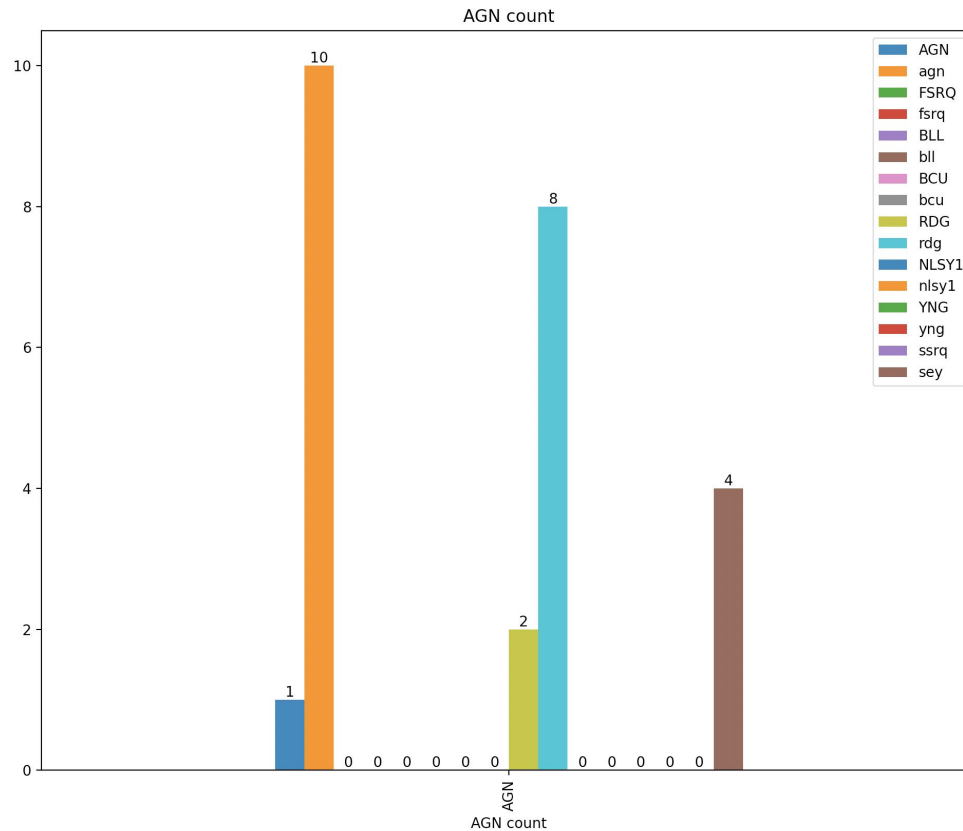
2FGL

2FGL Catalog

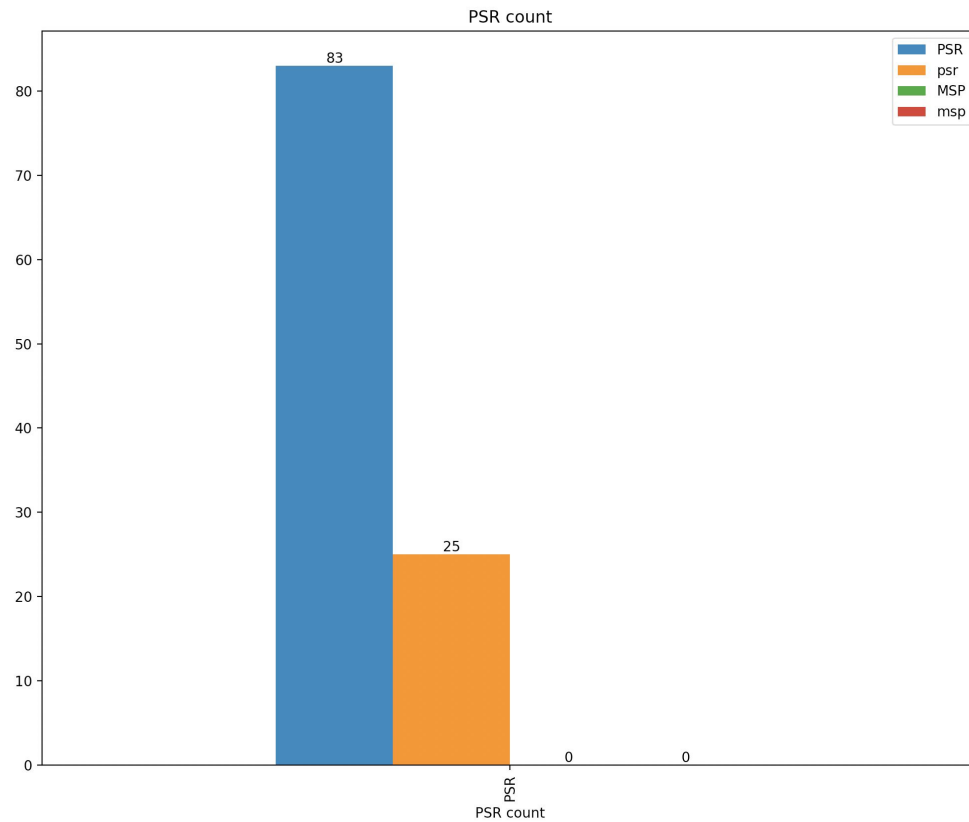


- 1873 Total
- 25 AGNs
- 211 agu
- 342 bzb
- 413 bzq
- 17 BZQ
- 108 Pulsars
- 0 UNK
- 652 Nulls

2FGL Catalog

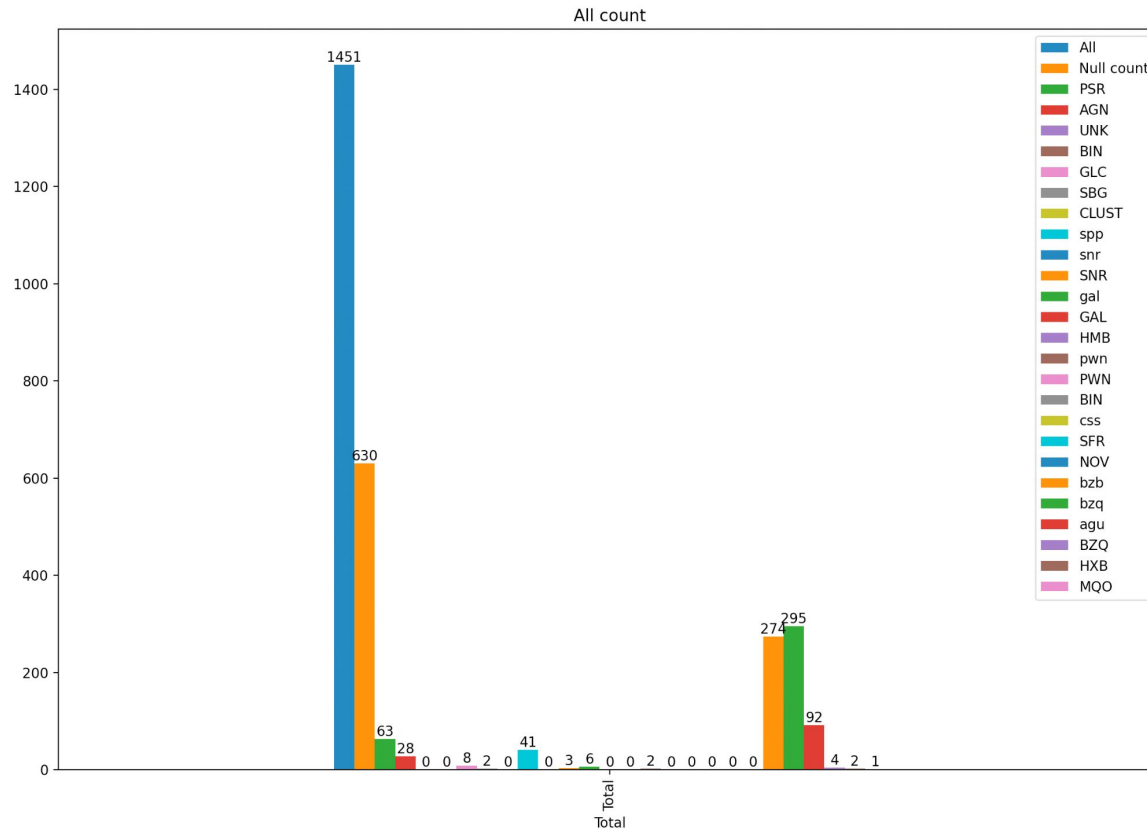


2FGL Catalog



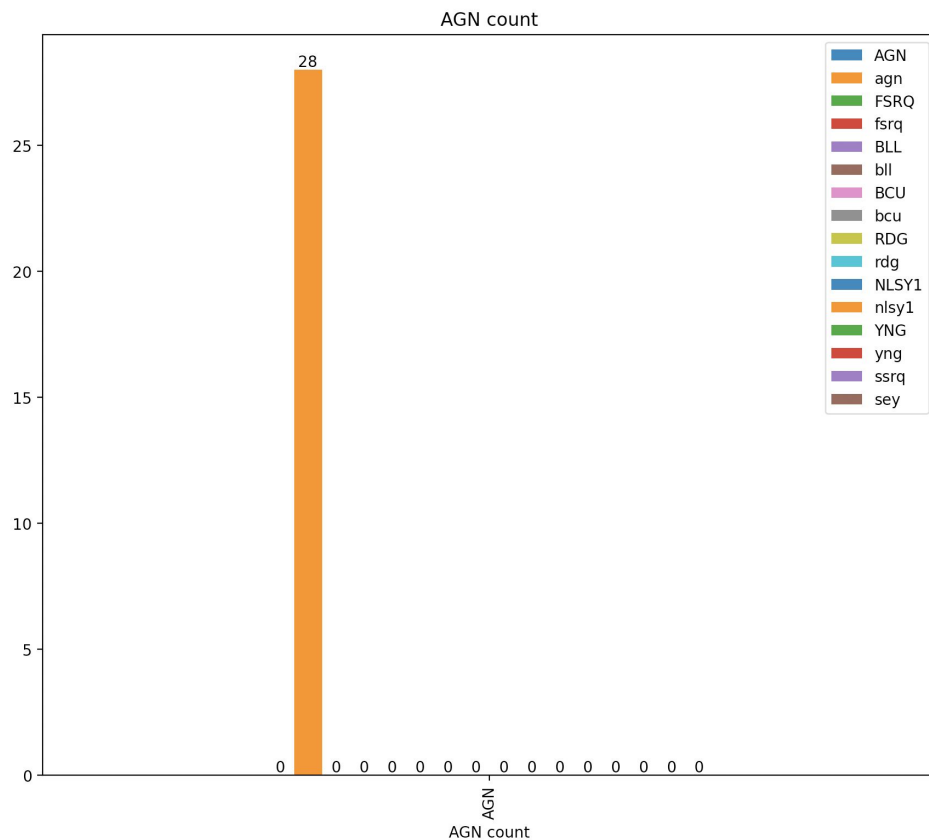
1FGL

1FGL Catalog

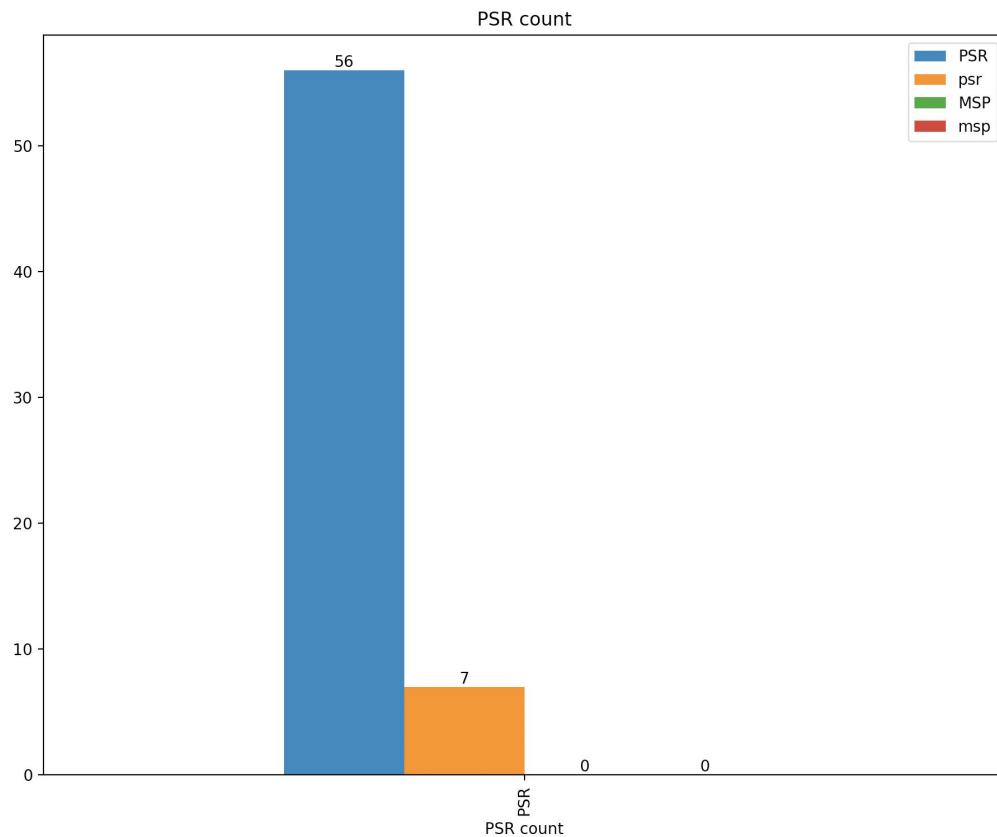


- 1451 Total
- 28 AGNs
- 92 agu
- 274 bzb
- 295 bzq
- 4 BZQ
- 63 Pulsars
- 0 UNK
- 630 Nulls

1FGL Catalog

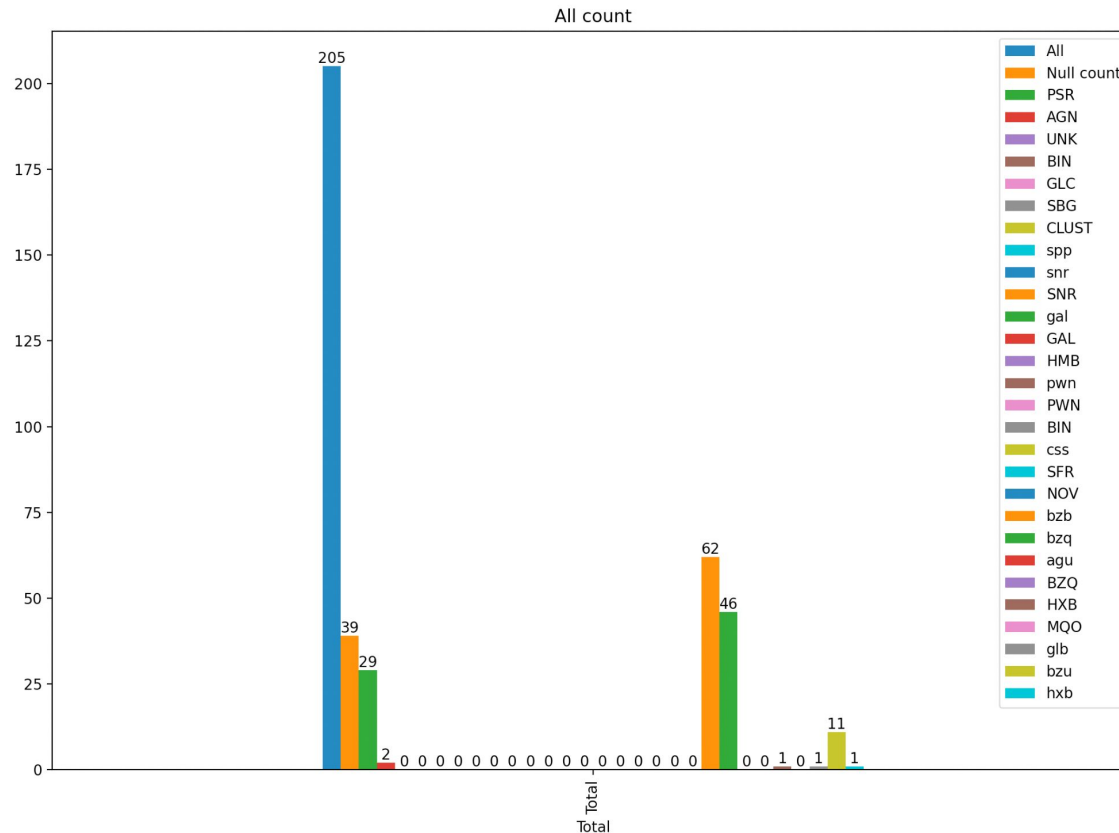


1FGL Catalog



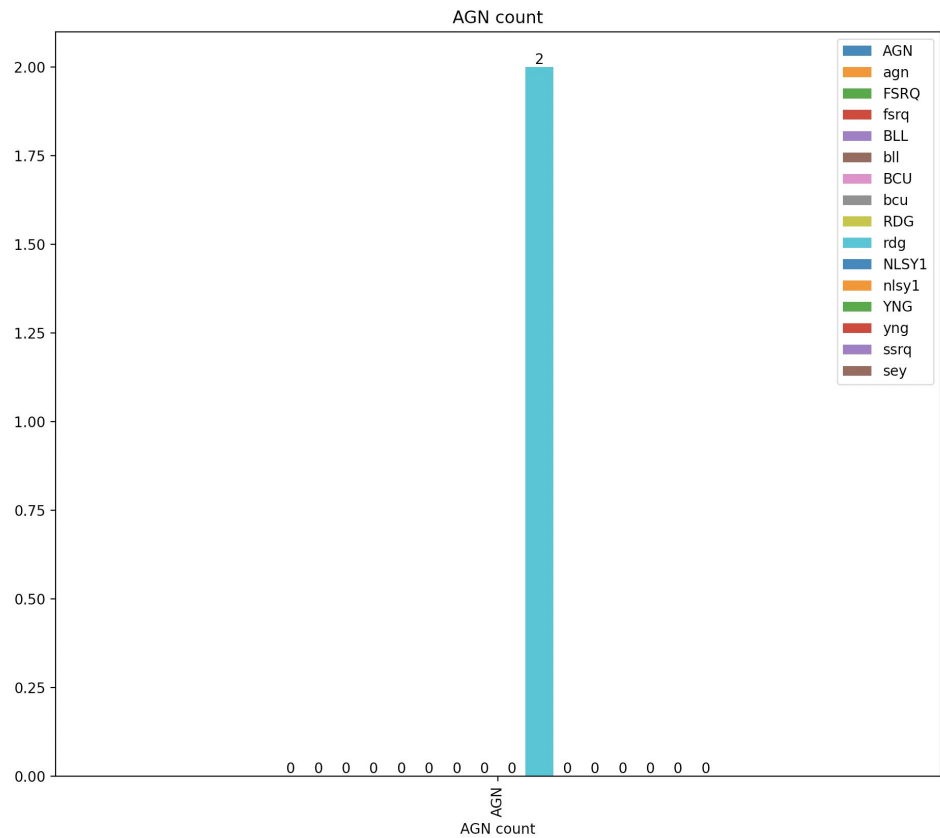
BSL

BSL Catalog

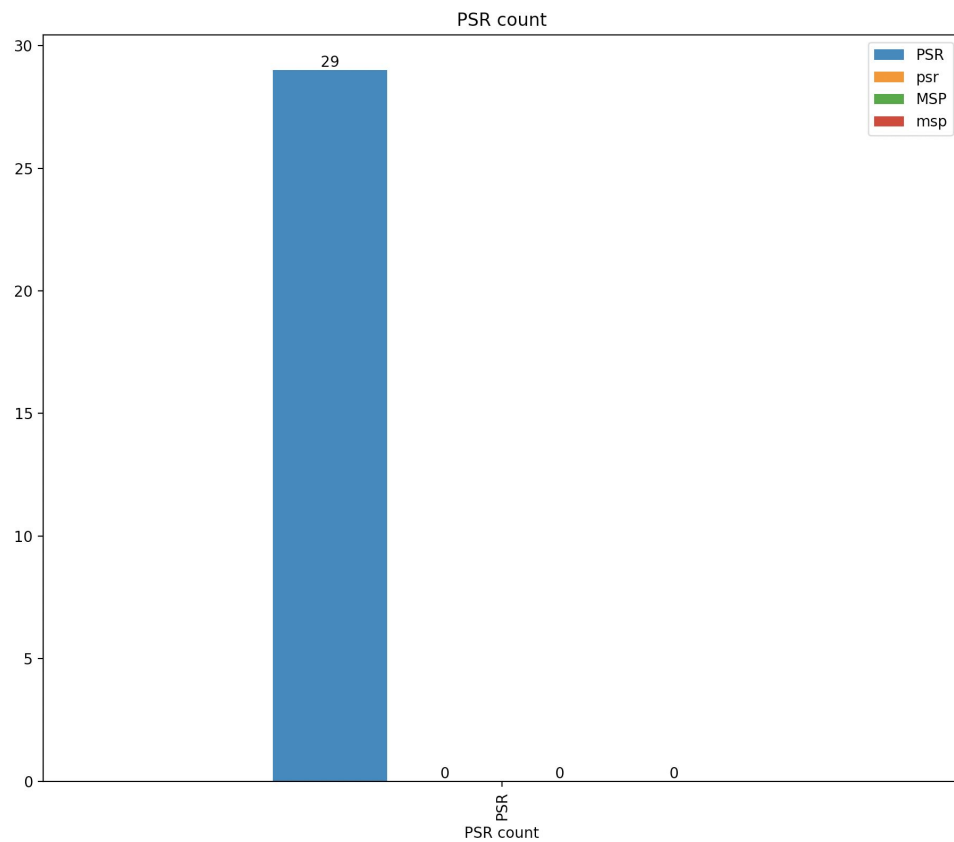


- 205 Total
- 2 AGNs
- 0 agu
- 62 bzb
- 46 bzq
- 11 bzu
- 0 BZQ
- 29 Pulsars
- 0 UNK
- 39 Nulls

BSL Catalog



BSL Catalog



Update: 07/12/2024

Today, I will discuss:

- Progress Summary
- Reading & Processing BSL, 1FGL, 2FGL, 3FGL datasets
- Joining datasets w/ Pandas dataframe merge
 - To join and merge datasets
- Sumamaring results

Progress Summary

- Joined BSL, 1FGL, 2FGL, & 3FGL datasets to get insights on changes since BSL
- Codes:
 - **Reporting/4FGL_Join_Report.py**
 - Reads datasets
 - Calls 4FGL_Dataset_Processing to get columns
 - Joins datasets using Pandas merge
 - **Utils/4FGL_Dataset_Processing.py**
 - Extracts needed columns from the dataset like catalog number, Class1, 0FGL name, 1FGL name, 2FGL name

Read & Process BSL dataset

- Read BSL (0FGL) dataset (Old)

```
# Read BSL file
path = os.path.join(os.getcwd(), "../../FITS/gll_psc3month_BSL_v2.fit")
mainfile = fits.open(path)
point_source_catalogue = mainfile[1]
columns = 0
keymap = {}
mapkey = {}
for key in list(point_source_catalogue.header.keys()):
    if "TTYPE" in key:
        keymap[int(key[5:len(key))]-1] = point_source_catalogue.header[key]
        mapkey[point_source_catalogue.header[key]] = int(key[5:len(key))]-1
        # print(columns, key, point_source_catalogue.header[key])
        columns += 1
keymap = OrderedDict(sorted(keymap.items()))
mapkey = OrderedDict(sorted(mapkey.items()))
```

path

Read using astropy.io.fits library

- Convert to Pandas dataframe

```
# Get BSL dataset
wanted_type_col = 17 # class1
dataset = FGL4Dataset_Process(point_source_catalogue, wanted_type_col, other_cols: [])
bsl_list = dataset.get_all_list()
# Convert into dataframe
bsl_df = pd.DataFrame(bsl_list, columns=["BSL", "CLASS1-BSL"])
# Save into csv
bsl_df.to_csv('../../CSV/bsl_df.csv')
```

Call processing method which returns list

Get Class1 column

Convert list to dataframe

Output dataframe to csv

Process Dataset Method

4 usages Krishiv Bhatia *

```
class FGL4Dataset_Process(Dataset):  
    all_list = []
```

Krishiv Bhatia *

```
def __init__(self, point_source_catalogue, wanted_type_col, other_cols):  
    self.all_list = []  
    for source in point_source_catalogue.data:  
        # source[0] is the catalog name & number like 0FGL J1413.1-6203  
        # source[0].split(' ')[1] gives J1413.1-6203  
        # add catalog number & class1 to lst  
        lst = [source[0].split(' ')[1]]  
        lst.append(source[wanted_type_col])  
        # now add other columns if they are present  
        for col in other_cols:  
            if source[col] is not None and source[col] != '':  
                lst.append(source[col].split()[1])  
            else:  
                lst.append('')  
        self.all_list.append(lst)
```

- Process the dataset read to return required columns as list

Add catalog number
& Class1 to list

Add other columns
passed via method
parameters to list

3 usages Krishiv Bhatia

```
def get_all_list(self):  
    return self.all_list
```

Read & Process 1FGL dataset

- Read 1FGL dataset

```
path = os.path.join(os.getcwd(), "../../FITS/gll_psc_v03_1FGL.fits")
mainfile = fits.open(path)
point_source_catalogue = mainfile[1]
columns = 0
keymap = {}
mapkey = {}
for key in list(point_source_catalogue.header.keys()):
    if "TTYPE" in key:
        keymap[int(key[5:len(key))]-1] = point_source_catalogue.header[key]
        mapkey[point_source_catalogue.header[key]] = int(key[5:len(key))]-1
        # print(columns, key, point_source_catalogue.header[key])
        columns += 1
```

Read using
astropy.io.fits library

- Convert into Pandas dataframe

```
# col 49 is 0FGL_Name: 78 TTYPE1 0FGL_Name
wanted_type_col = 54 # Class1
dataset = FGL4Dataset_Process(point_source_catalogue, wanted_type_col, other_cols: [49])
fgl1_lst = dataset.get_all_list()
fgl1_df = pd.DataFrame(fgl1_lst, columns = ["1FGL", "CLASS1-1FGL", "1FGL-BSL"])
fgl1_df.to_csv('../../CSV/fgl1_df.csv')
```

Call processing
method which
returns list

Get Class1 & 0FGL
name columns

Output
dataframe to csv

Convert list
to dataframe

Read & Process 2FGL dataset

- Read 2FGL dataset

```
columns = 0
keymap = {}
mapkey = {}
path = os.path.join(os.getcwd(), '../FITS/gll_psc_v09_2FGL.fit')
mainfile = fits.open(path)
point_source_catalogue = mainfile[1]
for key in list(point_source_catalogue.header.keys()):
    if "TTYPE" in key:
        keymap[int(key[5:len(key)])-1] = point_source_catalogue.header[key]
        mapkey[point_source_catalogue.header[key]] = int(key[5:len(key)])-1
        # print(columns, key, point_source_catalogue.header[key])
        columns += 1
```

Read using
astropy.io.fits library

- Convert into Pandas dataframe

Call processing
method which
returns list

Get Class1, 0FGL,
& 1FGL name
columns

```
# 145 TTYPE57 0FGL_Name, 146 TTYPE58 1FGL_Name
wanted_type_col = 63 # class1
dataset = FGL4Dataset_Process(point_source_catalogue, wanted_type_col, other_cols=[56,57])
fgl2_lst = dataset.get_all_list()
fgl2_df = pd.DataFrame(fgl2_lst, columns=["2FGL", "CLASS1-2FGL", "2FGL-BSL", "2FGL-1FGL"])
fgl2_df.to_csv('../CSV/fgl2_df.csv')
```

Output
dataframe to csv

Convert list
to dataframe

Merge/Join BSL, 1FGL, & 2FGL datasets

Join BSL and 1FGL
datasets using
pandas.merge

Left join on BSL column.
This will capture all rows
from BSL dataset but only
matching rows from 1FGL.

```
print("=====Merge-BSL-1FGL=====")
joined_BSL_1FGL = pd.merge(bsl_df, fgl1_df, left_on='BSL', right_on='1FGL-BSL', how='left')
joined_BSL_1FGL.to_csv('../..CSV/joined_BSL_1FGL.csv')
joined_BSL_1FGL_None_CLS1BSL = joined_BSL_1FGL[joined_BSL_1FGL["CLASS1-BSL"]=='']
joined_BSL_1FGL_None_CLS1BSL.to_csv('../..CSV/joined_BSL_1FGL_NoneCLS1_BSL.csv')

print("=====Merge BSL, 1FGL, 2FGL=====")
joined_BSL_1FGL_2FGL = pd.merge(joined_BSL_1FGL, fgl2_df, left_on='BSL', right_on='2FGL-BSL', how='left')
joined_BSL_1FGL_2FGL.to_csv('../..CSV/joined_BSL_1FGL_2FGL.csv')
joined_BSL_1FGL_2FGL_None_CLS1_BSL = joined_BSL_1FGL_2FGL[joined_BSL_1FGL_2FGL["CLASS1-BSL"]=='']
joined_BSL_1FGL_2FGL_None_CLS1_BSL.to_csv('../..CSV/joined_BSL_1FGL_2FGL_NoneCLS1_BSL.csv')
```

Join BSL, 1FGL
combined dataset
with 2FGL using
pandas.merge

Left join on BSL column.
This will capture all rows
from BSL dataset but only
matching rows from 2FGL.

Output CSV files

  main FermiLAT / CSV 		Go to file 	
 krishivbhatia for joining datasets and reporting		31a	
Name		Last commit message	
 ..			
 <code>__init__.py</code>		for joining datasets and reporting	
 <code>bsl_df.csv</code>		for joining datasets and reporting	
 <code>fgl1_df.csv</code>		for joining datasets and reporting	
 <code>fgl2_df.csv</code>		for joining datasets and reporting	
 <code>joined_BSL_1FGL.csv</code>		for joining datasets and reporting	
 <code>joined_BSL_1FGL_2FGL.csv</code>		for joining datasets and reporting	
 <code>joined_BSL_1FGL_2FGL_NONE_CS1_BSL.csv</code>		for joining datasets and reporting	
 <code>joined_BSL_1FGL_2FGL_upd.numbers</code>		for joining datasets and reporting	
 <code>joined_BSL_1FGL_2FGL_upd.xlsx</code>		for joining datasets and reporting	
 <code>joined_BSL_1FGL_NoneCLS1_BSL.csv</code>		for joining datasets and reporting	

Joined BSL, 1FGL, 2FGL datasets

1	BSL		1FGL			2FGL					
2	BSL Name	Cls1	1FGL Name	Cls1	BSL Name	2FGL Name	Cls1	BSL Name	1FGL Name		
3	J0007.4+7303	PSR	J0007.0+7303	PSR	J0007.4+7303	J0007.0+7303	PSR	J0007.4+7303	J0007.0+7303		No change
4	J0017.4-0503	bzq	J0017.4-0510	bzq	J0017.4-0503	J0017.6-0510	bzq	J0017.4-0503	J0017.4-0510		Change in source classification
5	J0025.1-7202	glb	J0023.9-7204	glc	J0025.1-7202	J0023.9-7204	glc	J0025.1-7202	J0023.9-7204		Source unidentified/unassociated throughout
6	J0030.3+0450	PSR	J0030.4+0451	PSR	J0030.3+0450	J0030.4+0450	PSR	J0030.3+0450	J0030.4+0451		Source later unidentified/unassociated/dropped (but maybe reassociated/identified later)
7	J0033.6-1921	bzb	J0033.5-1921	bzb	J0033.6-1921	J0033.5-1921	bzb	J0033.6-1921	J0033.5-1921		Source later identified/associated
8	J0036.7+5951	bzb	J0035.9+5951	bzb	J0036.7+5951	J0035.8+5951	bzb	J0036.7+5951	J0035.9+5951		
9	J0050.5-0928	bzb	J0050.6-0928	bzb	J0050.5-0928	J0050.6-0929	bzb	J0050.5-0928	J0050.6-0928		
10	J0051.1-0647	bzq	J0051.1-0649	bzq	J0051.1-0647	J0051.0-0648	bzq	J0051.1-0647	J0051.1-0649		
11	J0100.2+0750	bzu	J0100.2+0747	agu	J0100.2+0750	J0100.2+0746	bzb	J0100.2+0750	J0100.2+0747		
12	J0112.1+2247	bzb	J0112.0+2247	bzb	J0112.1+2247	J0112.1+2245	bzb	J0112.1+2247	J0112.0+2247		
13	J0118.7-2139	bzq	J0118.7-2137	bzq	J0118.7-2139	J0118.8-2142	bzq	J0118.7-2139	J0118.7-2137		
14	J0120.5-2703	bzb	J0120.5-2700	bzb	J0120.5-2703	J0120.4-2700	bzb	J0120.5-2703	J0120.5-2700		
15	J0136.6+3903	bzb	J0136.5+3905	bzb	J0136.6+3903	J0136.5+3905	bzb	J0136.6+3903	J0136.5+3905		
16	J0137.1+4751	bzq	J0137.0+4751	bzq	J0137.1+4751	J0136.9+4751	bzq	J0137.1+4751	J0137.0+4751		
17	J0144.5+2709	bzb	J0144.6+2703	bzb	J0144.5+2709	J0144.6+2704	bzb	J0144.5+2709	J0144.6+2703		
18	J0145.1-2728	bzq	J0144.9-2732	bzq	J0145.1-2728	J0145.1-2732	bzq	J0145.1-2728	J0144.9-2732		
19	J0204.8-1704	bzq	J0205.0-1702	bzq	J0204.8-1704	J0205.3-1657	bzq	J0204.8-1704	J0205.0-1702		
20	J0210.8-5100	bzq	J0210.6-5101	bzb	J0210.8-5100	J0210.7-5102	agu	J0210.8-5100	J0210.6-5101		
21	J0217.8+0146	bzq	J0217.9+0144	bzq	J0217.8+0146	J0217.9+0143	bzq	J0217.8+0146	J0217.9+0144		
22	J0220.9+3607	bzu	J0221.0+3555	bzq	J0220.9+3607	J0221.0+3555	bzq	J0220.9+3607	J0221.0+3555		
23	J0222.6+4302	bzb	J0222.6+4302	bzb	J0222.6+4302	J0222.6+4302	BZB	J0222.6+4302	J0222.6+4302		
24	J0229.5-3640	bzq	J0229.3-3644	bzq	J0229.5-3640	J0229.3-3644	bzq	J0229.5-3640	J0229.3-3644		
25	J0238.4+2855	bzq	J0237.9+2848	bzq	J0238.4+2855	J0237.8+2846	bzq	J0238.4+2855	J0237.9+2848		
26	J0238.6+1636	bzb	J0238.6+1637	bzb	J0238.6+1636	J0238.7+1637	BZB	J0238.6+1636	J0238.6+1637		
27	J0240.3+6113	HXB	J0240.5+6113	HXB	J0240.3+6113	J0240.5+6113	HMB	J0240.3+6113	J0240.5+6113		
28	J0245.6-4656	bzu	J0245.9-4652	bzq	J0245.6-4656	J0245.9-4652	bzq	J0245.6-4656	J0245.9-4652		
29	J0303.7-2410	bzb	J0303.5-2406	bzb	J0303.7-2410	J0303.4-2407	bzb	J0303.7-2410	J0303.5-2406		
30	J0320.0+4131	rdg	J0319.7+4130	agn	J0320.0+4131	J0319.8+4130	rdg	J0320.0+4131	J0319.7+4130		
31	J0334.1-4006	bzb	J0334.2-4010	bzb	J0334.1-4006	J0334.2-4008	bzb	J0334.1-4006	J0334.2-4010		
32	J0349.8-2102	bzq	J0349.9-2104	bzq	J0349.8-2102	J0350.0-2104	bzq	J0349.8-2102	J0349.9-2104		
33	J0357.5+3205	PSR	J0357.8+3205	PSR	J0357.5+3205	J0357.8+3205	PSR	J0357.5+3205	J0357.8+3205		
34	J0407.6-3829	bzu	J0407.4-3827	bzq	J0407.6-3829	J0407.3-3826	bzq	J0407.6-3829	J0407.4-3827		
35	J0412.9-5341	bzu	J0413.4-5334	bzq	J0412.9-5341	J0413.5-5332	bzq	J0412.9-5341	J0413.4-5334		
36	J0423.1-0112	bza	J0423.2-0118	bza	J0423.1-0112	J0423.2-0120	BZQ	J0423.1-0112	J0423.2-0118		

Joined BSL, 1FGL, 2FGL datasets

Green rows	109	No change
Yellow rows	36	Change in source classification
Blue rows	15	Source unidentified/unassociated throughout
Red rows	20	Source unidentified/unassociated/dropped (but maybe reassociated/identified later)
Gray rows	25	Source later identified/associated
Total rows	205	

Joined BSL, 1FGL, 2FGL datasets

Yellow rows: Change in source classification

Total: 36 rows

BSL	1FGL	2FGL	Count
glb	glc	glc	1
bzb	bzb	BZB	6
bzq	bzq	bzb	1
bzq	bzq	BZQ	6
bzq	BZQ	BZQ	4
bzq	agn	SEY	1
bzq	agu	bzq	1
bzq	bzb	agu	1

BSL	1FGL	2FGL	Count
x	SNR	SNR	1
x	PSR	PSR	1
bzu	agu	bzb	3
bzu	agu	bzq	2
bzu	bzq	bzq	4
HXB	HXB	HMB	1
hxb	HXB	HMB	1
rdg	agn	rdg	1
rdg	agn	RDG	1

Joined BSL, 1FGL, 2FGL datasets

Grey rows: Source later identified/associated

BSL	1FGL	2FGL	Count
		PSR	5
		psr	1
		spp	1
		agu	1
	agu	bzq	2
	agu	bzb	3
	agu	agu	1

Total: 25 rows

BSL	1FGL	2FGL	Count
	PSR	PST	3
	spp	SNR	1
x	SNR	SNR	1
x	spp	HMB	1
x	spp	SNR	1
x		spp	2
x		snr	1
x	psr	PSR	1

Joined BSL, 1FGL, 2FGL datasets

Red rows: Source later unidentified/unassociated/dropped
(but maybe reassociated/identified later)

Total: 20 rows

BSL	1FGL	2FGL	Count
	gal	No entry	1
bzb		bzb	1
	No entry		2
	No entry	No entry	3
	agu	No entry	1
bzb	No entry	No entry	1
bzu		agu	1
bzq	No entry	BZQ	1

BSL	1FGL	2FGL	Count
PSR	No entry	No entry	3
PSR	PSR	No entry	1
x	spp	No entry	1
x	SNR	No entry	1
		No entry	1
PSR	No entry	PSR	1
bzu		PSR	1

Joined BSL, 1FGL, 2FGL datasets

Blank in Class1

Blue rows	13	Source unidentified/unassociated throughout
Red rows	8	Source unidentified/unassociated/dropped (but maybe reassociated/identified later)
Gray rows	18	Source later identified/associated
Total rows	39	Blank in Class1

Joined BSL, 1FGL, 2FGL datasets

“x” in Class1

Blue rows	2	Source unidentified/unassociated throughout
Red rows	2	Source unidentified/unassociated/dropped (but maybe reassociated/identified later)
Gray rows	7	Source later identified/associated
Yellow rows	2	Change in source classification
Total rows	13	“x” in Class1

Added 3FGL also

- Read 3FGL dataset

```
columns = 0
keymap = {}
mapkey = {}
path = os.path.join(os.getcwd(), '../FITS/gl_psc_v16.fits')
mainfile = fits.open(path)
point_source_catalogue = mainfile[1]
for key in list(point_source_catalogue.header.keys()):
    if "CTYPE" in key:
        keymap[int(key[5:len(key))]-1] = point_source_catalogue.header[key]
        mapkey[point_source_catalogue.header[key]] = int(key[5:len(key))]-1
        print(columns, key, point_source_catalogue.header[key])
        columns += 1
```

Read using
astropy.io.fits library

- Convert into Pandas dataframe

```
# TTYPE65 0FGL_Name, TTYPE66 1FGL_Name, TTYPE67 2FGL_Name
wanted_type_col = 73 # class1
dataset = FGL4Dataset_Process(point_source_catalogue, wanted_type_col, other_cols: [64,65,66])
fgl3_lst = dataset.get_all_list()
fgl3_df = pd.DataFrame(fgl3_lst, columns = ["3FGL", "CLASS1-3FGL", "3FGL-BSL", "3FGL-1FGL",
                                           "3FGL-2FGL"])
fgl3_df.to_csv('../CSV/fgl3_df.csv')
```

Call processing
method which
returns list

Get Class1, 0FGL,
1FGL, & 2FGL
name columns

Convert list
to dataframe

Output
dataframe to csv

Merge/Join BSL, 1FGL, & 2FGL with 3FGL datasets

Join BSL and 1FGL
datasets using
pandas.merge

Left join on BSL column.
This will capture all rows
from BSL dataset but only
matching rows from 1FGL.

```
print("=====Merge BSL, 1FGL, 2FGL, with 3FGL=====")
joined_BSL_1FGL_2FGL_3FGL = pd.merge(joined_BSL_1FGL_2FGL, fgl3_df, left_on='BSL',
                                     right_on='3FGL-BSL', how='left')
joined_BSL_1FGL_2FGL_3FGL.to_csv('../CSV/joined_BSL_1FGL_2FGL_3FGL.csv')
joined_BSL_1FGL_2FGL_3FGL_NONE_CLS1_BSL = joined_BSL_1FGL_2FGL_3FGL[joined_BSL_1FGL_2FGL_3FGL["CLASS1-BSL"]=='']
joined_BSL_1FGL_2FGL_3FGL_NONE_CLS1_BSL.to_csv('../CSV/joined_BSL_1FGL_2FGL_3FGL_NONE_CLS1_BSL.csv')
```

Joined BSL, 1FGL, 2FGL, 3FGL datasets

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	U
BSL	CLS1	1FGL	CLS1	1FGL	2FGL	CLS1	2FGL-BSL	2FGL-1FGL	3FGL	CLS1	3FGL-BSL	3FGL-1FGL	3FGL-2FGL	No change					
J0007.4+7303	PSR	J0007.0+7303	PSR	J0007.4+7303	J0007.0+7303	PSR	J0007.4+7303	J0007.0+7303	J0007.0+7302	PSR	J0007.4+7303	J0007.0+7303	J0007.0+7303	Change in source classification					
J0017.4-0503	bzq	J0017.4-0510	bzq	J0017.4-0503	J0017.6-0510	bzq	J0017.4-0503	J0017.4-0510	J0017.6-0512	FSRQ	J0017.4-0503	J0017.4-0510	J0017.6-0510	Source unidentified/unassociated throughout					
J0025.1-7202	glb	J0023.9-7204	glc	J0025.1-7202	J0023.9-7204	glc	J0025.1-7202	J0023.9-7204	J0023.9-7203	glc	J0025.1-7202	J0023.9-7204	J0023.9-7204	Source later unidentified/unassociated/dropped					
J1553.4+1255	bzq	J1553.4+1255	bzq	J1553.4+1255	J1553.5+1255	bzq	J1553.4+1255	J1553.4+1255	J1553.5+1256	fsrq	J1553.4+1255	J1553.4+1255	J1553.5+1255	Source later identified/associated					
J1555.8+1110	bzb	J1555.7+1111	bzb	J1555.8+1110	J1555.7+1111	bzb	J1555.8+1110	J1555.7+1111	J1555.7+1111	BLL	J1555.8+1110	J1555.7+1111	J1555.7+1111						
J1604.0-4904		J1603.8-4903	agu	J1604.0-4904	J1603.8-4904	bzb	J1604.0-4904	J1603.8-4903	J1603.9-4903	bil	J1604.0-4904	J1603.8-4903	J1603.8-4904						
J1615.6-5049	x				J1615.0-5051	spp	J1615.6-5049		J1616.2-5054e	PWN	J1615.6-5049	J1617.5-5105c	J1615.0-5051						
J1622.4-4945																			
J1625.8-2527	bzq	J1625.7-2524	bzq	J1625.8-2527	J1625.7-2526	bzq	J1625.8-2527	J1625.7-2524	J1625.7-2527	fsrq	J1625.8-2527	J1625.7-2524	J1625.7-2526						
J1625.9-2423	bzu	J1625.8-2429c		J1625.9-2423	J1627.0-2425c	agu	J1625.9-2423		J1626.2-2428c		J1625.9-2423	J1625.8-2429c							
J1634.9-4737																			
J1635.2+3809	bzq	J1635.0+3808	bzq	J1635.2+3809	J1635.2+3810	bzq	J1635.2+3809	J1635.0+3808	J1635.2+3809	FSRQ	J1635.2+3809	J1635.0+3808	J1635.2+3810						
J1641.4+3939	bzq				J1640.7+3945	BZQ	J1641.4+3939												
J1648.1-4606	x	J1648.4-4609c	psr	J1648.1-4606	J1648.4-4612	PSR	J1648.1-4606	J1648.4-4609c	J1648.3-4611	PSR	J1648.1-4606	J1648.4-4609c	J1648.4-4612						
J1653.4-0200		J1653.6-0158		J1653.4-0200	J1653.6-0159		J1653.4-0200	J1653.6-0158	J1653.6-0158		J1653.4-0200	J1653.6-0158	J1653.6-0159						
J1653.9+3946	bzb	J1653.9+3945	bzb	J1653.9+3946	J1653.9+3945	BZB	J1653.9+3946	J1653.9+3945	J1653.9+3945	BLL	J1653.9+3946	J1653.9+3945	J1653.9+3945						
J1709.7-4428	PSR	J1709.7-4429	PSR	J1709.7-4428	J1709.7-4429	PSR	J1709.7-4428	J1709.7-4429	J1709.7-4429	PSR	J1709.7-4428	J1709.7-4429	J1709.7-4429						
J1714.7-3827	x	J1714.5-3830c		J1714.7-3827	J1714.5-3829	spp	J1714.7-3827	J1714.5-3830c	J1714.5-3832	snr	J1714.7-3827	J1714.5-3830c	J1714.5-3829						
J1719.3+1746	bzb	J1719.2+1745	bzb	J1719.3+1746	J1719.3+1744	bzb	J1719.3+1746	J1719.2+1745	J1719.2+1744	bil	J1719.3+1746	J1719.2+1745	J1719.3+1744						
J1732.8-3135	PSR																		
J1741.4-3046					J1740.4-3054c	spp	J1741.4-3046	J1740.3-3053c	J1741.1-3053	snr	J1741.4-3046		J1740.4-3054c						
J1742.1-2054	PSR	J1741.8-2101	PSR	J1742.1-2054	J1741.9-2054	PSR	J1742.1-2054		J1741.9-2054	PSR	J1742.1-2054	J1741.8-2101	J1741.9-2054						
J1746.0-2900																			
J1751.5+0935	bzb	J1751.5+0937	bzb	J1751.5+0935	J1751.5+0938	bzb	J1751.5+0935	J1751.5+0937	J1751.5+0939	bil	J1751.5+0935	J1751.5+0937	J1751.5+0938						
J1801.6-2327	x	J1801.3-2322c	spp	J1801.6-2327	J1801.3-2326e	SNR	J1801.6-2327		J1801.3-2326e	SNR	J1801.6-2327	J1801.3-2322c	J1801.3-2326e						

Joined BSL, 1FGL, 2FGL, 3FGL datasets

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	F
1	BSL			1FGL				2FGL					3FGL					
2	BSL	CLS1		1FGL	CLS1	1FGL		2FGL	CLS1	2FGL-BSL	2FGL-1FGL		3FGL	CLS1	3FGL-BSL	3FGL-1FGL	3FGL-2FGL	
3	J0007.4+7303	PSR		J0007.0+7303	PSR	J0007.4+7303		J0007.0+7303	PSR	J0007.4+7303	J0007.0+7303		J0007.0+7302	PSR	J0007.4+7303	J0007.0+7303	J0007.0+7303	
4	J0017.4-0503	bzq		J0017.4-0510	bzq	J0017.4-0503		J0017.6-0510	bzq	J0017.4-0503	J0017.4-0510		J0017.6-0512	FSRQ	J0017.4-0503	J0017.4-0510	J0017.6-0510	
5	J0025.1-7202	glb		J0023.9-7204	glc	J0025.1-7202		J0023.9-7204	glc	J0025.1-7202	J0023.9-7204		J0023.9-7203	glc	J0025.1-7202	J0023.9-7204	J0023.9-7204	
39	J0457.1-2325	bzq		J0457.0-2325	bzq	J0457.1-2325		J0457.0-2325	bzq	J0457.1-2325	J0457.0-2325		J0457.0-2324	FSRQ	J0457.1-2325	J0457.0-2325	J0457.0-2325	
40	J0507.9+6739	bzb		J0507.9+6738	bzb	J0507.9+6739		J0508.0+6737	bzb	J0507.9+6739	J0507.9+6738		J0508.0+6736	blI	J0507.9+6739	J0507.9+6738	J0508.0+6737	
41	J0516.2-6200	bzu		J0516.7-6207	agu	J0516.2-6200		J0516.8-6207	bzb	J0516.2-6200	J0516.7-6207		J0516.7-6207	blI	J0516.2-6200	J0516.7-6207	J0516.8-6207	
42	J0531.0+1331	bzq		J0531.0+1331	bzq	J0531.0+1331		J0530.8+1333	bzq	J0531.0+1331	J0531.0+1331		J0530.8+1330	FSRQ	J0531.0+1331	J0531.0+1331	J0530.8+1333	
43	J0534.6+2201	PSR		J0534.5+2200	PSR	J0534.6+2201		J0534.5+2201	PSR	J0534.6+2201	J0534.5+2200		J0534.5+2201i	PWN	J0534.6+2201		J0534.5+2201	
44	J0534.6+2201	PSR		J0534.5+2200	PSR	J0534.6+2201		J0534.5+2201	PSR	J0534.6+2201	J0534.5+2200		J0534.5+2201s	PWN	J0534.6+2201		J0534.5+2201	
45	J0534.6+2201	PSR		J0534.5+2200	PSR	J0534.6+2201		J0534.5+2201	PSR	J0534.6+2201	J0534.5+2200		J0534.5+2201	PSR	J0534.6+2201	J0534.5+2200	J0534.5+2201	
46	J0538.4-6856			J0538.9-6914	gal	J0538.4-6856							J0526.6-6825e	GAL	J0538.4-6856	J0538.9-6914	J0526.6-6825e	
47	J0538.8-4403	bzb		J0538.8-4404	bzb	J0538.8-4403		J0538.8-4405	BZB	J0538.8-4403	J0538.8-4404		J0538.8-4405	BLL	J0538.8-4403	J0538.8-4404	J0538.8-4405	
48	J0613.9-0202	PSR		J0613.7-0200	PSR	J0613.9-0202		J0613.8-0200	PSR	J0613.9-0202	J0613.7-0200		J0613.8-0201	PSR	J0613.9-0202	J0613.7-0200	J0613.8-0200	
49	J0614.3-3330			J0614.1-3328		J0614.3-3330		J0614.1-3329	PSR	J0614.3-3330	J0614.1-3328		J0614.1-3329	PSR	J0614.3-3330	J0614.1-3328	J0614.1-3329	
50	J0617.4+2234	x		J0617.2+2233	SNR	J0617.4+2234		J0617.2+2234e	SNR	J0617.4+2234	J0617.2+2233		J0617.2+2234e	SNR	J0617.4+2234	J0617.2+2233	J0617.2+2234e	
51	J0631.8+1034	PSR		J0631.4+1037	PSR	J0631.8+1034		J0631.5+1035	PSR	J0631.8+1034	J0631.4+1037		J0631.5+1036	PSR	J0631.8+1034	J0631.4+1037	J0631.5+1035	
52	J0633.5+0634	PSR		J0633.7+0632	PSR	J0633.5+0634		J0633.7+0633	PSR	J0633.5+0634	J0633.7+0632		J0633.7+0632	PSR	J0633.5+0634	J0633.7+0632	J0633.7+0633	
53	J0634.0+1745	PSR		J0633.9+1746	PSR	J0634.0+1745		J0633.9+1746	PSR	J0634.0+1745	J0633.9+1746		J0633.9+1746	PSR	J0634.0+1745	J0633.9+1746	J0633.9+1746	
54	J0643.2+0858			J0643.2+0859	agu	J0643.2+0858		J0643.2+0858	bzq	J0643.2+0858	J0643.2+0859		J0643.2+0859	fsrq	J0643.2+0858	J0643.2+0859	J0643.2+0858	
55	J0654.3+4513	bzq		J0654.3+4514	bzq	J0654.3+4513		J0654.2+4514	bzq	J0654.3+4513	J0654.3+4514		J0654.4+4514	FSRQ	J0654.3+4513	J0654.3+4514	J0654.2+4514	
56	J0654.3+5042	bzu		J0654.4+5042	agu	J0654.3+5042		J0654.5+5043	bzq	J0654.3+5042	J0654.4+5042		J0654.4+5042	fsrq	J0654.3+5042	J0654.4+5042	J0654.5+5043	
57	J0700.0-6611	bzu		J0700.4-6611	agu	J0700.0-6611		J0700.3-6611	bzb	J0700.0-6611	J0700.4-6611		J0700.6-6610	blI	J0700.0-6611	J0700.4-6611	J0700.3-6611	

Found
dup rows

Joined BSL, 1FGL, 2FGL, 3FGL datasets

Green rows	29	No change
Yellow rows	120	Change in source classification
Blue rows	4	Source unidentified/unassociated throughout
Red rows	19	Source unidentified/unassociated/dropped (but maybe reassociated/identified later)
Gray rows	35	Source later identified/associated
Total rows	207	Due to 2 extra rows

Joined BSL, 1FGL, 2FGL, 3FGL datasets

Yellow rows: Change in source classification

Total: 120 rows

BSL	1FGL	2FGL	3FGL	Count
bzq	bzq	bzq	FSRQ	20
bzq	bzq	bzq	fsrq	37
bzq	bzq	bzb	BLL	1
bzq	agn	SEY	NLSY1	1
bzq	agu	bzq	fsrq	1
bzq	bzb	agu	BCU	1
bzb	bzb	bzb	bll	29
bzb	bzb	bzb	BLL	9
bzb	bzb	BZB	BLL	6

BSL	1FGL	2FGL	3FGL	Count
bzu	agu	bzb	bll	2
bzu	agu	bzq	fsrq	2
bzu	agu	bzq	bll	1
bzu	bzq	bzq	FSRQ	1
bzu	bzq	bzq	fsrq	3
HXB	HXB	HMB	HMB	1
hxb	HXB	HMB	HMB	1
PSR	PSR	PSR	PWN	2
rdg	agn	rdg	RDG	1
rdg	agn	RDG	RDG	1

Joined BSL, 1FGL, 2FGL, 3FGL datasets

Red rows: Source later unidentified/unassociated/dropped
(but maybe reassociated/identified later)

Total: 19 rows

BSL	1FGL	2FGL	3FGL	Count
bzb	No entry	bzb	bll	1
		agu	No entry	1
	No entry			1
x				1
bzb	No entry	No entry	bll	1
bzu		agu		1
	No entry	No entry	No entry	2
bzq	No entry	BZQ	No entry	1

BSL	1FGL	2FGL	3FGL	Count
PSR	No entry	No entry	No entry	3
PSR	PSR	No entry	No entry	1
	No entry	No entry		2
			No entry	1
PSR		PSR	PSR	1
bzu		PSR	PSR	1
		No entry	No entry	1

Joined BSL, 1FGL, 2FGL, 3FGL datasets

Grey rows: Source later identified/associated

Total: 35 rows

BSL	1FGL	2FGL	3FGL	Count
			BIN	1
			PSR	3
			psr	1
		PSR	PSR	5
		psr	PSR	1
	No entry	spp	snr	1
	agu	No entry	bcu	1
	agu	bzb	bl	3
	agu	agu	bcu	1
	agu	bzq	fsrq	2
	gal	No entry	GAL	1
	PSR	PSR	PSR	3

BSL	1FGL	2FGL	3FGL	Count
	spp	SNR	SNR	1
x	SNR	SNR	SNR	2
x	spp	HMB	HMB	1
x	spp	SNR	SNR	1
x	spp	No entry	SNR	1
x	SNR	No entry	SNR	1
x		snr	snr	1
x	No entry	spp	PWN	1
x		spp	snr	1
x	PSR	PSR	PSR	1
x	psr	PSR	PSR	1

Update: 07/19/2024

Update

- Completed mapping from 3FGL to 4FGL-DR1, DR2, DR3, DR4
- <https://docs.google.com/spreadsheets/d/1zCkiX0jAg3HXRGd7OMma0AwSqxiUV39UADE2tSyR2Ro/edit?gid=1225144226#gid=1225144226>

Dataset row counts

Color	Description	Count
cyan	No change	1487
green	All present but have changes in associations	449
orange	Associated later	384
light magenta	All null class1	247
Yellow	Entry dropped	553
Red	Disassociated later	22
Light gray	Association dropped and reassociated	13
Total		3155